



STATE OF CALIFORNIA
DEPARTMENT OF TRANSPORTATION

**NOTICE TO BIDDERS
AND
SPECIAL PROVISIONS**

**FOR CONSTRUCTION ON STATE HIGHWAY IN HUMBOLDT COUNTY AT
VARIOUS LOCATIONS FROM 2.4 MILES NORTH OF THE MENDOCINO
COUNTY BORDER TO EEL RIVER BRIDGE 04-0016R**

In District 01 On Route 101

Under

Bid book dated September 30, 2024

Standard Specifications dated 2023

Project plans approved June 17, 2024

Standard Plans dated 2023

Identified by

Contract No. 01-0H6404

01-Hum-101-2.4/M53.9

Project ID 0117000245

Federal-Aid Project

ACPTNH-Q101(436)E

**Bids open Wednesday, November 13, 2024
Dated September 30, 2024**

SPECIAL NOTICES

- See sections 2-1.11 and 2-1.33B for in-use off-road diesel-fueled vehicle requirements regarding Certificate of Reported Compliance submittals.
- See sections 2 and 3 for contractors' registration requirements.
- The Department advises bidders that potential claim records must be submitted by the contractor using the Department's Internet potential claim system.
- See section 2 for submittal requirements for DBE quotes, DVBE quotes, and Non-Small Business Subcontractor Preference.
- For work plan for local material from (1) a noncommercial source or (2) a source not regulated under California jurisdiction, see section 6-1.03B(1).
- See section 7-1.02K(3) for the requirements for electronic submittal of certified payroll records using LCPTracker Pro.
- The flagging and temporary traffic control requirements have been revised. See sections 7-1.03, 7-1.04, and 12.
- See section 14-11.14 for changes to the management of treated wood waste.
- The schedules for the submittal of DBE forms have been revised. See section 2-1.33 for the submittal schedules.

CONTRACT NO. 01-0H6404

The Special Provisions contained herein have been prepared by or under the direction of the following Registered/Licensed Professionals.

HIGHWAY


06/28/2024
REGISTERED CIVIL ENGINEER DATE



TRAFFIC


6/24/2024
REGISTERED CIVIL ENGINEER DATE



LANDSCAPE

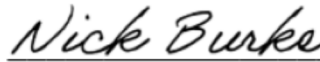

06-15-2024
LICENSED LANDSCAPE ARCHITECT DATE



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STANDARD PLANS LIST

The standard plan sheets applicable to this Contract include those listed below. The applicable revised standard plans (RSPs) listed below are included in the project plans.

A3A	Abbreviations (Sheet 1 of 3)
A3B	Abbreviations (Sheet 2 of 3)
A3C	Abbreviations (Sheet 3 of 3)
A10A	Legend - Lines and Symbols (Sheet 1 of 5)
A10B	Legend - Lines and Symbols (Sheet 2 of 5)
A10C	Legend - Lines and Symbols (Sheet 3 of 5)
A10D	Legend - Lines and Symbols (Sheet 4 of 5)
A10E	Legend - Lines and Symbols (Sheet 5 of 5)
A62A	Excavation and Backfill - Miscellaneous Details
A62D	Excavation and Backfill - Concrete Pipe Culverts
A62F	Excavation and Backfill - Metal and Plastic Culverts
A63A	Portable Concrete Barrier (Type 60K)
A73B	Markers
A87B	Hot Mix Asphalt Dikes
P52	Edge Drain Cleanout and Vent Details
P70	Hot Mix Asphalt Paving (Longitudinal Tapered Notched Wedge Joint)
D71	Drainage Inlet Markers
D73A	Precast Drainage Inlets - Types OS, OL and GOL
D73C	Precast Drainage Inlets - Types G1, G2, G3, G4, G5 and G6
D73E	Precast Drainage Inlets - Types GO and GDO
D73G	Precast Drainage Inlet Tables
D75A	Steel Pipe Inlets
D75C	Pipe Inlets - Ladder and Trash Rack Details
D77A	Grate Details No. 1
D77B	Grate Details No. 2
D78A	Gutter Depressions
D78C	Inlet Depressions - Hot Mix Asphalt Shoulders
RSP D86B	Pipe Culvert Headwalls, Endwalls and Warped Wingwalls
D87A	Corrugated Metal Pipe Downrain Details
D87C	Cable Anchorage System
D87D	Overside Drains

D88	Construction Loads on Culverts
D89A	Pipe Culvert Headwalls - Straight and "L"
D89B	Pipe Culvert Headwalls - Straight and "L"
RSP D90	Pipe Culvert Headwalls, Endwalls and Wingwalls - Types A, B and C
D93A	Pipe Riser Connections
D93B	Drainage Inlet Riser Connections
D94A	Metal and Plastic Flared End Sections
D94B	Concrete Flared End Sections
D97A	Corrugated Metal Pipe Coupling Details No. 1 - Annular Coupling Band Bar and Strap and Angle Connections
D97C	Corrugated Metal Pipe Coupling Details No. 3 - Helical and Universal Couplers
D97D	Corrugated Metal Pipe Coupling Details No. 4 - Hugger Coupling Bands
D97E	Corrugated Metal Pipe Coupling Details No. 5 - Standard Joint
D97F	Corrugated Metal Pipe Coupling Details No. 6 - Positive Joint
D97G	Corrugated Metal Pipe Coupling Details No. 7 - Downdrain
D98A	Slotted Corrugated Steel Pipe Drain Details
D102	Underdrains
H1	Landscape and Erosion Control Symbols
H2	Landscape Details
H3	Landscape Details
H51	Erosion Control Details - Fiber Roll and Compost Sock
H52	Rolled Erosion Control Product
RSP T1A	Temporary Crash Cushion, Sand Filled (Unidirectional)
T1A1	Temporary Crash Cushion, Sand Filled (Unidirectional)
T1B	Temporary Crash Cushion, Sand Filled (Bidirectional)
T2	Temporary Crash Cushion, Sand Filled (Shoulder Installations)
T3A	Temporary Railing (Type K)
T3B	Temporary Railing (Type K)
T3C	Temporary Barrier System (Cross Bolt)
T3D	Temporary Barrier System (Cross Bolt)
T3E	Temporary Barrier System (Cross Bolt)
T9	Traffic Control System Tables for Lane and Ramp Closures
T10	Traffic Control System for Lane Closure on Freeways and Expressways
T12	Traffic Control System for Half Road Closure on Multilane Conventional Highways and Expressways
T13	Traffic Control System with Reversible Control on Two Lane Conventional Highways

T13A	Traffic Control System - Two Lane Conventional Highways
T14	Traffic Control System for Ramp Closure
T15	Traffic Control System for Moving Lane Closure on Multilane Highways
T16	Traffic Control System for Moving Lane Closure on Multilane Highways
T17	Traffic Control System for Moving Lane Closure on Two Lane Highways
T18	Traffic Control System Construction Work Zone Speed Limit Reduction on Freeways and Expressways
T22	Traffic Control System for Construction Work Zone Speed Limit Reduction on Two Lane Conventional Highways
T51	Temporary Water Pollution Control Details (Temporary Silt Fence)
T53	Temporary Water Pollution Control Details (Temporary Cover)
T56	Temporary Water Pollution Control Details (Temporary Fiber Roll)
T57	Temporary Water Pollution Control Details (Temporary Check Dam)
T58	Temporary Water Pollution Control Details (Temporary Construction Entrance)
T59	Temporary Water Pollution Control Details (Temporary Concrete Washout Facility)
T61	Temporary Water Pollution Control Details (Temporary Drainage Inlet Protection)
T62	Temporary Water Pollution Control Details (Temporary Drainage Inlet Protection)
T63	Temporary Water Pollution Control Details (Temporary Drainage Inlet Protection)
T64	Temporary Water Pollution Control Details (Temporary Drainage Inlet Protection)
T65	Temporary Water Pollution Control Details (Temporary High-Visibility Fence)
T67	Temporary Water Pollution Control Details (Temporary Construction Roadway)
B11-47	Cable Railing
RS1	Roadside Signs - Typical Installation Details No. 1
RS2	Roadside Signs - Wood Post - Typical Installation Details No. 2
RS4	Roadside Signs - Typical Installation Details No. 4

CANCELED STANDARD PLANS LIST

The standard plan sheets listed below are canceled and not applicable to this contract.

Plan No.	Date Canceled	Plan No.	Date Canceled	Plan No.	Date Canceled
A77U1	01-22-24				
A77U2	01-22-24				
A77U3A	01-22-24				
A77U3B	01-22-24				
A77U4	01-22-24				
A77U5	01-22-24				
A77V1	01-22-24				
A77V2	01-22-24				
A78A	01-22-24				
A78B	01-22-24				
A78C1	01-22-24				
A78C2	01-22-24				
A78D1	01-22-24				
A78D2	01-22-24				
A78E1	01-22-24				
A78E2	01-22-24				
A78E3	01-22-24				
A78F1	01-22-24				
A78F2	01-22-24				
A78GA	01-22-24				
A78GB	01-22-24				
A78H	01-22-24				
A78J	01-22-24				
A78K	01-22-24				

NOTICE TO BIDDERS

Bids open Wednesday, November 13, 2024

Dated September 30, 2024

General work description: WSP trenchless installation, CSP, and erosion control.

The Department will receive sealed bids for CONSTRUCTION ON STATE HIGHWAY IN HUMBOLDT COUNTY AT VARIOUS LOCATIONS FROM 2.4 MILES NORTH OF THE MENDOCINO COUNTY BORDER TO EEL RIVER BRIDGE 04-0016R.

District-County-Route-Post Mile: 01-Hum-101-2.4/M53.9

Contract No. 01-0H6404

The Contractor must have either a Class A license or the following Class C license which constitutes a majority of the work: C-42.

The DBE Contract goal is 21 percent.

Federal-aid project no.:

ACPTNH-Q101(436)E

Bids must be on a cost+time basis.

Complete the work within the number of working days bid.

Do not bid less than 165 working days and not more than 235 working days.

The estimated cost of the project is \$13,900,000.

The Department will receive bids until 2:00 p.m. on the bid open date via Bid Express website. Bids received after this time will not be accepted. For more information refer to the Electronic Bidding Guide at the Office Engineer's website.

The Department will open and publicly read the bids through webcast/teleconference services immediately after the specified closing time.

For bid results go to:

<http://ppmoe.dot.ca.gov/des/oe/contractor-info.html>

Select *Electronic Bidding* under the *Bidding* tab.

District office addresses are provided in the *Standard Specifications*.

Present bidders' inquiries to the Department and view the Department's responses at:

<http://ppmoe.dot.ca.gov/des/oe/bid-inquiries.php>

Questions about alleged patent ambiguity of the plans, specifications, or estimate must be asked before bid opening. After bid opening, the Department does not consider these questions as bid protests.

Submit your bid with bidder's security equal to at least 10 percent of the bid.

Prevailing wages are required on this Contract. The Director of the California Department of Industrial Relations determines the general prevailing wage rates. Obtain the wage rates at the DIR website, <http://www.dir.ca.gov>, or from the Department's Labor Compliance Office of the district in which the work is located.

The federal minimum wage rates for this Contract as determined by the United States Secretary of Labor are available at <https://sam.gov/content/home>.

If the minimum wage rates as determined by the United States Secretary of Labor differs from the general prevailing wage rates determined by the Director of the California Department of Industrial Relations for similar classifications of labor, the Contractor and subcontractors must not pay less than the higher wage rate. The Department does not accept lower State wage rates not specifically included in the federal minimum wage determinations. This includes helper, or other classifications based on hours of experience, or any other classification not appearing in the federal wage determinations. Where federal wage determinations do not contain the State wage rate determination otherwise available for use by the Contractor and subcontractors, the Contractor and subcontractors must not pay less than the federal minimum wage rate that most closely approximates the duties of the employees in question.

The Department has made available Notices of Suspension and Proposed Debarment from the Federal Highway Administration. For a copy of the notices, go to http://www.dot.ca.gov/hq/esc/oe/contractor_info. Additional information is provided in the Excluded Parties List System at <https://sam.gov/content/home>.

Caltrans and the Construction Industry are committed to making partnering the way we do business. For more information, go to <http://www.dot.ca.gov/hq/construc/partnering.html>.

Department of Transportation

D01BA

BID ITEM LIST

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0001	070030	LEAD COMPLIANCE PLAN	LS	LUMP SUM
0002	080060	LEVEL 2 CRITICAL PATH METHOD SCHEDULE	LS	LUMP SUM
0003	090105	TIME-RELATED OVERHEAD (LS)	LS	LUMP SUM
0004	090200	DISPUTE RESOLUTION ADVISOR ON-SITE MEETING	EA	12
0005	090207	HOURLY OFF-SITE DISPUTE-RESOLUTION-ADVISOR-RELATED TASKS	HR	20
0006	100100	DEVELOP WATER SUPPLY	LS	LUMP SUM
0007	120090	CONSTRUCTION AREA SIGNS	LS	LUMP SUM
0008	120100	TRAFFIC CONTROL SYSTEM	LS	LUMP SUM
0009	120103	STATIONARY IMPACT ATTENUATOR VEHICLE	DAY	120
0010	010413	PORTABLE RADAR SPEED FEEDBACK SIGN SYSTEMS (LS)	LS	LUMP SUM
0011	128652	PORTABLE CHANGEABLE MESSAGE SIGN (LS)	LS	LUMP SUM
0012	129100	TEMPORARY CRASH CUSHION MODULE	EA	50
0013	129161	AUTOMATED FLAGGER ASSISTANCE DEVICE DAY	EA	10
0014	130100	JOB SITE MANAGEMENT	LS	LUMP SUM
0015	130201	WATER POLLUTION CONTROL PROGRAM	LS	LUMP SUM
0016	130510	TEMPORARY MULCH	SQYD	10,500
0017	130570	TEMPORARY COVER	SQYD	10,500
0018	130610	TEMPORARY CHECK DAM	LF	860
0019	130620	TEMPORARY DRAINAGE INLET PROTECTION	EA	64
0020	130640	TEMPORARY FIBER ROLL	LF	510

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0021	130650	TEMPORARY GRAVEL BAG BERM	LF	80
0022	130680	TEMPORARY SILT FENCE	LF	5,950
0023	130710	TEMPORARY CONSTRUCTION ENTRANCE	EA	38
0024	130720	TEMPORARY CONSTRUCTION ROADWAY	SQYD	5,920
0025	130730	STREET SWEEPING	LS	LUMP SUM
0026	130900	TEMPORARY CONCRETE WASHOUT	LS	LUMP SUM
0027	131103	WATER QUALITY SAMPLING AND ANALYSIS DAY	EA	66
0028	131104	WATER QUALITY MONITORING REPORT	EA	17
0029	131105	WATER QUALITY ANNUAL REPORT	EA	3
0030	131201	TEMPORARY CREEK DIVERSION SYSTEMS	LS	LUMP SUM
0031	146002	CONTRACTOR-SUPPLIED BIOLOGIST (LS)	LS	LUMP SUM
0032	146007	INVASIVE SPECIES CONTROL	LS	LUMP SUM
0033	013843	CONTRACTOR SUPPLIED ARBORIST	LS	LUMP SUM
0034	160110	TEMPORARY HIGH-VISIBILITY FENCE	LF	1,780
0035	160220	TEMPORARY CONSTRUCTION MAT (LS)	LS	LUMP SUM
0036	170103	CLEARING AND GRUBBING (LS)	LS	LUMP SUM
0037	194001	DITCH EXCAVATION	CY	90
0038	205035	WOOD MULCH	CY	1,930
0039	210010	MOVE-IN/MOVE-OUT (EROSION CONTROL)	EA	3
0040	037634	HYDRAULIC BIOTIC GROWTH MEDIUM	SQFT	84,500

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0041	210255	FIBER REINFORCED MATRIX (SQFT)	SQFT	84,500
0042	210270	ROLLED EROSION CONTROL PRODUCT (NETTING)	SQFT	130,000
0043	210350	FIBER ROLLS	LF	7,350
0044	210610	COMPOST (CY)	CY	560
0045	210630	INCORPORATE MATERIALS	SQFT	73,400
0046	260203	CLASS 2 AGGREGATE BASE (CY)	CY	10
0047	390132	HOT MIX ASPHALT (TYPE A)	TON	2.7
0048	394073	PLACE HOT MIX ASPHALT DIKE (TYPE A)	LF	78
0049	394077	PLACE HOT MIX ASPHALT DIKE (TYPE F)	LF	41
0050	394090	PLACE HOT MIX ASPHALT (MISCELLANEOUS AREA)	SQYD	13
0051	398100	REMOVE ASPHALT CONCRETE DIKE	LF	64
0052	510050	(F) - STRUCTURAL CONCRETE	CY	6
0053	510092	(F) - STRUCTURAL CONCRETE, HEADWALL	CY	42
0054	510094	(F) - STRUCTURAL CONCRETE, DRAINAGE INLET	CY	51
0055	510501	(F) - MINOR CONCRETE	CY	1
0056	600011	RAPID SETTING CONCRETE (PATCH)	CF	110
0057	610101	8" ALTERNATIVE PIPE CULVERT	LF	3
0058	610112	24" ALTERNATIVE PIPE CULVERT	LF	520
0059	610121	36" ALTERNATIVE PIPE CULVERT	LF	73
0060	610300	(F) - CONCRETE BACKFILL (PIPE TRENCH)	CY	87

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0061	015420	CONCRETE COLLAR	EA	1
0062	665008	10" CORRUGATED STEEL PIPE (.064" THICK)	LF	10
0063	665012	12" CORRUGATED STEEL PIPE (.079" THICK)	LF	24
0064	665018	18" CORRUGATED STEEL PIPE (.109" THICK)	LF	20
0065	665024	24" CORRUGATED STEEL PIPE (.109" THICK)	LF	1,840
0066	665037	36" CORRUGATED STEEL PIPE (.109" THICK)	LF	520
0067	665043	42" CORRUGATED STEEL PIPE (.138" THICK)	LF	710
0068	665053	54" CORRUGATED STEEL PIPE (.168" THICK)	LF	190
0069	665055	60" CORRUGATED STEEL PIPE (.109" THICK)	LF	95
0070	680406	8" PERFORATED STEEL PIPE UNDERDRAIN (.079" THICK)	LF	92
0071	680412	12" PERFORATED STEEL PIPE UNDERDRAIN (.079" THICK)	LF	10
0072	690106	8" CORRUGATED STEEL PIPE DOWNDRAIN (.079" THICK)	LF	5
0073	690124	24" CORRUGATED STEEL PIPE DOWNDRAIN (.109" THICK)	LF	1,290
0074	690132	30" CORRUGATED STEEL PIPE DOWNDRAIN (.109" THICK)	LF	5
0075	690137	36" CORRUGATED STEEL PIPE DOWNDRAIN (.109" THICK)	LF	370
0076	690142	42" CORRUGATED STEEL PIPE DOWNDRAIN (.109" THICK)	LF	180
0077	017097	60" CORRUGATED STEEL PIPE DOWNDRAIN (0.109" THICK)	LF	150
0078	013917	CABLE ANCHORAGE SYSTEM (MOD)	EA	22
0079	700615	CORRUGATED STEEL PIPE INLET	LF	71
0080	014911	36" WELDED STEEL PIPE TRENCHLESS INSTALLATION METHOD	LF	440

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0081	014912	42" WELDED STEEL PIPE TRENCHLESS INSTALLATION METHOD	LF	150
0082	017096	60" WELDED STEEL PIPE TRENCHLESS INSTALLATION METHOD	LF	680
0083	705015	24" STEEL FLARED END SECTION	EA	15
0084	705023	36" STEEL FLARED END SECTION	EA	1
0085	705027	42" STEEL FLARED END SECTION	EA	2
0086	705035	54" STEEL FLARED END SECTION	EA	1
0087	038552	60" STEEL FLARED END SECTION	EA	5
0088	710102	ABANDON CULVERT (LF)	LF	2,400
0089	710132	REMOVE CULVERT (LF)	LF	4,500
0090	710138	REMOVE DOWNDRAIN (EA)	EA	19
0091	710150	REMOVE INLET	EA	26
0092	710152	REMOVE HEADWALL	EA	4
0093	710160	REMOVE JUNCTION BOX	EA	3
0094	710165	REMOVE CONCRETE FLARED END SECTION (EA)	EA	1
0095	710167	REMOVE FLARED END SECTION (EA)	EA	15
0096	710366	CONCRETE INVERT PAVING	CY	35
0097	710368	CULVERT SLURRY-CEMENT BACKFILL	CY	610
0098	721431	CONCRETE (CONCRETE APRON)	CY	22
0099	013729	GRAVEL FILTER	CY	190
0100	723015	ROCK SLOPE PROTECTION (2T, CLASS IX, METHOD A)	CY	62

Item No.	Item Code	Item Description	Unit of Measure	Estimated Quantity
0101	723020	ROCK SLOPE PROTECTION (1T, CLASS VIII, METHOD A)	CY	330
0102	723030	ROCK SLOPE PROTECTION (1/2 T, CLASS VII, METHOD A) (CY)	CY	68
0103	723050	ROCK SLOPE PROTECTION (1/4 T, CLASS V, METHOD B) (CY)	CY	180
0104	723060	ROCK SLOPE PROTECTION (300 LB, CLASS IV, METHOD B) (CY)	CY	24
0105	723070	ROCK SLOPE PROTECTION (150 LB, CLASS III, METHOD B) (CY)	CY	45
0106	723080	ROCK SLOPE PROTECTION (60 LB, CLASS II, METHOD B) (CY)	CY	240
0107	723095	ROCK SLOPE PROTECTION (20 LB, CLASS I, METHOD B) (CY)	CY	11
0108	750001	(F) - MISCELLANEOUS IRON AND STEEL	LB	7,660
0109	820112	MARKER (CULVERT)	EA	66
0110	820113	TREATMENT BEST MANAGEMENT PRACTICE MARKER	EA	8
0111	820220	REMOVE MARKER	EA	66
0112	839521	(F) - CABLE RAILING	LF	88
0113	870009	MAINTAINING EXISTING TRAFFIC MANAGEMENT SYSTEM ELEMENTS DURING CONSTRUCTION	LS	LUMP SUM
0114	999990	MOBILIZATION	LS	LUMP SUM

SPECIAL PROVISIONS

ORGANIZATION

Special provisions are under headings that correspond with the main-section headings of the *Standard Specifications*. A main-section heading is a heading shown in the table of contents of the *Standard Specifications*.

Each special provision begins with a revision clause that describes or introduces a revision to the *Standard Specifications*.

Any paragraph added or deleted by a revision clause does not change the paragraph numbering of the *Standard Specifications* for any other reference to a paragraph of the *Standard Specifications*.

AA

DIVISION I GENERAL PROVISIONS

1 GENERAL

Add to section 1-1.01:

Bid Items and Applicable Sections

Item code	Item description	Applicable section
090200	DISPUTE RESOLUTION ADVISOR ON-SITE MEETING	5
090207	HOURLY OFF-SITE DISPUTE-RESOLUTION-ADVISOR-RELATED TASKS	5
010413	PORTABLE RADAR SPEED FEEDBACK SIGN SYSTEMS (LS)	12
013843	CONTRACTOR SUPPLIED ARBORIST	14
037634	HYDRAULIC BIOTIC GROWTH MEDIUM	21
015420	CONCRETE COLLAR	61
013917	CABLE ANCHORAGE SYSTEM (MOD)	69
017097	60" CORRUGATED STEEL PIPE DOWNDRAIN (0.109" THICK)	69
014911	36" WELDED STEEL PIPE TRENCHLESS INSTALLATION METHOD	70
014912	42" WELDED STEEL PIPE TRENCHLESS INSTALLATION METHOD	70
017096	60" WELDED STEEL PIPE TRENCHLESS INSTALLATION METHOD	70
038552	60" STEEL FLARED END SECTION	70
013729	GRAVEL FILTER	72
999990	MOBILIZATION	9

AA

2 BIDDING

Add between the 1st and 2nd paragraphs of section 2-1.06B:

The Department makes the following supplemental project information available:

Supplemental Project Information

Means	Description
Included in the <i>Information Handout</i>	1. PLAC - U.S. Army Corps of Engineers 404, South Pacific Division, Nationwide Permit No. 3, SPN-2023-00464, dated 5/31/2024 2. PLAC – California Department of Parks and Recreation, Right of Entry Permit, dated 3/11/24 3. PLAC - California Regional Water Quality Control Board, North Coast Region, Water Quality Certification, WDID No. 1B23161WNHU, dated 4/2/2024 4. PLAC - California Department of Fish and Wildlife, Northern Region, 1602 Streambed Alteration Agreement, No. EPIMS-HUM-39108-R1C, dated 2/16/2024 5. PLAC - California Department of Parks and Recreation, Section 4(f) de minimis Evaluation, dated 4/4/2023 6. Geotechnical Design Report, dated 4/5/2024 7. TMS Element Status and Loops Elements Status 8. Hydrology Analysis for Culverts for Temporary Creek Diversion System 9. Aerially Deposited Lead Site Investigation Report, dated May 2021 10. Tunnel Safety Orders
Available for inspection at: 5900 Folsom Blvd. Sacramento, CA 95819 Telephone no.: (916) 227-1000	Core from Geotechnical Borings

Replace section 2-1.11 with:

2-1.11 IN-USE OFF-ROAD DIESEL-FUELED VEHICLE LIST

Section 2-1.11 applies to non-informal-bid contracts.

Complete and submit the In-Use Off-Road Diesel-Fueled Vehicle List form under section 2-1.33.

On the In-Use Off-Road Diesel-Fueled Vehicle List form, list each fleet used by you or your subcontractor to perform work and is subject to 13 CCR § 2449 et seq. Submit a copy of a valid Certificate of Reported Compliance (13 CCR § 2449, subdivision (n)) for each fleet listed on the form within 10 days of bid opening. Failure to list a fleet used by you or your subcontractor to perform work on the In-Use Off-Road Diesel-Fueled Vehicle List form may result in a nonresponsive bid. Failure to submit the Certificate of Reported Compliance for a fleet listed on the In-Use Off-Road Diesel-Fueled Vehicle List form may result in a nonresponsive bid.

Replace section 2-1.18 with:

2-1.18 SMALL BUSINESS AND NON-SMALL BUSINESS SUBCONTRACTOR PREFERENCES

2-1.18A General

Section 2-1.18 applies to a non-federal-aid contract.

The Department applies small business preferences and non-small business preferences under Govt Code § 14835 et seq. and 2 CA Code of Regs § 1896 et seq.

Any contractor, subcontractor, supplier, or service provider who qualifies as a small business or small business for the purpose of public works is encouraged to apply for certification as an SBE by submitting its application to the Department of General Services, Office of Small Business and DVBE Services.

The Department of General Services determines whether an SBE was certified on the bid opening date. The Department of Transportation confirms an SBE's certification status before applying the small business or non-small business preference.

Contract award is based on the total bid, not the reduced bid.

Pursuant to Govt Code § 14838, SBE participation listed on the Small Business Enterprise – Commitment form used to meet the SBE participation goal requirement should not be listed on the Certified Small Business Listing for the Non-Small Business Preference form. Listing the same SBE participation on both the Small Business Enterprise – Commitment form and on the Certified Small Business Listing for the Non-Small Business Preference form may result in disallowance of the SBE's participation toward the small business or non-small business preference.

2-1.18B Small Business Preference

The Department allows a bidder certified as an SBE by the Department of General Services, Office of Small Business and DVBE Services, a preference if:

1. Bidder submitted a completed Request for Small Business Preference or Non-Small Business Preference form with its bid
2. Low bidder did not request the preference or is not certified as a small business

The Bidder's signature on the Request for Small Business Preference or Non-Small Business Preference form certifies that the Bidder is certified as an SBE at the date and time of bid or has submitted a complete application to the Department of General Services. The complete application and any required substantiating documentation must be received by the Department of General Services by 5:00 p.m. on the bid opening date.

The small business preference is a reduction for bid comparison in the total bid submitted by the small business contractor by the lesser of the following amounts:

1. 5 percent of the verified total bid of the low bidder
2. \$50,000

If the Department determines that a certified small business bidder is the low bidder after the application of the small business preference, the Department does not consider a request for non-small business preference.

2-1.18C Non-Small Business Subcontractor Preference

At time of bid, the Department allows a bidder not certified as an SBE by the Department of General Services, Office of Small Business and DVBE Services, the option of requesting the non-small business subcontractor preference. If a bidder elects to pursue that option at the time of bid, the bidder is required to meet the corresponding commitment.

For the non-small business subcontractor preference, submit the following:

1. Completed Request for Small Business Preference or Non-Small Business Preference form with your bid
2. Completed Certified Small Business Listing for the Non-Small Business Preference form which shows you are subcontracting at least 25 percent to certified SBEs

Each listed SBE must be certified as a small business or small business for the purpose of public works by 5:00 p.m. on the bid opening date. The complete application and any required substantiating documentation must be received by the Department of General Services by 5:00 p.m. on the bid opening date.

Failure to submit a completed Certified Small Business Listing for the Non–Small Business Preference form by 4 p.m. on the 4th business day after bid opening will result in a nonresponsive bid.

The non–small business subcontractor preference is a reduction for bid comparison in the total bid submitted by the non–small business contractor requesting the preference by the lesser of the following amounts:

1. 5 percent of the verified total bid of the low bidder
2. \$50,000

Replace section 2-1.33B with:

2-1.33B Bid Form Submittal Schedules

2-1.33B(1) General

The *Bid* book includes forms specific to the Contract. The deadlines for the submittal of the forms vary depending on the requirements of each Contract. Determine the requirements of the Contract and submit the forms based on the applicable schedule specified in section 2-1.33B.

Bid forms and information on the form that are due after the time of bid may be submitted at the time of bid.

2-1.33B(2) Federal-Aid Contracts

2-1.33B(2)(a) General

Section 2-1.33B(2) applies to a federal-aid contract.

2-1.33B(2)(b) Contracts with a DBE Goal

2-1.33B(2)(b)(i) General

Section 2-1.33B(2)(b) applies if a DBE goal is shown on the *Notice to Bidders*.

2-1.33B(2)(b)(ii) Bid Form Submittal

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Federal-Aid Contract with a DBE Goal

Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number
Copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid except for the public works contractor registration number
Copy of the Subcontractor List as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Small Business Status	Time of bid
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid
DBE Commitment	No later than 4 p.m. on the 5th day after bid opening ^b
DBE Confirmation	No later than 4 p.m. on the 5th day after bid opening ^b
DBE Good Faith Efforts Documentation	No later than 4 p.m. on the 5th day after bid opening ^b

^aSubmit only if you choose the option.

^bIf the last day for submitting the bid form falls on a Saturday or holiday, it may be submitted on the next business day with the same effect as if it had been submitted on the day specified.

2-1.33B(2)(b)(iii) Reserved**2-1.33B(2)(c) Contracts without a DBE Goal****2-1.33B(2)(c)(i) General**

Section 2-1.33B(2)(c) applies if a DBE goal is not shown on the *Notice to Bidders*.

2-1.33B(2)(c)(ii) Bid Form Schedule

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Federal-Aid Contract without a DBE Goal	
Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number
Copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid except for the public works contractor registration number
Copy of the Subcontractor List as submitted at the time of bid with the public works contractor registration numbers	10 days after bid opening
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Small Business Status	Time of bid
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid

^aSubmit only if you choose the option.

2-1.33B(2)(c)(iii) Reserved**2-1.33B(2)(d)–2-1.33B(2)(h) Reserved****2-1.33B(3) Non-Federal-Aid Contracts****2-1.33B(3)(a) General**

Section 2-1.33B(3) applies to non-federal-aid contracts.

2-1.33B(3)(b) Contracts with a DVBE Goal**2-1.33B(3)(b)(i) General**

Section 2-1.33B(3)(b) applies if a DVBE goal is shown on the *Notice to Bidders*.

2-1.33B(3)(b)(ii) Bid Form Submittal

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Non-Federal-Aid Contract with a DVBE Goal

Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number for a joint-venture contract
For a joint-venture contract, copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid
Certified DVBE Summary	No later than 4 p.m. on the 4th business day after bid opening
California Company Preference	Time of bid
Small Business Enterprise – Commitment	No later than 4 p.m. on the 4th business day after bid opening
Small Business Enterprise – Confirmation	No later than 4 p.m. on the 4th business day after bid opening
Request for Small Business Preference or Non–Small Business Preference ^a	Time of bid
Certified Small Business Listing for the Non–Small Business Preference ^a	No later than 4 p.m. on the 4th business day after bid opening

^aSubmit only if you choose the option or preference.

2-1.33B(3)(b)(iii) Reserved

2-1.33B(3)(c) Contracts without a DVBE Goal

2-1.33B(3)(c)(i) General

Section 2-1.33B(3)(c) applies if a DVBE goal is not shown on the *Notice to Bidders*.

2-1.33B(3)(c)(ii) Bid Form Submittal

Submit the bid forms according to the schedule shown in the following table:

Bid Form Submittal Schedule for a Non-Federal-Aid Contract without a DVBE Goal

Form	Submittal deadline
Bid to the Department of Transportation	Time of bid except for the public works contractor registration number for a joint-venture contract
For a joint-venture contract, copy of the Bid to the Department of Transportation as submitted at the time of bid with the public works contractor registration number	10 days after bid opening
Subcontractor List	Time of bid
In-Use Off-Road Diesel-Fueled Vehicle List	10 days after bid opening
Opt Out of Payment Adjustments for Price Index Fluctuations ^a	Time of bid
California Company Preference	Time of bid
Certified DVBE Summary ^b	No later than 4 p.m. on the 4th business day after bid opening
Small Business Enterprise – Commitment	No later than 4 p.m. on the 4th business day after bid opening
Small Business Enterprise – Confirmation	No later than 4 p.m. on the 4th business day after bid opening
Request for Small Business Preference or Non–Small Business Preference ^a	Time of bid
Certified Small Business Listing for the Non–Small Business Preference ^a	No later than 4 p.m. on the 4th business day after bid opening

^aSubmit only if you choose the option or preference.

^bSubmit only if you obtain DVBE participation or you are the apparent low bidder, 2nd low bidder, or 3rd low bidder and you choose to receive the specified incentive.

2-1.33B(3)(c)(iii) Reserved

2-1.33B(3)(d)–2-1.33B(3)(h) Reserved

2-1.33B(4)–2-1.33B(9) Reserved

AA

5 CONTROL OF WORK

Replace section 5-1.13B with:

5-1.13B Disadvantaged Business Enterprises

5-1.13B(1) General

Section 5-1.13B applies to a federal-aid contract.

Use each DBE as listed on the DBE Commitment form unless you receive Department prior authorization for termination under section 5-1.13B(2)(c). Ensure that all subcontracts and agreements with DBEs to supply labor or materials are performed under 49 CFR 26.

Maintain records of subcontracts made with DBE subcontractors and records of materials purchased from DBE suppliers. Include in the records:

1. Name and business address of each DBE subcontractor, DBE vendor, and DBE trucking company, regardless of tier
2. Date of payment and total amount paid to each DBE business

If you are a DBE contractor, include the date of work performed by your own forces and the corresponding value of the work.

Before the 15th day of each month for the previous month's work, submit:

1. Monthly DBE Trucking Verification form
2. Monthly DBE Payment form

If a DBE is decertified before completing its work, the business must notify you in writing of the decertification date within 15 days of decertification. Notify the Engineer and submit the DBE's decertification notice within 2 business days of your receipt. Upon work completion, complete a Disadvantage Business Enterprises (DBE) Certification Status Change form and submit within 10 days of Contract acceptance.

Upon work completion, complete a Final Report – Utilization of Disadvantaged Business Enterprises (DBE), First-Tier Subcontractors form and submit within 10 days of Contract acceptance. The Department withholds the greater of 10 percent of the DBE commitment or \$10,000 until the form is submitted. The Department releases the withhold upon submission of the completed form. If additional payments are made to a DBE after submittal of the completed form, submit an updated form to reflect such payments.

Failure to carry out requirements of 49 CFR 26 is a material breach of the Contract, which may result in the termination of the Contract or other remedy as the Department deems appropriate, such as:

1. Withholding monthly progress payments
2. Assessing sanctions
3. Applying liquidated damages
4. Disqualification from future bidding as nonresponsive

5-1.13B(2) Disadvantaged Business Enterprises

5-1.13B(2)(a) General

Section 5-1.13(B)(2) applies if a DBE goal is shown on the *Notice to Bidders*.

Certification as a DBE identifies if the business has the means to perform its work under assigned North American Industry Classification System codes and work codes applicable to the type of work the DBE will perform on the Contract. Certification does not ensure the DBE will perform a commercially useful function on the Contract.

You are responsible for ensuring each DBE listed on the DBE Commitment form performs:

1. The description and value of the subcontracted work or material supplied as committed
2. A commercially useful function under 49 CFR 26.55 for committed work or materials

For DBE committed work, the Department only pays for work performed or supplied by the listed DBE and if a commercially useful function was performed by the listed DBE.

You are responsible to remediate noncompliant DBE work to meet your DBE commitment. Submit a DBE commitment remediation plan within 5 business days of the Engineer's request.

Pay your DBEs in conformance with section 5-1.13E.

Failure to promptly pay DBEs may result in a withholds corresponding to the value of the DBE's committed work from future progress payments. In addition, unpaid DBE amounts will not count towards your DBE commitment, which may result in equivalent withholds or deductions and a 2 percent penalty on the unpaid amount for every month payment is not made.

5-1.13B(2)(b) Commercially Useful Function

DBEs must perform a commercially useful function under 49 CFR 26.55 when performing work or supplying materials listed on the DBE Commitment form. The DBEs value of work will only count toward the DBE commitment if the DBE performs a commercially useful function under 49 CFR 26.55.

Provide written notification to the Engineer at least 15 days in advance of each DBE's initial performance of work or supplying materials for the Contract. Include the DBE's name, contract work to be performed, and the location, date, and time of where their work will take place.

Within 10 days of a DBE initially performing work or supplying materials on the Contract, submit your initial evaluation and validation of their performance of a commercially useful function using DBE Commercially Useful Function Evaluation form. Include the following supporting information with your submittal:

1. Subcontract agreement with the DBE
2. Purchase orders
3. Bills of lading
4. Invoices
5. Proof of payment

Monitor your DBEs' performance of commercially useful function with quarterly evaluations and validations throughout their duration of work on the Contract using DBE Commercially Useful Function Evaluation form. Submit your quarterly evaluation and validation DBE Commercially Useful Function Evaluation forms by the 5th of the month for the previous three month's work. Include any additional supplemental supporting information with your submittal. If your DBE's work-start and -end dates for the Contract exceed a three-month period, regardless of time not on the Contract, quarterly evaluations and validations are required.

Notify the Engineer immediately if you believe the DBE may not be performing a commercially useful function.

The Department will verify your DBEs performance of commercially useful functions by reviewing your initial and quarterly DBE Commercially Useful Function Evaluation forms, your submitted supporting information, field observations, and through select Department evaluations. The Department may evaluate DBEs and their commercially useful function performance at any time during the Contract. In such instances, the Department will provide written notice to you and your DBE at least 2 business days prior to the evaluation. You and your DBE must participate in the evaluation. Upon completing the evaluation, the Department will share the evaluation results with you and your DBE. The evaluation results may include items that must be remedied upon your receipt. If the Department determines the DBE is not performing a commercially function you must suspend performance of the noncompliant work.

You and your DBEs must submit any additional commercially useful function related records and documents within 5 business days of Department request such as:

1. Proof of ownership or lease and rental agreements for equipment
2. Tax records
3. Employee rosters
4. Certified payroll records
5. Inventory rosters

Failure to submit required DBE Commercially Useful Function Evaluation forms or requested records and documents will result in withhold of payment for the value of work completed by the DBE.

If you and or the Department determine a listed DBE is not performing a commercially useful function in performance of their DBE committed work, suspend performance of the noncompliant portion of the work. Submit a corrective action plan within 5 days of the noncompliant commercially useful function determination. The plan must identify how you will remediate when feasible or demonstrate commercially useful function compliance for the remaining portion of the DBE's work. Allow 5 days for plan review. The corrective actions must be implemented within 5 days of Engineer's authorization of your plan and prior to resumption of the noncompliant portion of the DBE's committed work.

If corrective actions cannot be accomplished to assure the DBE will perform a commercially useful function on the Contract, you may have good cause to request termination of the DBE under section 5-1.13B(2)(c).

5-1.13B(2)(c) Termination

Termination of a DBE may be allowable for good cause reasons under 49 CFR 26.53(f)(3) with prior written authorization from the Department. You must provide documentation supporting good cause reasoning with your termination request. If the termination request is authorized by the Department, you

must then either replace the DBE with another DBE or demonstrate good faith efforts to do so under 5-1.13B(2)(d).

Use the following procedure to request the termination of a DBE or portion of their work:

1. Provide written notice to the DBE of your intent to use other forces or material sources and include one or more of the good cause reasons under 49 CFR 26.53(f)(3). Simultaneously send a copy of this written notice to the Engineer. Your written notice to the DBE must request they provide any response to both you and the Engineer.
2. Provide the DBE with 5 business days to respond to your written notice by either acknowledging their agreement or documenting their reasoning as to why the use of other forces or sources of materials should not occur. If the DBE does not respond within 5 business days, you may move forward with the request process as if the DBE had agreed to your written notice.
3. Submit your DBE termination request by written letter to the Engineer and include:
 - 3.1. One or more good cause reasons identified under 49 CFR 26.53(f)(3) along with supporting documentation.
 - 3.2. Your written notice to the DBE regarding the request, including proof of transmission and tracking documentation of your written notice.
 - 3.3. The DBE's response to your written notice, if received. If a written response was not provided, provide a statement to that effect.

The Department will respond to your complete DBE termination request as follows:

1. Where the DBE has agreed in writing or fails to timely respond to your written notice, the Department will respond within 2 business days from receipt of your request.
2. Where the DBE has disagreed in writing with your written notice, the Department will meet with you and the DBE within 5 business days from receipt of your request. The Department will respond to your request within 5 business days from this meeting.
3. If you fail to provide a complete request for DBE termination the Department will identify deficiencies within 5 business days from receipt of your request.

If the Department authorizes your DBE termination request it will do so in writing.

Work performed by a firm other than the committed DBE or authorized replacement DBE without first obtaining Department authorization for termination will be a violation of these specifications and DBE federal regulations. Such violations will result in payment deductions for the value of the work associated with the noncompliant DBE commitment. In addition, if the committed DBE is also a listed subcontractor, the Department applies an additional penalty up to 10 percent of the value of the subject work as a permanent deduction.

5-1.13B(2)(d) Replacement

After receiving Department written authorization of your DBE termination request, you must obtain separate Department authorization of your replacement plan.

Your replacement plan must identify DBE replacement firms to perform the work or demonstrate that you have made a good faith effort to use DBE replacement firms. DBE replacement firms must:

1. Perform at least the same dollar amount of work as the terminated DBE to the extent needed to meet the DBE commitment
2. Possess certifications for the most specific available North American Industry Classification System codes and work codes applicable to the work the firm will perform on the Contract
3. Perform a commercially useful function under 49 CFR 26.55

Use the following procedure to request authorization of your replacement plan:

1. Submit a request to replace a DBE with other forces or material sources by written letter to the Department which must include:
 - 1.1. Description of remaining uncommitted item work made available for replacement DBE solicitation and participation.
 - 1.2. The proposed DBE replacement firm's business information, the work they have agreed to perform, and the following:

- 1.2.1. Quote for bid item work and description of work to be performed
- 1.2.2. Proposed subcontract agreement and written confirmation of agreement to perform on the Contract
- 1.2.3. Revised Subcontracting Request form
2. If you have not identified a DBE replacement firm, submit documentation of your good faith efforts to use DBE replacement firms within 7 days of Department's authorization to terminate the DBE. You may request the Department's approval to extend this submittal period to a total of 14 days. The Department considers your documented actions taken to identify a DBE replacement firm in determining whether a good faith effort was made under 49 CFR 26 app A. Submit documentation of actions taken to find a DBE replacement firm, such as:
 - 2.1. Search results of certified DBEs available to perform the original DBE work identified and or other work you had intended to self-perform, to the extent needed to meet your DBE commitment
 - 2.2. Solicitations of DBEs for performance of work identified in 2.1
 - 2.3. Correspondence with interested DBEs that may have included contract details and requirements
 - 2.4. Negotiation efforts with DBEs that reflect why an agreement was not reached
 - 2.5. If a DBE's quote was rejected, provide your reasoning for the rejection, such as why the DBE was unqualified for the work, or why the price quote was unreasonable or excessive
 - 2.6. Copies of each DBE's and non-DBE's price quotes for work identified in 2.1, as the Department may contact the firms to verify solicitation efforts and determine if the DBE quotes are substantially higher
 - 2.7. Additional documentation that you believe supports your good faith effort

The Department will respond to your complete replacement plan as follows:

1. If a DBE replacement firm has been identified and required documentation has been provided, the Department will respond within 2 business days from receipt of your plan
2. If a DBE replacement firm has not been identified, but good faith effort documents have been provided, the Department will respond within 5 business days from receipt of your plan
3. If you fail to provide a complete replacement plan, the Department will return your request and identify deficiencies within 5 business days from receipt of your plan

If the Department authorizes your replacement plan it will do so in writing.

Submit a revised Subcontracting Request form if your replacement plan is authorized.

DBE committed work performed by a nonauthorized firm, will be a violation of these specifications and DBE federal regulations. Such violations will result in payment deductions for the value of the work associated with the DBE commitment. The Department will take a permanent deduction for the value of the DBE work that was not performed by the authorized DBE. In addition, if the associated work was also to be performed by a listed subcontractor, the Department applies an additional penalty up to 10 percent of the value of the subject work as a permanent deduction.

5-1.13B(3) Use of Joint Checks

You may use a joint check between the Contractor or lower-tier subcontractor and a DBE subcontractor purchasing materials from a material supplier if you obtain prior approval from the Department for your proposed use of joint checks upon submittal of a DBE Joint Check Agreement Request form.

To use a joint check, the following conditions must be met:

1. All parties, including the Contractor, must agree in writing to the use of a joint check
2. Entity issuing the joint check acts solely to guarantee payment
3. DBE must release the check to the material supplier
4. Department must authorize the request before implementation
5. Any party to the agreement must provide requested documentation within 10 days of the Department's request for the documentation
6. Agreement to use a joint check must be short-term, not to exceed 1 year, allowing sufficient time needed to establish or increase a credit line with the material supplier

A request for a joint check agreement may be initiated by any party.

If a joint check is used, the DBE remains responsible for all elements of 49 CFR 26.55(c)(1).

Failure to comply with section 5-1.13B(3) disqualifies DBE participation and results in no credit and no payment to the Contractor for DBE participation.

A joint check may not be used between the Contractor or subcontractor and a DBE regular dealer, bulk material supplier, manufacturer, wholesaler, broker, trucker, packager, manufacturer's representative, or other persons who arrange or expedite transactions.

Replace section 5-1.13D with:

5-1.13D Non-Small Business Preference

5-1.13D(1) General

Section 5-1.13D applies to a non-federal-aid contract when you have obtained a non-small business subcontractor preference under section 2-1.18.

Use each SBE as shown on the Certified Small Business Listing for the Non-Small Business Preference form unless you receive authorization for a substitution under section 5-1.13D(3).

Maintain records of subcontracts and agreements made with each SBE. Include in the records:

1. SBE subcontract or agreement
2. Name and business address of each business
3. Proof of each payment made to an SBE

Upon reasonable notice and during normal business hours, permit access to the SBE premises for the purpose of:

1. Interviewing employees.
2. Inspecting and copying books, records, accounts, and other material that may be relevant to a matter under investigation.

Upon work completion, complete a Final Report – Utilization of Small Business Enterprise form and submit within 10 days of Contract acceptance. The Department withholds the greater of 10 percent of the SBE work amount or \$10,000 until the form is submitted. The Department releases the withhold upon submission of the completed form. If additional payments are made to an SBE after submittal of the completed form, submit an updated form to reflect such payments.

5-1.13D(2) Commercially Useful Function

SBEs must perform a commercially useful function under 2 CA Code Regs § 1896.15 when performing tasks listed on the Certified Small Business Listing for the Non-Small Business Preference form and as a Department authorized substitute SBE. The SBE's value of work will only count toward the non-small business preference if the SBE performs a commercially useful function under 2 CA Code Regs § 1896.15.

Provide written notification to the Engineer at least 15 days in advance of each SBE's initial performance of work on the Contract. Include the SBE's name, contract work to be performed, and the location, date, and time of where the SBE's work will take place.

Monitor each SBE's performance of a commercially useful function throughout their duration of work on the Contract. Maintain supporting documentation of commercially useful function related records. Provide copies of the following to the Department within 5 business days of a specific SBE request:

1. Subcontract or agreement with the SBE
2. Purchase orders
3. Bills of lading
4. Invoices
5. Proof of payment

You and your SBEs must submit any commercially useful function related records and documents within 5 business days of Department request such as:

1. Proof of ownership or lease and rental agreements for equipment

2. Tax records
3. Employee rosters
4. Certified payroll records
5. Inventory rosters

Notify the Engineer immediately if you believe an SBE may not be performing a commercially useful function.

Failure to submit requested records and documents, or failure of an SBE to perform a commercially useful function, will be a violation of these specifications. Such violations will result in referral to the Department of General Services, and you may be subject to sanctions under Govt Code § 14842 et seq.

5-1.13D(3) Substitutions

Substitution of an SBE may be requested for the reasons listed in Pub Cont Code § 4107. The Department will consider your SBE substitution request under 2 CA Code Regs § 1896.10. The substitute must be another SBE.

The requirement that an SBE be certified by the bid opening date does not apply to SBE substitutions after contract award.

Include in your substitution request:

1. Copy of the written notice issued to the SBE with proof of delivery
2. Copy of the SBE response to the notice
3. Your explanation of the reason for the substitution request
4. Names and certification numbers of the listed SBE and the proposed substitute SBE
5. Description of the work to be performed including dollar amount or percentage of overall contract the proposed substitute SBE will perform
6. If an SBE substitution cannot occur, written justification and the steps that were taken to try to acquire a new SBE and how that non-small business preference portion of the Contract will be fulfilled

Upon receipt of the substitution request, the listed SBE will be notified by the Department of the substitution request. The listed SBE will have 5 business days to respond. Where the listed SBE objects to the substitution, the Department will schedule a hearing.

If the Department authorizes your SBE substitution request, it will do so in writing.

Work performed by a firm other than the listed SBE or authorized substitute will be a violation of these specifications. Such violations will result in notification of the Department of General Services, and you may be subject to sanctions under Govt Code § 14842 et seq.

Replace section 5-1.13E with:

5-1.13E Prompt Payment

Section 5-1.13E applies to all contracts.

Pay your subcontractors within 7 days of receipt of each progress payment under Pub Cont Code §§ 10262 and 10262.5. Pay duly authorized motor carriers of property in dump trucks for transportation charges under Bus & Prof Code § 7108.6. Pay other entities, such as material suppliers, within 30 days of receipt of each progress payment.

Each month, after the 15th and prior to 20th, submit the following payment information through the Department's prompt payment monitoring system at <https://caltrans.dbesystem.com>:

1. Subcontractor's or entity's business name
2. Description of work performed
 - 2.1. Bid item numbers or change order numbers
 - 2.2. Written narrative of work performed
3. Value of work performed
4. Amount paid to subcontractor or entity
5. Withhold amount, if applicable

6. Explanation of withhold reasoning, if applicable

Your subcontractors and other entities may validate payments received using the prompt payment monitoring system.

If a subcontractor's or other entity's work is in dispute, provide a written withhold notification to the subcontractor or entity and the Engineer no later than 7 days after receipt of the corresponding progress payment that includes the following:

1. Value of the disputed work
2. Amount of the withhold being taken
3. Bid item numbers or change order numbers associated with the disputed work
4. Explanation of the deficiencies of the disputed work and how the corresponding value was calculated
5. Corrective actions to be taken for release of withheld amount

The Department may request additional documentation from you to evaluate whether you applied the withhold in good faith. Submit requested documents within 10 days of receipt of request.

The Department may withhold the same amount of your withhold from a future progress pay estimate if the Department determines any of the following has occurred:

1. Withhold was not applied in good faith
2. Requested additional withhold documentation records were not provided
3. Payment information was not submitted through the prompt payment monitoring system
4. Required withhold notification was not provided

The Department may also apply a 2 percent penalty on the withhold amount for every month payment is not made.

Replace section 5-1.15 with:

5-1.15 SAFETY REPRESENTATIVE

5-1.15A General

Allocate a full-time, on-site safety representative to project safety for the duration of this Contract. The safety representative must be available after work hours as needed. The safety representative is not required to be on-site during a plant establishment period or when no work is being performed during project temporary suspensions.

Develop a written site-specific safety program that incorporates known hazards associated with the project.

5-1.15B Submittals

Submit a resume of the safety representative before starting work on the project, including education, heavy construction safety experience, and completed safety training.

Submit a copy of the safety representative's certificates for Cal/OSHA 30-hour construction training course and traffic control supervisor training as an informational submittal. The certificates must include:

1. Names of the Department authorized entity providing certification
2. Name of the individual receiving certification
3. Date when the certification was provided
4. Expiration date

Submit a copy of the site-specific safety program as an informational submittal:

1. At least 15 days before starting work
2. At least 48 hours before starting work whenever a revision is made to the site-specific safety program

5-1.15C Quality Assurance

5-1.15C(1) General

Not Used

5-1.15C(2) Certifications

The safety representative must have completed training and received a certificate of completion for:

1. Cal/OSHA 30-hour construction training course.
2. Traffic control supervisor certification. A list of Department authorized traffic control supervisor training providers is available at:

<https://dot.ca.gov/programs/construction/safety-traffic/safety-training-courses>

The safety representative must have one of the following certifications from the Board of Certified Safety Professionals:

1. Certified Safety Professional
2. Construction Health and Safety Technician
3. Occupational Hygiene and Safety Technician

5-1.15C(3) Qualifications

The safety representative must be a competent and qualified person, as defined by Cal/OSHA, and must provide safety oversight on the project. The safety representative must have:

1. At least 5 years of experience in heavy construction safety
2. Knowledge in the area of safety procedure for all types of work being performed on this project
3. Knowledge of Department specifications and plans, Cal/OSHA policy and procedures, and 8 CA Code of Regs safety requirements

The safety representative must have completed safety training and maintain valid certifications for:

1. First aid
2. Cardiopulmonary resuscitation

5-1.15D Construction

The safety representative must be available by:

1. Cellular telephone
2. Two-way radio
3. Mobile internet access

The safety representative must:

1. Develop the site-specific safety program
2. Confirm each worker's compliance with the Cal/OSHA drug testing requirements
3. Conduct the on-site new project orientation for each worker assigned to the project
4. Review the construction work plans for each subcontractor before starting work
5. Conduct or attend pre-planning sessions for high hazard work such as excavations, demolition, confined space entry, falsework, crane hoisting, high-risk utilities such as, high pressure pipelines with fuel or natural gas, and tunneling or boring
6. Conduct weekly job site safety meetings
7. Review work schedules, including subcontractors' work schedules, to ensure adequacy of job hazard analyses and address all safety concerns and issues
8. Ensure safety data sheets are available on-site for all materials and have a hazard communication program for informing workers of material hazards
9. Verify compliance with hazardous waste requirements under section 14-11, including spill prevention and control measures
10. Conduct daily safety inspection of the job site for hazards and compliance with safety requirements in the specifications and 8 CA Code of Regs
11. Report all incidents to the Engineer immediately, including the investigation of any close calls

When monitoring work zone, take immediate corrective action and notify the Engineer if an imminent danger is identified.

If the Engineer determines the safety representative is not meeting the intended duties and responsibilities, you must replace the safety representative within 30 days of written notice from the Engineer. If you need to replace the safety representative, notify the Engineer and provide replacement safety representative resume and required certifications at least 15 days before the replacement.

Add to the end of section 5-1.20A:

During the progress of the work under this Contract, work under the following contracts may be in progress at or near the job site of this Contract:

Coincident or Adjacent Contracts

Contract no.	County–Route–Post Mile	Location	Type of work
01-0H6414	Hum-101-0.9	Near Cooks Valley	Construct bridge
01-0M9104	Hum-101-R8.5/M54.0	Various	Pavement delineation
01-0L1504	Hum-101-R11.8, R51.8, R52.9	Near Rio Dell	Bridge deck
01-0A1104	Hum-101-27.7	Near Myers Flat	Bridge retrofit
01-0L9504	Hum-101-R43.0, R46.2	Near Redcrest	Bridge deck
01-0A1114	Hum-101-M53.7/M54.2	Near Rio Dell	Bridge retrofit
01-0H2404	Hum-254-18.5/23.5	Near Weott	Culvert
01-0H2414	Hum-254-20.1/20.4	Near Weott	Culvert
01-464804	Hum-101-1.1/2.2	Cooks Valley	Roadway alignment
01-0M9504	Hum-101-R5.6, 23.0, R46.2, M54.0	Various	Bridge deck
01-0L6404	Hum-101-11.6/76.7	Various	Install fiber optic cable

Coordinate lane closures and traffic handling with the Engineer and with contractors of coincident or adjacent projects. Potential conflicts may not be limited to the contracts listed above.

Add to the end of section 5-1.20B(1):

The Department will communicate with PLAC agencies unless communication between you and the PLAC agency is authorized.

AA

6 CONTROL OF MATERIALS

Replace section 6-1.03B with:

6-1.03B Submittals

6-1.03B(1) General

Not Used

6-1.03B(2) Work Plan

For local material, such as rock, gravel, earth, structure backfill, pervious backfill, imported borrow, and culvert bedding, obtained from a (1) noncommercial source, or (2) source not regulated under California jurisdiction, submit a local material plan for each material at least 60 days before placing the material. The local material plan must include:

1. Certification signed by you and an engineer who is registered as a civil engineer in the State or a professional geologist licensed as a professional geologist by the State stating:

I am aware local material from a noncommercial source or a source not regulated under CA jurisdiction must be sampled and analyzed for pH and lead and may require sampling and analysis under section 6-1.03B(3) for other constituents of concern based on the land use history. I am aware that local material sources must not contain ADL at concentrations greater than 80 mg/kg total lead or equal to or greater than 5 mg/L soluble lead as determined by the Waste Extraction Test (WET) Procedures, 22 CA Code of Regs § 66261.24(a)(2) App II. I am aware that a maximum quantity of material may be excavated at the site based on the minimum number of samples taken before excavating at the site under section 6-1.03B(3).

2. Land use history of the local material location and surrounding property
3. Sampling protocol
4. Number of samples per volume of local material
5. QA and QC requirements and procedures
6. Qualifications of sampling personnel
7. Stockpile history
8. Name and address of the analytical laboratory that will perform the chemical analyses
9. Analyses that will be performed for lead and pH
10. Other analyses that will be performed for possible hazardous constituents based on:
 - 10.1. Source property history
 - 10.2. Land use adjacent to source property
 - 10.3. Constituents of concern in the ground water basin where the job site is located

The plan must be sealed and signed by an engineer who is registered as a civil engineer in the State or a professional geologist licensed as a professional geologist by the State.

If the plan requires revisions, the Engineer provides comments. Submit a revised plan within 7 days of receiving comments. Allow 7 days for the review.

6-1.03B(3) Analytical Test Results

At least 15 days before placing local material, submit analytical test results for each local material obtained from a noncommercial source or a source not regulated under CA jurisdiction. The analytical test results must include:

1. Certification signed by an engineer who is registered as a civil engineer in the State or a professional geologist licensed as a professional geologist by the State stating:

The analytical testing described in the local material plan has been performed. I performed a statistical analysis of the test results using the US EPA's ProUCL software with the applicable 95 percent upper confidence limit. I certify that the material from the local material source is suitable for unrestricted use at the job site, it has a pH above 5.0, does not contain soluble lead in concentrations equal to or greater than 5mg/l as determined by the Waste Extraction Test (WET) Procedures, 22 CA Code of Regs § 66261.24(a)(2) App II, does not contain lead in concentrations above 80 mg/kg total lead, is free from all other contaminants identified in the local material plan, and will comply with the job site's basin plan and water quality objectives of the RWQCB.

2. Chain of custody of samples
3. Analytical results no older than 1 year
4. Statistical analysis of the data using US EPA's ProUCL software with a 95 percent upper confidence limit
5. Comparison of sample results to hazardous waste concentration thresholds and the RWQCB's basin plan requirements and water quality objectives for the job site location

6-1.03B(4) Sample and Analysis

Sample and analyze local material from a (1) noncommercial source or (2) a source not regulated under CA jurisdiction:

1. Before bringing the local material to the job site
2. As described in the local material plan
3. Under US EPA Test Methods for Evaluating Solid Waste, Physical/Chemical Methods (SW-846)

The sample collection must be designed to generate a data set representative of the entire volume of proposed local material.

Before excavating at the (1) noncommercial material source or (2) a source not regulated under CA jurisdiction, collect the minimum number of samples and perform the minimum number of analytical tests for the corresponding maximum volume of local material as shown in the following table:

Minimum Number of Samples and Analytical Tests for Local Material

Maximum volume of imported borrow (cu yd)	Minimum number of samples and analytical tests
< 5,000	8
5,000–10,000	12 for the first 5,000 cu yd plus 1 for each additional 1,000 cu yd or portion thereof
10,000–20,000	17 for the first 10,000 cu yd plus 1 for each additional 2,500 cu yd or portion thereof
20,000–40,000	21 for the first 20,000 cu yd plus 1 for each additional 5,000 cu yd or portion thereof
40,000–80,000	25 for the first 40,000 cu yd plus 1 for each additional 10,000 cu yd or portion thereof
> 80,000	29 for the first 80,000 cu yd plus 1 for each additional 20,000 cu yd or portion thereof

Do not collect composite samples or mix individual samples to form a composite sample.

Analyze the samples using the US EPA's ProUCL software with a 95 percent upper confidence limit. All chemical analysis must be performed by a laboratory certified by the SWRCB's Environmental Laboratory Accreditation Program (ELAP).

The analytical test results must demonstrate that the local material:

1. Is not a hazardous waste
2. Has a pH above 5.0
3. Has an average total lead concentration, based upon the 95 percent upper confidence limit, at or below 80 mg/kg
4. Is free of possible contaminants identified in the local material plan
5. Complies with the RWQCB's basin plan for the job site location
6. Complies with the RWQCB's water quality objectives for the job site location

6-1.03C Local Material Management

Do not place local material until authorized.

If the Engineer determines the appearance, odor, or texture of any delivered local material suggests possible contamination, sample and analyze the material. The sampling and analysis is change order work unless (1) hazardous waste is discovered or (2) the analytical test results indicate the material does not comply with section 6-1.03B(3).

Dispose of noncompliant local material at an appropriately permitted CA Class I, CA Class II or CA Class III facility. You are the generator of noncompliant local materials.

Replace section 6-1.04 with:

6-1.04 BUY AMERICA

6-1.04A General

Buy America requirements do not apply to the following:

1. Tools and construction equipment used in performing the work
2. Temporary work that is not incorporated into the finished project

6-1.04B Crumb Rubber (Pub Res Code § 42703(d))

Furnish crumb rubber with a certificate of compliance. Crumb rubber must be:

1. Produced in the United States
2. Derived from waste tires taken from vehicles owned and operated in the United States

6-1.04C Steel and Iron Materials

Steel and iron materials must be melted and manufactured in the United States except:

1. Foreign pig iron and processed, pelletized, and reduced iron ore may be used in the domestic production of the steel and iron materials
2. If the total combined cost of the materials produced outside the United States does not exceed the greater of 0.1 percent of the total bid or \$2,500, the material may be used if authorized

Furnish steel and iron materials to be incorporated into the work with certificates of compliance and certified mill test reports. Mill test reports must indicate where the steel and iron were melted and manufactured.

All melting and manufacturing processes for these materials, including an application of a coating, must occur in the United States. Coating includes all processes that protect or enhance the value of the material to which the coating is applied.

6-1.04D Manufactured Products

Iron and steel used in precast concrete manufactured products must meet the requirements of section 6-1.04C regardless of the amount used.

Iron and steel used in other manufactured products must meet the requirements of section 6-1.04C if the weight of steel and iron components constitute 90 percent or more of the total weight of the manufactured product.

6-1.04E Construction Materials

The following construction materials must be produced in the United States under standards in 2 CFR 184.6:

1. Non-ferrous metals
2. Plastic and polymer-based products such as:
 - 2.1. Polyvinylchloride
 - 2.2. Composite building materials
3. Glass
4. Fiber optic cable including drop cable
5. Optical fiber
6. Lumber
7. Engineered wood
8. Drywall

All manufacturing processes for these materials as defined in 2 CFR 184.6 must occur in the United States.

Furnish construction materials to be incorporated into the work with certificates of compliance with each project delivery. Manufacturer's certificate of compliance must identify where the construction material was manufactured and attest specifically to compliance with its 2 CFR 184.6 standard.

Minor additions of articles, materials, supplies, or binding agents to these construction materials do not change the categorization of the construction material.

7 LEGAL RELATIONS AND RESPONSIBILITY TO THE PUBLIC

Replace Section 7-1.02K(6)(j)(iii) with:

7-1.02K(6)(j)(iii) Unregulated Earth Material Containing Lead

Section 7-1.02K(6)(j)(iii) includes specifications for handling, removing, and disposing of unregulated earth material containing lead. Management of this material exposes workers to health hazards that must be addressed in your lead compliance plan. This material contains average lead concentrations below 80 mg/kg total lead and below 5 mg/L soluble lead and is not regulated by DTSC as a hazardous substance or a hazardous waste. This material does not require disposal at a permitted landfill or solid waste disposal facility. The RWQCB has jurisdiction over reuse of this material at locations outside the job site limits.

Unregulated earth material exists throughout the job site.

Lead is typically found within the top 2 feet of material within the highway. Reuse all of the excavated material on the right-of-way.

Handle the material under all applicable laws, rules, and regulations, including those of the following agencies:

1. Cal/OSHA
2. CA RWQCB, Region-1, North Coast

Add to the end of section 7-1.02M(2):

Obtain the emergency phone numbers of the California Department of Forestry and Fire Protection unit headquarters, United States Forest Service ranger district office, and U.S. Department of Interior Bureau of Land Management field offices. Submit these phone numbers to the Engineer before the start of job site activities. Post the agencies names and emergency phone numbers at a prominent place at the job site.

Hydrocarbon-fueled engines, both stationary and mobile, must be equipped with spark arresters pursuant to Pub Res Code § 4442 except for either of the following:

1. Motor trucks, truck tractors, buses, or passenger vehicles, if equipped with a muffler as defined in the Vehicle Code
2. Equipment powered by turbocharged engines if:
 - 2.1 Exhausted gases pass through the rotating turbine wheel
 - 2.2 There is no exhaust bypass to the atmosphere
 - 2.3 The turbocharger is in effective mechanical condition

Each toilet must have a metal ashtray at least 6 inches in diameter by 8 inches deep, half-filled with sand, and within easy reach of anyone accessing the facility.

Locate flammable materials at least 50 feet away from equipment service, parking, and gas or oil storage areas. Each small mobile or stationary engine site must be cleared of flammable material for a radius of at least 15 feet from the engine.

Before clearing and grubbing, clear a fire break at the outer limits of the areas to be cleared and grubbed. Where clearing and grubbing limits allow, use a minimum fire break width of 20 feet. Each area to be cleared and grubbed must be cleared and kept clear of flammable material such as dry grass, weeds, brush, downed trees, oily rags and waste, paper, cartons, and plastic waste.

Furnish the following fire tools:

1. 1 shovel and 1 fully charged fire extinguisher UL rated at 4B:C or more on each truck, personnel vehicle, tractor, grader, or other heavy equipment.

2. 1 shovel and one 5-gallon water-filled backpack fire pump for each welder.
3. 1 shovel or 1 chemical pressurized fire extinguisher, fully charged, for each gasoline-powered tool, including chain saws, soil augers, and rock drills. The fire tools must always be within 25 feet from the point of operation of the power tool. Each fire extinguisher must be of the type and size required by the California Department of Forestry and Fire Protection.

Each shovel must be size O or larger and at least 46 inches long.

Furnish a pickup truck and operator for the sole purpose of fire control during working hours. The truck must be equipped with:

1. 10 shovels, 5 axes, two 5-gallon water-filled backpack fire pumps
2. 100-gallon tank of water with a gasoline motor powered pump and 100 feet of 3/4-inch hose on a reel

In addition to being available at the site of the work, the truck and operator must patrol the area of construction from noon until at least 1/2 hour after job site activities have ended. If the fire danger rating is "very high" or "extreme" or "fire weather watches" or "red flag warning" is issued, the truck and operator must patrol the area of construction while work is being done and for at least 1/2 hour after job site activities have ended.

Cal Fire, USFS, and BLM have established the following adjective class ratings for 5 levels of fire danger for use in public information releases and fire protection signing: "low," "moderate," "high," "very high," "extreme." Obtain the fire danger rating daily for the project area from the nearest Cal Fire unit headquarters, USFS ranger district office, or BLM field office. Monitor the National Weather Service daily forecasts for "fire weather watches" and "red flag warnings" covering the project's locations.

Arrangements have been made with Cal Fire, USFS, and BLM to notify the Department when the fire danger rating is "very high" or "extreme." This information will be furnished to the Engineer who will notify you for dissemination and action in the area affected. If a discrepancy between this notice and the fire danger rating obtained from the nearest office of either Cal Fire or USFS exists, you must conduct operations according to the higher of the two fire danger ratings.

If the fire danger rating is "very high" or a "fire weather watch" is issued, then:

1. Falling of dead trees or snags must be discontinued.
2. No open burning is permitted and fires must be extinguished.
3. Welding must be discontinued except in an enclosed building or within an area cleared of flammable material for a radius of 25 feet.
4. Blasting must be discontinued.
5. Smoking is allowed only in automobiles and cabs of trucks equipped with an ashtray or in cleared areas immediately surrounded by a fire break unless prohibited by other authority.
6. Vehicular travel is restricted to cleared areas except in case of emergency.

If the fire danger rating is "extreme" or a "red flag warning" is issued, take the precautions specified for a "very high" fire danger rating or a "fire weather watch" issuance, except:

1. Smoking is only allowed in automobiles and cabs of trucks equipped with an ashtray.
2. Work of a nature that could start a fire requires that properly equipped fire guards be assigned to such operation for the duration of the work.

The Engineer may suspend work wholly or in part due to hazardous fire conditions. If the controlling activity is suspended due to hazardous fire conditions, the associated days are considered adverse weather-related conditions and working days will not be assessed. If field and weather conditions become such that the work is suspended, section 7-1.02M(2) will not be enforced for the period of the suspension.

Delete the 24th paragraph of section 7-1.04.

Replace section 7-1.11B with:

7-1.11B FHWA-1273

FHWA-1273 – Revised October 23, 2023

REQUIRED CONTRACT PROVISIONS FEDERAL-AID CONSTRUCTION CONTRACTS

- I. General
- II. Nondiscrimination
- III. Non-segregated Facilities
- IV. Davis-Bacon and Related Act Provisions
- V. Contract Work Hours and Safety Standards Act Provisions
- VI. Subletting or Assigning the Contract
- VII. Safety: Accident Prevention
- VIII. False Statements Concerning Highway Projects
- IX. Implementation of Clean Air Act and Federal Water Pollution Control Act
- X. Certification Regarding Debarment, Suspension, Ineligibility and Voluntary Exclusion
- XI. Certification Regarding Use of Contract Funds for Lobbying
- XII. Use of United States-Flag Vessels:

ATTACHMENTS

A. Employment and Materials Preference for Appalachian Development Highway System or Appalachian Local Access Road Contracts (included in Appalachian contracts only)

I. GENERAL

1. Form FHWA-1273 must be physically incorporated in each construction contract funded under title 23, United States Code, as required in 23 CFR 633.102(b) (excluding emergency contracts solely intended for debris removal). The contractor (or subcontractor) must insert this form in each subcontract and further require its inclusion in all lower tier subcontracts (excluding purchase orders, rental agreements and other agreements for supplies or services). 23 CFR 633.102(e).

The applicable requirements of Form FHWA-1273 are incorporated by reference for work done under any purchase order, rental agreement or agreement for other services. The prime contractor shall be responsible for compliance by any subcontractor, lower-tier subcontractor or service provider. 23 CFR 633.102(e).

Form FHWA-1273 must be included in all Federal-aid design-build contracts, in all subcontracts and in lower tier subcontracts (excluding subcontracts for design services, purchase orders, rental agreements and other agreements for supplies or services) in accordance with 23 CFR 633.102. The design-builder shall be responsible for compliance by any subcontractor, lower-tier subcontractor or service provider.

Contracting agencies may reference Form FHWA-1273 in solicitation-for-bids or request-for-proposals documents, however, the Form FHWA-1273 must be physically incorporated (not referenced) in all contracts, subcontracts and lower-tier subcontracts (excluding purchase orders, rental agreements and other agreements for supplies or services related to a construction contract). 23 CFR 633.102(b).

2. Subject to the applicability criteria noted in the following sections, these contract provisions shall apply to all work

performed on the contract by the contractor's own organization and with the assistance of workers under the contractor's immediate superintendence and to all work performed on the contract by piecework, station work, or by subcontract. 23 CFR 633.102(d).

3. A breach of any of the stipulations contained in these Required Contract Provisions may be sufficient grounds for withholding of progress payments, withholding of final payment, termination of the contract, suspension / debarment or any other action determined to be appropriate by the contracting agency and FHWA.

4. Selection of Labor: During the performance of this contract, the contractor shall not use convict labor for any purpose within the limits of a construction project on a Federal-aid highway unless it is labor performed by convicts who are on parole, supervised release, or probation. 23 U.S.C. 114(b). The term Federal-aid highway does not include roadways functionally classified as local roads or rural minor collectors. 23 U.S.C. 101(a).

II. NONDISCRIMINATION (23 CFR 230.107(a); 23 CFR Part 230, Subpart A, Appendix A; EO 11246)

The provisions of this section related to 23 CFR Part 230, Subpart A, Appendix A are applicable to all Federal-aid construction contracts and to all related construction subcontracts of \$10,000 or more. The provisions of 23 CFR Part 230 are not applicable to material supply, engineering, or architectural service contracts.

In addition, the contractor and all subcontractors must comply with the following policies: Executive Order 11246, 41 CFR Part 60, 29 CFR Parts 1625-1627, 23 U.S.C. 140, Section 504 of the Rehabilitation Act of 1973, as amended (29 U.S.C. 794), Title VI of the Civil Rights Act of 1964, as amended (42 U.S.C. 2000d et seq.), and related regulations including 49 CFR Parts 21, 26, and 27; and 23 CFR Parts 200, 230, and 633.

The contractor and all subcontractors must comply with: the requirements of the Equal Opportunity Clause in 41 CFR 60-1.4(b) and, for all construction contracts exceeding \$10,000, the Standard Federal Equal Employment Opportunity Construction Contract Specifications in 41 CFR 60-4.3.

Note: The U.S. Department of Labor has exclusive authority to determine compliance with Executive Order 11246 and the policies of the Secretary of Labor including 41 CFR Part 60, and 29 CFR Parts 1625-1627. The contracting agency and the FHWA have the authority and the responsibility to ensure compliance with 23 U.S.C. 140, Section 504 of the Rehabilitation Act of 1973, as amended (29 U.S.C. 794), and Title VI of the Civil Rights Act of 1964, as amended (42 U.S.C. 2000d et seq.), and related regulations including 49 CFR Parts 21, 26, and 27; and 23 CFR Parts 200, 230, and 633.

The following provision is adopted from 23 CFR Part 230, Subpart A, Appendix A, with appropriate revisions to conform to the U.S. Department of Labor (US DOL) and FHWA requirements.

1. Equal Employment Opportunity: Equal Employment Opportunity (EEO) requirements not to discriminate and to take affirmative action to assure equal opportunity as set forth under laws, executive orders, rules, regulations (see 28 CFR Part 35, 29 CFR Part 1630, 29 CFR Parts 1625-1627, 41 CFR Part 60 and 49 CFR Part 27) and orders of the Secretary of Labor as modified by the provisions prescribed herein, and imposed pursuant to 23 U.S.C. 140, shall constitute the EEO and specific affirmative action standards for the contractor's project activities under this contract. The provisions of the Americans with Disabilities Act of 1990 (42 U.S.C. 12101 et seq.) set forth under 28 CFR Part 35 and 29 CFR Part 1630 are incorporated by reference in this contract. In the execution of this contract, the contractor agrees to comply with the following minimum specific requirement activities of EEO:

a. The contractor will work with the contracting agency and the Federal Government to ensure that it has made every good faith effort to provide equal opportunity with respect to all of its terms and conditions of employment and in their review of activities under the contract. 23 CFR 230.409 (g)(4) & (5).

b. The contractor will accept as its operating policy the following statement:

"It is the policy of this Company to assure that applicants are employed, and that employees are treated during employment, without regard to their race, religion, sex, sexual orientation, gender identity, color, national origin, age or disability. Such action shall include: employment, upgrading, demotion, or transfer; recruitment or recruitment advertising; layoff or termination; rates of pay or other forms of compensation; and selection for training, including apprenticeship, pre-apprenticeship, and/or on-the-job training."

2. EEO Officer: The contractor will designate and make known to the contracting officers an EEO Officer who will have the responsibility for and must be capable of effectively administering and promoting an active EEO program and who must be assigned adequate authority and responsibility to do so.

3. Dissemination of Policy: All members of the contractor's staff who are authorized to hire, supervise, promote, and discharge employees, or who recommend such action or are substantially involved in such action, will be made fully cognizant of and will implement the contractor's EEO policy and contractual responsibilities to provide EEO in each grade and classification of employment. To ensure that the above agreement will be met, the following actions will be taken as a minimum:

a. Periodic meetings of supervisory and personnel office employees will be conducted before the start of work and then not less often than once every six months, at which time the contractor's EEO policy and its implementation will be reviewed and explained. The meetings will be conducted by the EEO Officer or other knowledgeable company official.

b. All new supervisory or personnel office employees will be given a thorough indoctrination by the EEO Officer, covering all major aspects of the contractor's EEO obligations within thirty days following their reporting for duty with the contractor.

c. All personnel who are engaged in direct recruitment for the project will be instructed by the EEO Officer in the contractor's procedures for locating and hiring minorities and women.

d. Notices and posters setting forth the contractor's EEO policy will be placed in areas readily accessible to employees, applicants for employment and potential employees.

e. The contractor's EEO policy and the procedures to implement such policy will be brought to the attention of employees by means of meetings, employee handbooks, or other appropriate means.

4. Recruitment: When advertising for employees, the contractor will include in all advertisements for employees the notation: "An Equal Opportunity Employer." All such advertisements will be placed in publications having a large circulation among minorities and women in the area from which the project work force would normally be derived.

a. The contractor will, unless precluded by a valid bargaining agreement, conduct systematic and direct recruitment through public and private employee referral sources likely to yield qualified minorities and women. To meet this requirement, the contractor will identify sources of potential minority group employees and establish with such identified sources procedures whereby minority and women applicants may be referred to the contractor for employment consideration.

b. In the event the contractor has a valid bargaining agreement providing for exclusive hiring hall referrals, the contractor is expected to observe the provisions of that agreement to the extent that the system meets the contractor's compliance with EEO contract provisions. Where implementation of such an agreement has the effect of discriminating against minorities or women, or obligates the contractor to do the same, such implementation violates Federal nondiscrimination provisions.

c. The contractor will encourage its present employees to refer minorities and women as applicants for employment. Information and procedures with regard to referring such applicants will be discussed with employees.

5. Personnel Actions: Wages, working conditions, and employee benefits shall be established and administered, and personnel actions of every type, including hiring, upgrading, promotion, transfer, demotion, layoff, and termination, shall be taken without regard to race, color, religion, sex, sexual orientation, gender identity, national origin, age or disability. The following procedures shall be followed:

a. The contractor will conduct periodic inspections of project sites to ensure that working conditions and employee facilities do not indicate discriminatory treatment of project site personnel.

b. The contractor will periodically evaluate the spread of wages paid within each classification to determine any evidence of discriminatory wage practices.

c. The contractor will periodically review selected personnel actions in depth to determine whether there is evidence of discrimination. Where evidence is found, the contractor will promptly take corrective action. If the review indicates that the discrimination may extend beyond the actions reviewed, such corrective action shall include all affected persons.

d. The contractor will promptly investigate all complaints of alleged discrimination made to the contractor in connection with its obligations under this contract, will attempt to resolve such complaints, and will take appropriate corrective action.

within a reasonable time. If the investigation indicates that the discrimination may affect persons other than the complainant, such corrective action shall include such other persons. Upon completion of each investigation, the contractor will inform every complainant of all of their avenues of appeal.

6. Training and Promotion:

a. The contractor will assist in locating, qualifying, and increasing the skills of minorities and women who are applicants for employment or current employees. Such efforts should be aimed at developing full journey level status employees in the type of trade or job classification involved.

b. Consistent with the contractor's work force requirements and as permissible under Federal and State regulations, the contractor shall make full use of training programs (i.e., apprenticeship and on-the-job training programs for the geographical area of contract performance). In the event a special provision for training is provided under this contract, this subparagraph will be superseded as indicated in the special provision. The contracting agency may reserve training positions for persons who receive welfare assistance in accordance with 23 U.S.C. 140(a).

c. The contractor will advise employees and applicants for employment of available training programs and entrance requirements for each.

d. The contractor will periodically review the training and promotion potential of employees who are minorities and women and will encourage eligible employees to apply for such training and promotion.

7. Unions: If the contractor relies in whole or in part upon unions as a source of employees, the contractor will use good faith efforts to obtain the cooperation of such unions to increase opportunities for minorities and women. 23 CFR 230.409. Actions by the contractor, either directly or through a contractor's association acting as agent, will include the procedures set forth below:

a. The contractor will use good faith efforts to develop, in cooperation with the unions, joint training programs aimed toward qualifying more minorities and women for membership in the unions and increasing the skills of minorities and women so that they may qualify for higher paying employment.

b. The contractor will use good faith efforts to incorporate an EEO clause into each union agreement to the end that such union will be contractually bound to refer applicants without regard to their race, color, religion, sex, sexual orientation, gender identity, national origin, age, or disability.

c. The contractor is to obtain information as to the referral practices and policies of the labor union except that to the extent such information is within the exclusive possession of the labor union and such labor union refuses to furnish such information to the contractor, the contractor shall so certify to the contracting agency and shall set forth what efforts have been made to obtain such information.

d. In the event the union is unable to provide the contractor with a reasonable flow of referrals within the time limit set forth in the collective bargaining agreement, the contractor will, through independent recruitment efforts, fill the employment vacancies without regard to race, color, religion, sex, sexual orientation, gender identity, national origin, age, or disability; making full efforts to obtain qualified and/or qualifiable minorities and women. The failure of a union to provide

sufficient referrals (even though it is obligated to provide exclusive referrals under the terms of a collective bargaining agreement) does not relieve the contractor from the requirements of this paragraph. In the event the union referral practice prevents the contractor from meeting the obligations pursuant to Executive Order 11246, as amended, and these special provisions, such contractor shall immediately notify the contracting agency.

8. Reasonable Accommodation for Applicants / Employees with Disabilities: The contractor must be familiar with the requirements for and comply with the Americans with Disabilities Act and all rules and regulations established thereunder. Employers must provide reasonable accommodation in all employment activities unless to do so would cause an undue hardship.

9. Selection of Subcontractors, Procurement of Materials and Leasing of Equipment: The contractor shall not discriminate on the grounds of race, color, religion, sex, sexual orientation, gender identity, national origin, age, or disability in the selection and retention of subcontractors, including procurement of materials and leases of equipment. The contractor shall take all necessary and reasonable steps to ensure nondiscrimination in the administration of this contract.

a. The contractor shall notify all potential subcontractors, suppliers, and lessors of their EEO obligations under this contract.

b. The contractor will use good faith efforts to ensure subcontractor compliance with their EEO obligations.

10. Assurances Required:

a. The requirements of 49 CFR Part 26 and the State DOT's FHWA-approved Disadvantaged Business Enterprise (DBE) program are incorporated by reference.

b. The contractor, subrecipient or subcontractor shall not discriminate on the basis of race, color, national origin, or sex in the performance of this contract. The contractor shall carry out applicable requirements of 49 CFR part 26 in the award and administration of DOT-assisted contracts. Failure by the contractor to carry out these requirements is a material breach of this contract, which may result in the termination of this contract or such other remedy as the recipient deems appropriate, which may include, but is not limited to:

- (1) Withholding monthly progress payments;
- (2) Assessing sanctions;
- (3) Liquidated damages; and/or
- (4) Disqualifying the contractor from future bidding as non-responsible.

c. The Title VI and nondiscrimination provisions of U.S. DOT Order 1050.2A at Appendixes A and E are incorporated by reference. 49 CFR Part 21.

11. Records and Reports: The contractor shall keep such records as necessary to document compliance with the EEO requirements. Such records shall be retained for a period of three years following the date of the final payment to the contractor for all contract work and shall be available at reasonable times and places for inspection by authorized representatives of the contracting agency and the FHWA.

a. The records kept by the contractor shall document the following:

(1) The number and work hours of minority and non-minority group members and women employed in each work classification on the project;

(2) The progress and efforts being made in cooperation with unions, when applicable, to increase employment opportunities for minorities and women; and

(3) The progress and efforts being made in locating, hiring, training, qualifying, and upgrading minorities and women.

b. The contractors and subcontractors will submit an annual report to the contracting agency each July for the duration of the project indicating the number of minority, women, and non-minority group employees currently engaged in each work classification required by the contract work. This information is to be reported on [Form FHWA-1391](#). The staffing data should represent the project work force on board in all or any part of the last payroll period preceding the end of July. If on-the-job training is being required by special provision, the contractor will be required to collect and report training data. The employment data should reflect the work force on board during all or any part of the last payroll period preceding the end of July.

III. NONSEGREGATED FACILITIES

This provision is applicable to all Federal-aid construction contracts and to all related construction subcontracts of more than \$10,000. 41 CFR 60-1.5.

As prescribed by 41 CFR 60-1.8, the contractor must ensure that facilities provided for employees are provided in such a manner that segregation on the basis of race, color, religion, sex, sexual orientation, gender identity, or national origin cannot result. The contractor may neither require such segregated use by written or oral policies nor tolerate such use by employee custom. The contractor's obligation extends further to ensure that its employees are not assigned to perform their services at any location under the contractor's control where the facilities are segregated. The term "facilities" includes waiting rooms, work areas, restaurants and other eating areas, time clocks, restrooms, washrooms, locker rooms and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing provided for employees. The contractor shall provide separate or single-user restrooms and necessary dressing or sleeping areas to assure privacy between sexes.

IV. DAVIS-BACON AND RELATED ACT PROVISIONS

This section is applicable to all Federal-aid construction projects exceeding \$2,000 and to all related subcontracts and lower-tier subcontracts (regardless of subcontract size), in accordance with 29 CFR 5.5. The requirements apply to all projects located within the right-of-way of a roadway that is functionally classified as Federal-aid highway. 23 U.S.C. 113. This excludes roadways functionally classified as local roads or rural minor collectors, which are exempt. 23 U.S.C. 101. Where applicable law requires that projects be treated as a project on a Federal-aid highway, the provisions of this subpart will apply regardless of the location of the project. Examples include: Surface Transportation Block Grant Program projects funded under 23 U.S.C. 133 [excluding recreational trails projects], the Nationally Significant Freight and Highway

Projects funded under 23 U.S.C. 117, and National Highway Freight Program projects funded under 23 U.S.C. 167.

The following provisions are from the U.S. Department of Labor regulations in 29 CFR 5.5 "Contract provisions and related matters" with minor revisions to conform to the FHWA-1273 format and FHWA program requirements.

1. Minimum wages (29 CFR 5.5)

a. *Wage rates and fringe benefits.* All laborers and mechanics employed or working upon the site of the work (or otherwise working in construction or development of the project under a development statute), will be paid unconditionally and not less often than once a week, and without subsequent deduction or rebate on any account (except such payroll deductions as are permitted by regulations issued by the Secretary of Labor under the Copeland Act ([29 CFR part 3](#))), the full amount of basic hourly wages and bona fide fringe benefits (or cash equivalents thereof) due at time of payment computed at rates not less than those contained in the wage determination of the Secretary of Labor which is attached hereto and made a part hereof, regardless of any contractual relationship which may be alleged to exist between the contractor and such laborers and mechanics. As provided in paragraphs (d) and (e) of 29 CFR 5.5, the appropriate wage determinations are effective by operation of law even if they have not been attached to the contract. Contributions made or costs reasonably anticipated for bona fide fringe benefits under the Davis-Bacon Act ([40 U.S.C. 3141\(2\)\(B\)](#)) on behalf of laborers or mechanics are considered wages paid to such laborers or mechanics, subject to the provisions of paragraph 1.e. of this section; also, regular contributions made or costs incurred for more than a weekly period (but not less often than quarterly) under plans, funds, or programs which cover the particular weekly period, are deemed to be constructively made or incurred during such weekly period. Such laborers and mechanics must be paid the appropriate wage rate and fringe benefits on the wage determination for the classification(s) of work actually performed, without regard to skill, except as provided in paragraph 4. of this section. Laborers or mechanics performing work in more than one classification may be compensated at the rate specified for each classification for the time actually worked therein: *Provided*, That the employer's payroll records accurately set forth the time spent in each classification in which work is performed. The wage determination (including any additional classifications and wage rates conformed under paragraph 1.c. of this section) and the Davis-Bacon poster (WH-1321) must be posted at all times by the contractor and its subcontractors at the site of the work in a prominent and accessible place where it can be easily seen by the workers.

b. *Frequently recurring classifications.* (1) In addition to wage and fringe benefit rates that have been determined to be prevailing under the procedures set forth in [29 CFR part 1](#), a wage determination may contain, pursuant to § 1.3(f), wage and fringe benefit rates for classifications of laborers and mechanics for which conformance requests are regularly submitted pursuant to paragraph 1.c. of this section, provided that:

(i) The work performed by the classification is not performed by a classification in the wage determination for which a prevailing wage rate has been determined;

(ii) The classification is used in the area by the construction industry; and

(iii) The wage rate for the classification bears a reasonable relationship to the prevailing wage rates contained in the wage determination.

(2) The Administrator will establish wage rates for such classifications in accordance with paragraph 1.c.(1)(iii) of this section. Work performed in such a classification must be paid at no less than the wage and fringe benefit rate listed on the wage determination for such classification.

c. *Conformance.* (1) The contracting officer must require that any class of laborers or mechanics, including helpers, which is not listed in the wage determination and which is to be employed under the contract be classified in conformance with the wage determination. Conformance of an additional classification and wage rate and fringe benefits is appropriate only when the following criteria have been met:

(i) The work to be performed by the classification requested is not performed by a classification in the wage determination; and

(ii) The classification is used in the area by the construction industry; and

(iii) The proposed wage rate, including any bona fide fringe benefits, bears a reasonable relationship to the wage rates contained in the wage determination.

(2) The conformance process may not be used to split, subdivide, or otherwise avoid application of classifications listed in the wage determination.

(3) If the contractor and the laborers and mechanics to be employed in the classification (if known), or their representatives, and the contracting officer agree on the classification and wage rate (including the amount designated for fringe benefits where appropriate), a report of the action taken will be sent by the contracting officer by email to DBAconformance@dol.gov. The Administrator, or an authorized representative, will approve, modify, or disapprove every additional classification action within 30 days of receipt and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(4) In the event the contractor, the laborers or mechanics to be employed in the classification or their representatives, and the contracting officer do not agree on the proposed classification and wage rate (including the amount designated for fringe benefits, where appropriate), the contracting officer will, by email to DBAconformance@dol.gov, refer the questions, including the views of all interested parties and the recommendation of the contracting officer, to the Administrator for determination. The Administrator, or an authorized representative, will issue a determination within 30 days of receipt and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(5) The contracting officer must promptly notify the contractor of the action taken by the Wage and Hour Division

under paragraphs 1.c.(3) and (4) of this section. The contractor must furnish a written copy of such determination to each affected worker or it must be posted as a part of the wage determination. The wage rate (including fringe benefits where appropriate) determined pursuant to paragraph 1.c.(3) or (4) of this section must be paid to all workers performing work in the classification under this contract from the first day on which work is performed in the classification.

d. *Fringe benefits not expressed as an hourly rate.* Whenever the minimum wage rate prescribed in the contract for a class of laborers or mechanics includes a fringe benefit which is not expressed as an hourly rate, the contractor may either pay the benefit as stated in the wage determination or may pay another bona fide fringe benefit or an hourly cash equivalent thereof.

e. *Unfunded plans.* If the contractor does not make payments to a trustee or other third person, the contractor may consider as part of the wages of any laborer or mechanic the amount of any costs reasonably anticipated in providing bona fide fringe benefits under a plan or program, *Provided*, That the Secretary of Labor has found, upon the written request of the contractor, in accordance with the criteria set forth in § 5.28, that the applicable standards of the Davis-Bacon Act have been met. The Secretary of Labor may require the contractor to set aside in a separate account assets for the meeting of obligations under the plan or program.

f. *Interest.* In the event of a failure to pay all or part of the wages required by the contract, the contractor will be required to pay interest on any underpayment of wages.

2. Withholding (29 CFR 5.5)

a. *Withholding requirements.* The contracting agency may, upon its own action, or must, upon written request of an authorized representative of the Department of Labor, withhold or cause to be withheld from the contractor so much of the accrued payments or advances as may be considered necessary to satisfy the liabilities of the prime contractor or any subcontractor for the full amount of wages and monetary relief, including interest, required by the clauses set forth in this section for violations of this contract, or to satisfy any such liabilities required by any other Federal contract, or federally assisted contract subject to Davis-Bacon labor standards, that is held by the same prime contractor (as defined in § 5.2). The necessary funds may be withheld from the contractor under this contract, any other Federal contract with the same prime contractor, or any other federally assisted contract that is subject to Davis-Bacon labor standards requirements and is held by the same prime contractor, regardless of whether the other contract was awarded or assisted by the same agency, and such funds may be used to satisfy the contractor liability for which the funds were withheld. In the event of a contractor's failure to pay any laborer or mechanic, including any apprentice or helper working on the site of the work all or part of the wages required by the contract, or upon the contractor's failure to submit the required records as discussed in paragraph 3.d. of this section, the contracting agency may on its own initiative and after written notice to the contractor, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds until such violations have ceased.

b. *Priority to withheld funds.* The Department has priority to funds withheld or to be withheld in accordance with paragraph

2 a. of this section or Section V, paragraph 3 a., or both, over claims to those funds by:

- (1) A contractor's surety(ies), including without limitation performance bond sureties and payment bond sureties;
- (2) A contracting agency for its procurement costs;
- (3) A trustee(s) (either a court-appointed trustee or a U.S. trustee, or both) in bankruptcy of a contractor, or a contractor's bankruptcy estate;
- (4) A contractor's assignee(s);
- (5) A contractor's successor(s); or
- (6) A claim asserted under the Prompt Payment Act, [31 U.S.C. 3901-3907](#).

3. Records and certified payrolls (29 CFR 5.5)

a. *Basic record requirements* (1) *Length of record retention.* All regular payrolls and other basic records must be maintained by the contractor and any subcontractor during the course of the work and preserved for all laborers and mechanics working at the site of the work (or otherwise working in construction or development of the project under a development statute) for a period of at least 3 years after all the work on the prime contract is completed.

(2) *Information required.* Such records must contain the name; Social Security number; last known address, telephone number, and email address of each such worker; each worker's correct classification(s) of work actually performed; hourly rates of wages paid (including rates of contributions or costs anticipated for bona fide fringe benefits or cash equivalents thereof of the types described in [40 U.S.C. 3141\(2\)\(B\)](#) of the Davis-Bacon Act); daily and weekly number of hours actually worked in total and on each covered contract; deductions made; and actual wages paid.

(3) *Additional records relating to fringe benefits.* Whenever the Secretary of Labor has found under paragraph 1.e. of this section that the wages of any laborer or mechanic include the amount of any costs reasonably anticipated in providing benefits under a plan or program described in [40 U.S.C. 3141\(2\)\(B\)](#) of the Davis-Bacon Act, the contractor must maintain records which show that the commitment to provide such benefits is enforceable, that the plan or program is financially responsible, and that the plan or program has been communicated in writing to the laborers or mechanics affected, and records which show the costs anticipated or the actual cost incurred in providing such benefits.

(4) *Additional records relating to apprenticeship.* Contractors with apprentices working under approved programs must maintain written evidence of the registration of apprenticeship programs, the registration of the apprentices, and the ratios and wage rates prescribed in the applicable programs.

b. *Certified payroll requirements* (1) *Frequency and method of submission.* The contractor or subcontractor must submit weekly, for each week in which any DBA- or Related Acts-covered work is performed, certified payrolls to the contracting

agency. The prime contractor is responsible for the submission of all certified payrolls by all subcontractors. A contracting agency or prime contractor may permit or require contractors to submit certified payrolls through an electronic system, as long as the electronic system requires a legally valid electronic signature; the system allows the contractor, the contracting agency, and the Department of Labor to access the certified payrolls upon request for at least 3 years after the work on the prime contract has been completed; and the contracting agency or prime contractor permits other methods of submission in situations where the contractor is unable or limited in its ability to use or access the electronic system.

(2) *Information required.* The certified payrolls submitted must set out accurately and completely all of the information required to be maintained under paragraph 3 a.(2) of this section, except that full Social Security numbers and last known addresses, telephone numbers, and email addresses must not be included on weekly transmittals. Instead, the certified payrolls need only include an individually identifying number for each worker (e.g., the last four digits of the worker's Social Security number). The required weekly certified payroll information may be submitted using Optional Form WH-347 or in any other format desired. Optional Form WH-347 is available for this purpose from the Wage and Hour Division website at <https://www.dol.gov/sites/dolgov/files/WH/legacy/files/wh347.pdf> or its successor website. It is not a violation of this section for a prime contractor to require a subcontractor to provide full Social Security numbers and last known addresses, telephone numbers, and email addresses to the prime contractor for its own records, without weekly submission by the subcontractor to the contracting agency.

(3) *Statement of Compliance.* Each certified payroll submitted must be accompanied by a "Statement of Compliance," signed by the contractor or subcontractor, or the contractor's or subcontractor's agent who pays or supervises the payment of the persons working on the contract, and must certify the following:

(i) That the certified payroll for the payroll period contains the information required to be provided under paragraph 3.b. of this section, the appropriate information and basic records are being maintained under paragraph 3.a. of this section, and such information and records are correct and complete;

(ii) That each laborer or mechanic (including each helper and apprentice) working on the contract during the payroll period has been paid the full weekly wages earned, without rebate, either directly or indirectly, and that no deductions have been made either directly or indirectly from the full wages earned, other than permissible deductions as set forth in [29 CFR part 3](#); and

(iii) That each laborer or mechanic has been paid not less than the applicable wage rates and fringe benefits or cash equivalents for the classification(s) of work actually performed, as specified in the applicable wage determination incorporated into the contract.

(4) *Use of Optional Form WH-347.* The weekly submission of a properly executed certification set forth on the reverse side of Optional Form WH-347 will satisfy the requirement for submission of the "Statement of Compliance" required by paragraph 3.b.(3) of this section.

(5) *Signature.* The signature by the contractor, subcontractor, or the contractor's or subcontractor's agent must be an original handwritten signature or a legally valid electronic signature.

(6) *Falsification.* The falsification of any of the above certifications may subject the contractor or subcontractor to civil or criminal prosecution under [18 U.S.C. 1001](#) and [31 U.S.C. 3729](#).

(7) *Length of certified payroll retention.* The contractor or subcontractor must preserve all certified payrolls during the course of the work and for a period of 3 years after all the work on the prime contract is completed.

c. *Contracts, subcontracts, and related documents.* The contractor or subcontractor must maintain this contract or subcontract and related documents including, without limitation, bids, proposals, amendments, modifications, and extensions. The contractor or subcontractor must preserve these contracts, subcontracts, and related documents during the course of the work and for a period of 3 years after all the work on the prime contract is completed.

d. *Required disclosures and access* (1) *Required record disclosures and access to workers.* The contractor or subcontractor must make the records required under paragraphs 3.a. through 3.c. of this section, and any other documents that the contracting agency, the State DOT, the FHWA, or the Department of Labor deems necessary to determine compliance with the labor standards provisions of any of the applicable statutes referenced by § 5.1, available for inspection, copying, or transcription by authorized representatives of the contracting agency, the State DOT, the FHWA, or the Department of Labor, and must permit such representatives to interview workers during working hours on the job.

(2) *Sanctions for non-compliance with records and worker access requirements.* If the contractor or subcontractor fails to submit the required records or to make them available, or refuses to permit worker interviews during working hours on the job, the Federal agency may, after written notice to the contractor, sponsor, applicant, owner, or other entity, as the case may be, that maintains such records or that employs such workers, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds. Furthermore, failure to submit the required records upon request or to make such records available, or to permit worker interviews during working hours on the job, may be grounds for debarment action pursuant to § 5.12. In addition, any contractor or other person that fails to submit the required records or make those records available to WHD within the time WHD requests that the records be produced will be precluded from introducing as evidence in an administrative proceeding under [29 CFR part 6](#) any of the required records that were not provided or made available to WHD. WHD will take into consideration a reasonable request from the contractor or person for an extension of the time for submission of records. WHD will determine the reasonableness of the request and may consider, among other things, the location of the records and the volume of production.

(3) *Required information disclosures.* Contractors and subcontractors must maintain the full Social Security number and last known address, telephone number, and email address

of each covered worker, and must provide them upon request to the contracting agency, the State DOT, the FHWA, the contractor, or the Wage and Hour Division of the Department of Labor for purposes of an investigation or other compliance action.

4. Apprentices and equal employment opportunity (29 CFR 5.5)

a. *Apprentices* (1) *Rate of pay.* Apprentices will be permitted to work at less than the predetermined rate for the work they perform when they are employed pursuant to and individually registered in a bona fide apprenticeship program registered with the U.S. Department of Labor, Employment and Training Administration, Office of Apprenticeship (OA), or with a State Apprenticeship Agency recognized by the OA. A person who is not individually registered in the program, but who has been certified by the OA or a State Apprenticeship Agency (where appropriate) to be eligible for probationary employment as an apprentice, will be permitted to work at less than the predetermined rate for the work they perform in the first 90 days of probationary employment as an apprentice in such a program. In the event the OA or a State Apprenticeship Agency recognized by the OA withdraws approval of an apprenticeship program, the contractor will no longer be permitted to use apprentices at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(2) *Fringe benefits.* Apprentices must be paid fringe benefits in accordance with the provisions of the apprenticeship program. If the apprenticeship program does not specify fringe benefits, apprentices must be paid the full amount of fringe benefits listed on the wage determination for the applicable classification. If the Administrator determines that a different practice prevails for the applicable apprentice classification, fringe benefits must be paid in accordance with that determination.

(3) *Apprenticeship ratio.* The allowable ratio of apprentices to journeymen on the job site in any craft classification must not be greater than the ratio permitted to the contractor as to the entire work force under the registered program or the ratio applicable to the locality of the project pursuant to paragraph 4.a.(4) of this section. Any worker listed on a payroll at an apprentice wage rate, who is not registered or otherwise employed as stated in paragraph 4.a.(1) of this section, must be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any apprentice performing work on the job site in excess of the ratio permitted under this section must be paid not less than the applicable wage rate on the wage determination for the work actually performed.

(4) *Reciprocity of ratios and wage rates.* Where a contractor is performing construction on a project in a locality other than the locality in which its program is registered, the ratios and wage rates (expressed in percentages of the journeyworker's hourly rate) applicable within the locality in which the construction is being performed must be observed. If there is no applicable ratio or wage rate for the locality of the project, the ratio and wage rate specified in the contractor's registered program must be observed.

b. *Equal employment opportunity.* The use of apprentices and journeymen under this part must be in conformity with

the equal employment opportunity requirements of Executive Order 11246, as amended, and [29 CFR part 30](#).

c. Apprentices and Trainees (programs of the U.S. DOT).

Apprentices and trainees working under apprenticeship and skill training programs which have been certified by the Secretary of Transportation as promoting EEO in connection with Federal-aid highway construction programs are not subject to the requirements of paragraph 4 of this Section IV. 23 CFR 230.111(e)(2). The straight time hourly wage rates for apprentices and trainees under such programs will be established by the particular programs. The ratio of apprentices and trainees to journeymen shall not be greater than permitted by the terms of the particular program.

5. Compliance with Copeland Act requirements. The contractor shall comply with the requirements of 29 CFR part 3, which are incorporated by reference in this contract as provided in 29 CFR 5.5.

6. Subcontracts. The contractor or subcontractor must insert FHWA-1273 in any subcontracts, along with the applicable wage determination(s) and such other clauses or contract modifications as the contracting agency may by appropriate instructions require, and a clause requiring the subcontractors to include these clauses and wage determination(s) in any lower tier subcontracts. The prime contractor is responsible for the compliance by any subcontractor or lower tier subcontractor with all the contract clauses in this section. In the event of any violations of these clauses, the prime contractor and any subcontractor(s) responsible will be liable for any unpaid wages and monetary relief, including interest from the date of the underpayment or loss, due to any workers of lower-tier subcontractors, and may be subject to debarment, as appropriate. 29 CFR 5.5.

7. Contract termination: debarment. A breach of the contract clauses in 29 CFR 5.5 may be grounds for termination of the contract, and for debarment as a contractor and a subcontractor as provided in 29 CFR 5.12.

8. Compliance with Davis-Bacon and Related Act requirements. All rulings and interpretations of the Davis-Bacon and Related Acts contained in 29 CFR parts 1, 3, and 5 are herein incorporated by reference in this contract as provided in 29 CFR 5.5.

9. Disputes concerning labor standards. As provided in 29 CFR 5.5, disputes arising out of the labor standards provisions of this contract shall not be subject to the general disputes clause of this contract. Such disputes shall be resolved in accordance with the procedures of the Department of Labor set forth in 29 CFR parts 5, 6, and 7. Disputes within the meaning of this clause include disputes between the contractor (or any of its subcontractors) and the contracting agency, the U.S. Department of Labor, or the employees or their representatives.

10. Certification of eligibility. a. By entering into this contract, the contractor certifies that neither it nor any person or firm who has an interest in the contractor's firm is a person or firm ineligible to be awarded Government contracts by virtue of [40 U.S.C. 3144\(b\)](#) or § 5.12(a).

b. No part of this contract shall be subcontracted to any person or firm ineligible for award of a Government contract by virtue of [40 U.S.C. 3144\(b\)](#) or § 5.12(a).

c. The penalty for making false statements is prescribed in the U.S. Code, Title 18 Crimes and Criminal Procedure, [18 U.S.C. 1001](#).

11. Anti-retaliation. It is unlawful for any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate against, or to cause any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate against, any worker or job applicant for:

a. Notifying any contractor of any conduct which the worker reasonably believes constitutes a violation of the DBA, Related Acts, this part, or [29 CFR part 1](#) or [3](#);

b. Filing any complaint, initiating or causing to be initiated any proceeding, or otherwise asserting or seeking to assert on behalf of themselves or others any right or protection under the DBA, Related Acts, this part, or [29 CFR part 1](#) or [3](#);

c. Cooperating in any investigation or other compliance action, or testifying in any proceeding under the DBA, Related Acts, this part, or [29 CFR part 1](#) or [3](#); or

d. Informing any other person about their rights under the DBA, Related Acts, this part, or [29 CFR part 1](#) or [3](#).

V. CONTRACT WORK HOURS AND SAFETY STANDARDS ACT

Pursuant to 29 CFR 5.5(b), the following clauses apply to any Federal-aid construction contract in an amount in excess of \$100,000 and subject to the overtime provisions of the Contract Work Hours and Safety Standards Act. These clauses shall be inserted in addition to the clauses required by 29 CFR 5.5(a) or 29 CFR 4.6. As used in this paragraph, the terms laborers and mechanics include watchpersons and guards.

1. Overtime requirements. No contractor or subcontractor contracting for any part of the contract work which may require or involve the employment of laborers or mechanics shall require or permit any such laborer or mechanic in any workweek in which he or she is employed on such work to work in excess of forty hours in such workweek unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of forty hours in such workweek. 29 CFR 5.5.

2. Violation; liability for unpaid wages; liquidated damages. In the event of any violation of the clause set forth in paragraph 1. of this section the contractor and any subcontractor responsible therefor shall be liable for the unpaid wages and interest from the date of the underpayment. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such District or to such territory), for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or

mechanic, including watchpersons and guards, employed in violation of the clause set forth in paragraph 1. of this section, in the sum currently provided in 29 CFR 5.5(b)(2)* for each calendar day on which such individual was required or permitted to work in excess of the standard workweek of forty hours without payment of the overtime wages required by the clause set forth in paragraph 1. of this section.

* \$31 as of January 15, 2023 (See 88 FR 88 FR 2210) as may be adjusted annually by the Department of Labor, pursuant to the Federal Civil Penalties Inflation Adjustment Act of 1990.

3. Withholding for unpaid wages and liquidated damages

a. *Withholding process.* The FHWA or the contracting agency may, upon its own action, or must, upon written request of an authorized representative of the Department of Labor, withhold or cause to be withheld from the contractor so much of the accrued payments or advances as may be considered necessary to satisfy the liabilities of the prime contractor or any subcontractor for any unpaid wages; monetary relief, including interest; and liquidated damages required by the clauses set forth in this section on this contract, any other Federal contract with the same prime contractor, or any other federally assisted contract subject to the Contract Work Hours and Safety Standards Act that is held by the same prime contractor (as defined in § 5.2). The necessary funds may be withheld from the contractor under this contract, any other Federal contract with the same prime contractor, or any other federally assisted contract that is subject to the Contract Work Hours and Safety Standards Act and is held by the same prime contractor, regardless of whether the other contract was awarded or assisted by the same agency, and such funds may be used to satisfy the contractor liability for which the funds were withheld.

b. *Priority to withheld funds.* The Department has priority to funds withheld or to be withheld in accordance with Section IV paragraph 2.a. or paragraph 3.a. of this section, or both, over claims to those funds by:

- (1) A contractor's surety(ies), including without limitation performance bond sureties and payment bond sureties;
- (2) A contracting agency for its procurement costs;
- (3) A trustee(s) (either a court-appointed trustee or a U.S. trustee, or both) in bankruptcy of a contractor, or a contractor's bankruptcy estate;
- (4) A contractor's assignee(s);
- (5) A contractor's successor(s); or
- (6) A claim asserted under the Prompt Payment Act, [31 U.S.C. 3901](#)–3907.

4. **Subcontracts.** The contractor or subcontractor must insert in any subcontracts the clauses set forth in paragraphs 1. through 5. of this section and a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor is responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in paragraphs 1. through 5. In the

event of any violations of these clauses, the prime contractor and any subcontractor(s) responsible will be liable for any unpaid wages and monetary relief, including interest from the date of the underpayment or loss, due to any workers of lower-tier subcontractors, and associated liquidated damages and may be subject to debarment, as appropriate.

5. **Anti-retaliation.** It is unlawful for any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate against, or to cause any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate against, any worker or job applicant for:

a. Notifying any contractor of any conduct which the worker reasonably believes constitutes a violation of the Contract Work Hours and Safety Standards Act (CWHSSA) or its implementing regulations in this part;

b. Filing any complaint, initiating or causing to be initiated any proceeding, or otherwise asserting or seeking to assert on behalf of themselves or others any right or protection under CWHSSA or this part;

c. Cooperating in any investigation or other compliance action, or testifying in any proceeding under CWHSSA or this part; or

d. Informing any other person about their rights under CWHSSA or this part.

VI. SUBLETTING OR ASSIGNING THE CONTRACT

This provision is applicable to all Federal-aid construction contracts on the National Highway System pursuant to 23 CFR 635.116.

1. The contractor shall perform with its own organization contract work amounting to not less than 30 percent (or a greater percentage if specified elsewhere in the contract) of the total original contract price, excluding any specialty items designated by the contracting agency. Specialty items may be performed by subcontract and the amount of any such specialty items performed may be deducted from the total original contract price before computing the amount of work required to be performed by the contractor's own organization (23 CFR 635.116).

a. The term "perform work with its own organization" in paragraph 1 of Section VI refers to workers employed or leased by the prime contractor, and equipment owned or rented by the prime contractor, with or without operators. Such term does not include employees or equipment of a subcontractor or lower tier subcontractor, agents of the prime contractor, or any other assignees. The term may include payments for the costs of hiring leased employees from an employee leasing firm meeting all relevant Federal and State regulatory requirements. Leased employees may only be included in this term if the prime contractor meets all of the following conditions: (based on longstanding interpretation)

- (1) the prime contractor maintains control over the supervision of the day-to-day activities of the leased employees;
- (2) the prime contractor remains responsible for the quality of the work of the leased employees;

(3) the prime contractor retains all power to accept or exclude individual employees from work on the project; and
(4) the prime contractor remains ultimately responsible for the payment of predetermined minimum wages, the submission of payrolls, statements of compliance and all other Federal regulatory requirements.

b. "Specialty Items" shall be construed to be limited to work that requires highly specialized knowledge, abilities, or equipment not ordinarily available in the type of contracting organizations qualified and expected to bid or propose on the contract as a whole and in general are to be limited to minor components of the overall contract. 23 CFR 635.102.

2. Pursuant to 23 CFR 635.116(a), the contract amount upon which the requirements set forth in paragraph (1) of Section VI is computed includes the cost of material and manufactured products which are to be purchased or produced by the contractor under the contract provisions.

3. Pursuant to 23 CFR 635.116(c), the contractor shall furnish (a) a competent superintendent or supervisor who is employed by the firm, has full authority to direct performance of the work in accordance with the contract requirements, and is in charge of all construction operations (regardless of who performs the work) and (b) such other of its own organizational resources (supervision, management, and engineering services) as the contracting officer determines is necessary to assure the performance of the contract.

4. No portion of the contract shall be sublet, assigned or otherwise disposed of except with the written consent of the contracting officer, or authorized representative, and such consent when given shall not be construed to relieve the contractor of any responsibility for the fulfillment of the contract. Written consent will be given only after the contracting agency has assured that each subcontract is evidenced in writing and that it contains all pertinent provisions and requirements of the prime contract. (based on long-standing interpretation of 23 CFR 635.116).

5. The 30-percent self-performance requirement of paragraph (1) is not applicable to design-build contracts; however, contracting agencies may establish their own self-performance requirements. 23 CFR 635.116(d).

VII. SAFETY: ACCIDENT PREVENTION

This provision is applicable to all Federal-aid construction contracts and to all related subcontracts.

1. In the performance of this contract the contractor shall comply with all applicable Federal, State, and local laws governing safety, health, and sanitation (23 CFR Part 635). The contractor shall provide all safeguards, safety devices and protective equipment and take any other needed actions as it determines, or as the contracting officer may determine, to be reasonably necessary to protect the life and health of employees on the job and the safety of the public and to protect property in connection with the performance of the work covered by the contract. 23 CFR 635.108.

2. It is a condition of this contract, and shall be made a condition of each subcontract, which the contractor enters into pursuant to this contract, that the contractor and any subcontractor shall not permit any employee, in performance of the contract, to work in surroundings or under conditions which are unsanitary, hazardous or dangerous to his/her health or safety, as determined under construction safety and

health standards (29 CFR Part 1926) promulgated by the Secretary of Labor, in accordance with Section 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 3704). 29 CFR 1926.10.

3. Pursuant to 29 CFR 1926.3, it is a condition of this contract that the Secretary of Labor or authorized representative thereof, shall have right of entry to any site of contract performance to inspect or investigate the matter of compliance with the construction safety and health standards and to carry out the duties of the Secretary under Section 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 3704).

VIII. FALSE STATEMENTS CONCERNING HIGHWAY PROJECTS

This provision is applicable to all Federal-aid construction contracts and to all related subcontracts.

In order to assure high quality and durable construction in conformity with approved plans and specifications and a high degree of reliability on statements and representations made by engineers, contractors, suppliers, and workers on Federal-aid highway projects, it is essential that all persons concerned with the project perform their functions as carefully, thoroughly, and honestly as possible. Willful falsification, distortion, or misrepresentation with respect to any facts related to the project is a violation of Federal law. To prevent any misunderstanding regarding the seriousness of these and similar acts, Form FHWA-1022 shall be posted on each Federal-aid highway project (23 CFR Part 635) in one or more places where it is readily available to all persons concerned with the project:

18 U.S.C. 1020 reads as follows:

"Whoever, being an officer, agent, or employee of the United States, or of any State or Territory, or whoever, whether a person, association, firm, or corporation, knowingly makes any false statement, false representation, or false report as to the character, quality, quantity, or cost of the material used or to be used, or the quantity or quality of the work performed or to be performed, or the cost thereof in connection with the submission of plans, maps, specifications, contracts, or costs of construction on any highway or related project submitted for approval to the Secretary of Transportation; or

Whoever knowingly makes any false statement, false representation, false report or false claim with respect to the character, quality, quantity, or cost of any work performed or to be performed, or materials furnished or to be furnished, in connection with the construction of any highway or related project approved by the Secretary of Transportation; or

Whoever knowingly makes any false statement or false representation as to material fact in any statement, certificate, or report submitted pursuant to provisions of the Federal-aid Roads Act approved July 11, 1916, (39 Stat. 355), as amended and supplemented;

Shall be fined under this title or imprisoned not more than 5 years or both."

IX. IMPLEMENTATION OF CLEAN AIR ACT AND FEDERAL WATER POLLUTION CONTROL ACT (42 U.S.C. 7606; 2 CFR 200.88; EO 11738)

This provision is applicable to all Federal-aid construction contracts in excess of \$150,000 and to all related subcontracts. 48 CFR 2.101; 2 CFR 200.327.

By submission of this bid/proposal or the execution of this contract or subcontract, as appropriate, the bidder, proposer, Federal-aid construction contractor, subcontractor, supplier, or vendor agrees to comply with all applicable standards, orders or regulations issued pursuant to the Clean Air Act (42 U.S.C. 7401-7671q) and the Federal Water Pollution Control Act, as amended (33 U.S.C. 1251-1387). Violations must be reported to the Federal Highway Administration and the Regional Office of the Environmental Protection Agency. 2 CFR Part 200, Appendix II.

The contractor agrees to include or cause to be included the requirements of this Section in every subcontract, and further agrees to take such action as the contracting agency may direct as a means of enforcing such requirements. 2 CFR 200.327.

X. CERTIFICATION REGARDING DEBARMENT, SUSPENSION, INELIGIBILITY AND VOLUNTARY EXCLUSION

This provision is applicable to all Federal-aid construction contracts, design-build contracts, subcontracts, lower-tier subcontracts, purchase orders, lease agreements, consultant contracts or any other covered transaction requiring FHWA approval or that is estimated to cost \$25,000 or more – as defined in 2 CFR Parts 180 and 1200. 2 CFR 180.220 and 1200.220.

1. Instructions for Certification – First Tier Participants:

a. By signing and submitting this proposal, the prospective first tier participant is providing the certification set out below.

b. The inability of a person to provide the certification set out below will not necessarily result in denial of participation in this covered transaction. The prospective first tier participant shall submit an explanation of why it cannot provide the certification set out below. The certification or explanation will be considered in connection with the department or agency's determination whether to enter into this transaction. However, failure of the prospective first tier participant to furnish a certification or an explanation shall disqualify such a person from participation in this transaction. 2 CFR 180.320.

c. The certification in this clause is a material representation of fact upon which reliance was placed when the contracting agency determined to enter into this transaction. If it is later determined that the prospective participant knowingly rendered an erroneous certification, in addition to other remedies available to the Federal Government, the contracting agency may terminate this transaction for cause of default. 2 CFR 180.325.

d. The prospective first tier participant shall provide immediate written notice to the contracting agency to whom this proposal is submitted if any time the prospective first tier participant learns that its certification was erroneous when submitted or has become erroneous by reason of changed circumstances. 2 CFR 180.345 and 180.350.

e. The terms "covered transaction," "debarred," "suspended," "ineligible," "participant," "person," "principal," and "voluntarily excluded," as used in this clause, are defined in 2 CFR Parts 180, Subpart I, 180.900-180.1020, and 1200. "First Tier Covered Transactions" refers to any covered transaction between a recipient or subrecipient of Federal funds and a participant (such as the prime or general contract). "Lower Tier Covered Transactions" refers to any covered transaction under a First Tier Covered Transaction (such as subcontracts). "First Tier Participant" refers to the participant who has entered into a covered transaction with a recipient or subrecipient of Federal funds (such as the prime or general contractor). "Lower Tier Participant" refers any participant who has entered into a covered transaction with a First Tier Participant or other Lower Tier Participants (such as subcontractors and suppliers).

f. The prospective first tier participant agrees by submitting this proposal that, should the proposed covered transaction be entered into, it shall not knowingly enter into any lower tier covered transaction with a person who is debarred, suspended, declared ineligible, or voluntarily excluded from participation in this covered transaction, unless authorized by the department or agency entering into this transaction. 2 CFR 180.330.

g. The prospective first tier participant further agrees by submitting this proposal that it will include the clause titled "Certification Regarding Debarment, Suspension, Ineligibility and Voluntary Exclusion-Lower Tier Covered Transactions," provided by the department or contracting agency, entering into this covered transaction, without modification, in all lower tier covered transactions and in all solicitations for lower tier covered transactions exceeding the \$25,000 threshold. 2 CFR 180.220 and 180.300.

h. A participant in a covered transaction may rely upon a certification of a prospective participant in a lower tier covered transaction that is not debarred, suspended, ineligible, or voluntarily excluded from the covered transaction, unless it knows that the certification is erroneous. 2 CFR 180.300; 180.320, and 180.325. A participant is responsible for ensuring that its principals are not suspended, debarred, or otherwise ineligible to participate in covered transactions. 2 CFR 180.335. To verify the eligibility of its principals, as well as the eligibility of any lower tier prospective participants, each participant may, but is not required to, check the System for Award Management website (<https://www.sam.gov/>). 2 CFR 180.300, 180.320, and 180.325.

i. Nothing contained in the foregoing shall be construed to require the establishment of a system of records in order to render in good faith the certification required by this clause. The knowledge and information of the prospective participant is not required to exceed that which is normally possessed by a prudent person in the ordinary course of business dealings.

j. Except for transactions authorized under paragraph (f) of these instructions, if a participant in a covered transaction knowingly enters into a lower tier covered transaction with a person who is suspended, debarred, ineligible, or voluntarily excluded from participation in this transaction, in addition to other remedies available to the Federal Government, the department or agency may terminate this transaction for cause or default. 2 CFR 180.325.

* * * * *

2. Certification Regarding Debarment, Suspension, Ineligibility and Voluntary Exclusion – First Tier Participants:

a. The prospective first tier participant certifies to the best of its knowledge and belief, that it and its principals:

(1) Are not presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from participating in covered transactions by any Federal department or agency, 2 CFR 180.335;.

(2) Have not within a three-year period preceding this proposal been convicted of or had a civil judgment rendered against them for commission of fraud or a criminal offense in connection with obtaining, attempting to obtain, or performing a public (Federal, State, or local) transaction or contract under a public transaction; violation of Federal or State antitrust statutes or commission of embezzlement, theft, forgery, bribery, falsification or destruction of records, making false statements, or receiving stolen property, 2 CFR 180.800;

(3) Are not presently indicted for or otherwise criminally or civilly charged by a governmental entity (Federal, State or local) with commission of any of the offenses enumerated in paragraph (a)(2) of this certification, 2 CFR 180.700 and 180.800; and

(4) Have not within a three-year period preceding this application/proposal had one or more public transactions (Federal, State or local) terminated for cause or default. 2 CFR 180.335(d).

(5) Are not a corporation that has been convicted of a felony violation under any Federal law within the two-year period preceding this proposal (USDOT Order 4200.6 implementing appropriations act requirements); and

(6) Are not a corporation with any unpaid Federal tax liability that has been assessed, for which all judicial and administrative remedies have been exhausted, or have lapsed, and that is not being paid in a timely manner pursuant to an agreement with the authority responsible for collecting the tax liability (USDOT Order 4200.6 implementing appropriations act requirements).

b. Where the prospective participant is unable to certify to any of the statements in this certification, such prospective participant should attach an explanation to this proposal. 2 CFR 180.335 and 180.340.

3. Instructions for Certification - Lower Tier Participants:

(Applicable to all subcontracts, purchase orders, and other lower tier transactions requiring prior FHWA approval or estimated to cost \$25,000 or more - 2 CFR Parts 180 and 1200). 2 CFR 180.220 and 1200.220.

a. By signing and submitting this proposal, the prospective lower tier participant is providing the certification set out below.

b. The certification in this clause is a material representation of fact upon which reliance was placed when this transaction was entered into. If it is later determined that the prospective lower tier participant knowingly rendered an erroneous certification, in addition to other remedies available to the Federal Government, the department, or agency with which

this transaction originated may pursue available remedies, including suspension and/or debarment.

c. The prospective lower tier participant shall provide immediate written notice to the person to which this proposal is submitted if at any time the prospective lower tier participant learns that its certification was erroneous by reason of changed circumstances. 2 CFR 180.365.

d. The terms "covered transaction," "debarred," "suspended," "ineligible," "participant," "person," "principal," and "voluntarily excluded," as used in this clause, are defined in 2 CFR Parts 180, Subpart I, 180.900 – 180.1020, and 1200. You may contact the person to which this proposal is submitted for assistance in obtaining a copy of those regulations. "First Tier Covered Transactions" refers to any covered transaction between a recipient or subrecipient of Federal funds and a participant (such as the prime or general contractor). "Lower Tier Covered Transactions" refers to any covered transaction under a First Tier Covered Transaction (such as subcontracts). "First Tier Participant" refers to the participant who has entered into a covered transaction with a recipient or subrecipient of Federal funds (such as the prime or general contractor). "Lower Tier Participant" refers any participant who has entered into a covered transaction with a First Tier Participant or other Lower Tier Participants (such as subcontractors and suppliers).

e. The prospective lower tier participant agrees by submitting this proposal that, should the proposed covered transaction be entered into, it shall not knowingly enter into any lower tier covered transaction with a person who is debarred, suspended, declared ineligible, or voluntarily excluded from participation in this covered transaction, unless authorized by the department or agency with which this transaction originated. 2 CFR 1200.220 and 1200.332.

f. The prospective lower tier participant further agrees by submitting this proposal that it will include this clause titled "Certification Regarding Debarment, Suspension, Ineligibility and Voluntary Exclusion-Lower Tier Covered Transaction," without modification, in all lower tier covered transactions and in all solicitations for lower tier covered transactions exceeding the \$25,000 threshold. 2 CFR 180.220 and 1200.220.

g. A participant in a covered transaction may rely upon a certification of a prospective participant in a lower tier covered transaction that is not debarred, suspended, ineligible, or voluntarily excluded from the covered transaction, unless it knows that the certification is erroneous. A participant is responsible for ensuring that its principals are not suspended, debarred, or otherwise ineligible to participate in covered transactions. To verify the eligibility of its principals, as well as the eligibility of any lower tier prospective participants, each participant may, but is not required to, check the System for Award Management website (<https://www.sam.gov/>), which is compiled by the General Services Administration. 2 CFR 180.300, 180.320, 180.330, and 180.335.

h. Nothing contained in the foregoing shall be construed to require establishment of a system of records in order to render in good faith the certification required by this clause. The knowledge and information of participant is not required to exceed that which is normally possessed by a prudent person in the ordinary course of business dealings.

i. Except for transactions authorized under paragraph e of these instructions, if a participant in a covered transaction knowingly enters into a lower tier covered transaction with a person who is suspended, debarred, ineligible, or voluntarily

excluded from participation in this transaction, in addition to other remedies available to the Federal Government, the department or agency with which this transaction originated may pursue available remedies, including suspension and/or debarment. 2 CFR 180.325.

4. Certification Regarding Debarment, Suspension, Ineligibility and Voluntary Exclusion--Lower Tier Participants:

a. The prospective lower tier participant certifies, by submission of this proposal, that neither it nor its principals:

(1) is presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from participating in covered transactions by any Federal department or agency, 2 CFR 180.355;

(2) is a corporation that has been convicted of a felony violation under any Federal law within the two-year period preceding this proposal (USDOT Order 4200.6 implementing appropriations act requirements); and

(3) is a corporation with any unpaid Federal tax liability that has been assessed, for which all judicial and administrative remedies have been exhausted, or have lapsed, and that is not being paid in a timely manner pursuant to an agreement with the authority responsible for collecting the tax liability. (USDOT Order 4200.6 implementing appropriations act requirements)

b. Where the prospective lower tier participant is unable to certify to any of the statements in this certification, such prospective participant should attach an explanation to this proposal.

XI. CERTIFICATION REGARDING USE OF CONTRACT FUNDS FOR LOBBYING

This provision is applicable to all Federal-aid construction contracts and to all related subcontracts which exceed \$100,000. 49 CFR Part 20, App. A.

1. The prospective participant certifies, by signing and submitting this bid or proposal, to the best of his or her knowledge and belief, that:

a. No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of any Federal agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.

b. If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any Federal agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or

cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure Form to Report Lobbying," in accordance with its instructions.

2. This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by 31 U.S.C. 1352. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

3. The prospective participant also agrees by submitting its bid or proposal that the participant shall require that the language of this certification be included in all lower tier subcontracts, which exceed \$100,000 and that all such recipients shall certify and disclose accordingly.

XII. USE OF UNITED STATES-FLAG VESSELS:

This provision is applicable to all Federal-aid construction contracts, design-build contracts, subcontracts, lower-tier subcontracts, purchase orders, lease agreements, or any other covered transaction. 46 CFR Part 381.

This requirement applies to material or equipment that is acquired for a specific Federal-aid highway project. 46 CFR 381.7. It is not applicable to goods or materials that come into inventories independent of an FHWA funded-contract.

When oceanic shipments (or shipments across the Great Lakes) are necessary for materials or equipment acquired for a specific Federal-aid construction project, the bidder, proposer, contractor, subcontractor, or vendor agrees:

1. To utilize privately owned United States-flag commercial vessels to ship at least 50 percent of the gross tonnage (computed separately for dry bulk carriers, dry cargo liners, and tankers) involved, whenever shipping any equipment, material, or commodities pursuant to this contract, to the extent such vessels are available at fair and reasonable rates for United States-flag commercial vessels. 46 CFR 381.7.

2. To furnish within 20 days following the date of loading for shipments originating within the United States or within 30 working days following the date of loading for shipments originating outside the United States, a legible copy of a rated, 'on-board' commercial ocean bill-of-lading in English for each shipment of cargo described in paragraph (b)(1) of this section to both the Contracting Officer (through the prime contractor in the case of subcontractor bills-of-lading) and to the Office of Cargo and Commercial Sealift (MAR-620), Maritime Administration, Washington, DC 20590. (MARAD requires copies of the ocean carrier's (master) bills of lading, certified onboard, dated, with rates and charges. These bills of lading may contain business sensitive information and therefore may be submitted directly to MARAD by the Ocean Transportation Intermediary on behalf of the contractor). 46 CFR 381.7.

**ATTACHMENT A - EMPLOYMENT AND MATERIALS
PREFERENCE FOR APPALACHIAN DEVELOPMENT
HIGHWAY SYSTEM OR APPALACHIAN LOCAL ACCESS
ROAD CONTRACTS (23 CFR 633, Subpart B, Appendix B)**
This provision is applicable to all Federal-aid projects funded
under the Appalachian Regional Development Act of 1965.

1. During the performance of this contract, the contractor undertaking to do work which is, or reasonably may be, done as on-site work, shall give preference to qualified persons who regularly reside in the labor area as designated by the DOL wherein the contract work is situated, or the subregion, or the Appalachian counties of the State wherein the contract work is situated, except:

a. To the extent that qualified persons regularly residing in the area are not available.

b. For the reasonable needs of the contractor to employ supervisory or specially experienced personnel necessary to assure an efficient execution of the contract work.

c. For the obligation of the contractor to offer employment to present or former employees as the result of a lawful collective bargaining contract, provided that the number of nonresident persons employed under this subparagraph (1c) shall not exceed 20 percent of the total number of employees employed by the contractor on the contract work, except as provided in subparagraph (4) below.

2. The contractor shall place a job order with the State Employment Service indicating (a) the classifications of the laborers, mechanics and other employees required to perform the contract work, (b) the number of employees required in each classification, (c) the date on which the participant estimates such employees will be required, and (d) any other pertinent information required by the State Employment Service to complete the job order form. The job order may be placed with the State Employment Service in writing or by telephone. If during the course of the contract work, the information submitted by the contractor in the original job order is substantially modified, the participant shall promptly notify the State Employment Service.

3. The contractor shall give full consideration to all qualified job applicants referred to him by the State Employment Service. The contractor is not required to grant employment to any job applicants who, in his opinion, are not qualified to perform the classification of work required.

4. If, within one week following the placing of a job order by the contractor with the State Employment Service, the State Employment Service is unable to refer any qualified job applicants to the contractor, or less than the number requested, the State Employment Service will forward a certificate to the contractor indicating the unavailability of applicants. Such certificate shall be made a part of the contractor's permanent project records. Upon receipt of this certificate, the contractor may employ persons who do not normally reside in the labor area to fill positions covered by the certificate, notwithstanding the provisions of subparagraph (1c) above.

5. The provisions of 23 CFR 633.207(e) allow the contracting agency to provide a contractual preference for the use of mineral resource materials native to the Appalachian region.

6. The contractor shall include the provisions of Sections 1 through 4 of this Attachment A in every subcontract for work which is, or reasonably may be, done as on-site work.

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8 PROSECUTION AND PROGRESS

Replace the row for *Schedule* in the table in the 2nd paragraph of section 8-1.03 with:

Schedule	Baseline schedule and weekly statement of working days report
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Replace section 8-1.04C with:

8-1.04C Delayed Start

Section 8-1.04B does not apply.

Start job site activities within 55 days after receiving notice that the Contract has been approved by the Attorney General or the attorney appointed and authorized to represent the Department.

Do not start job site activities until the Department authorizes or accepts your submittal for:

- 1. Contractor-supplied biologist
- 2. Biological resource information program
- 3. CPM baseline schedule
- 4. WPCP or SWPPP, whichever applies
- 5. Notification of DRA or DRB nominee and disclosure statement
- 6. Contingency plan for opening closures to traffic
- 7. SSPC QP certifications

If the submittals for Contractor-supplied biologist and biological resource information program are authorized, you may enter the job site only to measure controlling field dimensions and locate utilities.

Do not start other job site activities until all the submittals from the above list are authorized or accepted and the following information is received by the Engineer:

- 1. Notice of Materials To Be Used form.
- 2. Written statement from the vendor that the order for the sign panels has been received and accepted by the vendor. The statement must show the dates that the materials will be shipped.

You may start job site activities before the 55th day after Contract approval if you:

- 1. Obtain specified authorization or acceptance for each submittal before the 55th day
- 2. Receive authorization to start

Submit a notice 72 hours before starting job site activities. If the project has more than 1 location of work, submit a separate notice for each location.

Replace the 4th and 5th paragraphs of section 8-1.05 with:

The Engineer issues a weekly statement of working days report by the end of the following week. If you disagree with a weekly statement of working days report, submit an RFI within 5 business days of receipt of the report.

The weekly statement of working days report shows:

- 1. Working days and non-working days during the reporting week
- 2. Time adjustments
- 3. Calculations of work completion dates, including working days remaining
- 4. Controlling activities

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Replace the 1st paragraph of section 9-1.11D with:

[illegible]

Replace the 1st sentence in the 3rd paragraph of section 10-6 with:

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Replace section 12-3.03 with:

1. Be orange low-density polyethylene (LDPE)
2. Be flexible and collapsible upon vehicle impact
3. Have 6-inch-wide Type V or VIII, white and orange retroreflective stripes

4. Have a weighted base to maintain an upright position and prevent displacement by passing high-speed truck traffic
5. Have a height such that the top of the drum is at least 39 inches above the traveled way

Retroreflective stripes must be visible from 1,500 feet, during the hours of darkness under an illumination of legal high-beam headlights by persons with 20/20 vision or vision corrected to 20/20.

The weighted base must:

1. Be a preformed rubber base or a tire sidewall base
2. Be detachable
3. Have a maximum 50-lb weight and a minimum weight of:
 - 3.1. 22 lb for a posted speed limit less than 30 mph
 - 3.2. 30 lb for a posted speed limit 30 to 50 mph
 - 3.3. 40 lb for a posted speed greater than 50 mph
4. Be shaped to prevent rolling upon impact
5. Have a maximum 36-inch outside diameter

A preformed rubber base must:

1. Be manufactured to fit the traffic drum
2. Have a 4-inch maximum height above the ground surface

A tire sidewall base must:

1. Be no more than 2 tire sidewalls
2. Have a combined 5-inch maximum height above the ground surface
3. Have circumferential contact with the road surface

12-3.03C Construction

Use one type of traffic drum on the project.

Use the same type and brand of retroreflective sheeting for all traffic drums used on the project.

Do not use sandbags or sand filled ballasts.

Place traffic drum weighted base edge outside of the lane width open to traffic.

Move traffic drums to the unpaved surface or remove at the end of each work period.

Check for traffic drums that are displaced or cease to operate or function as specified at least once per day during non-working days. Immediately replace or repair them to their original condition and place back in their original locations.

12-3.03D Payment

Not Used

Add to section 12-3.11B(5)(b):

Provide two C50B(CA) construction project funding identification signs.

Legend for the type of project must be one line and read as follows:

IMPROVED DRAINAGE SYSTEM

Legend for the year of completion on project funding sign must read as follows:

YEAR OF COMPLETION 2026

Replace section 12-3.20 with:

12-3.20 TEMPORARY BARRIER SYSTEMS

12-3.20A General

12-3.20A(1) Summary

Section 12-3.20 includes specifications for placing, maintaining, repairing, and removing temporary barrier systems.

Temporary barrier system consists of:

1. New or undamaged used interconnected barrier segments
2. Segment connection hardware
3. Stakes and anchor bolts

12-3.20A(2) Definitions

clear area width: Minimum width throughout the length of the barrier system that must be maintained clear of obstructions, objects, and work resources during non-working hours. The width is measured perpendicular from the non-traffic side toe.

set back distance: Space measured between the closest toe of temporary barrier and the edge of traveled way for each direction of traffic.

12-3.20A(3) Submittals

Submit as informational submittal for each type of temporary barrier system:

1. Certificate of compliance.
2. Manufacturer's installation instructions except for temporary concrete barriers with loop and pin and temporary concrete barriers with cross bolt.
3. Manufacturer's QC test results and daily production log, through the Data Interchange for Materials Engineering (DIME) website. QC test results must include the concrete mix design number, barrier stamped ID, and must be submitted within 3 business days of QC test completion.

Submit test reports for cross bolts that certify compliance with the applicable ASTM requirements. The test reports must be from a laboratory that is accredited to International Standards Organization/International Electrotechnical Commission 17025 by the American Association for Laboratory Accreditation (A2LA) or the ANSI-ASQ National Accreditation Board.

Submit a signed manufacturer's replacement evaluation report within 10 days of damage to a temporary steel barrier system.

12-3.20A(4) Quality Assurance

12-3.20A(4)(a) General

Temporary barrier systems must comply with MASH Test Level 3 except for Type K temporary railing.

Except for temporary concrete barriers with loop and pin and temporary concrete barriers with cross bolt, temporary barrier systems must:

1. Be on the Authorized Materials List for highway safety features
2. Comply with the manufacturer's drawings shown on the Department's Division of Safety Programs website and the manufacturer's installation instructions

If a discrepancy exists, governing ranking in descending order is:

1. These specifications
2. Manufacturer's drawings
3. Manufacturer's installation instructions

QC sampling, testing, and inspection personnel must have an ACI Concrete Field-Testing Technician, Grade I certification.

Temporary concrete barrier segments must:

1. Comply with the requirements for tier 3 precast concrete in section 90-4
2. Be fabricated at a plant on the Authorized Facility Audit List

Concrete must be sampled and tested at the minimum frequencies shown in the following table.

Concrete QC Tests		
Quality characteristic	Test method	Minimum testing frequency
Compressive strength	ASTM C172/C172M, ASTM C31/C31M, and ASTM C39/C39M	Once per 300 cu yd of concrete cast, or every day of casting, whichever is more frequent
Slump	ASTM C143/C143M	
Temperature at time of mixing	ASTM C1064/C1064M	
Density	ASTM C138	Once per 600 cu yd of concrete cast or every 7 days of batching, whichever is more frequent
Air content	ASTM C231/C231M or ASTM C173/C173M	If concrete is air entrained, once for each set of cylinders, and when conditions warrant

A daily production log of PC activities must be maintained under section 90-4.01C(4).

12-3.20A(4)(b) Quality Control

Replace damaged temporary concrete barrier segments with exposed reinforcing steel or concrete spalls 1-1/2 inches in depth and 4 inches in width or greater.

Replace damaged temporary steel barrier segments with permanent bends, tearing, or buckling as described in the signed manufacturer's replacement evaluation report.

Realign temporary barrier system within 2 days of impact or displacement when displaced more than 3 inches except when the temporary barrier system is displaced into a traveled lane realign immediately.

12-3.20B Materials

12-3.20B(1) General

Temporary barrier segment must:

1. Be a minimum 31-1/2 inches in height
2. Have at least two lifting holes
3. Be designed to be used with temporary traffic screen when required

Temporary barrier segment may have your name or logo on each barrier segment. The name or logo must be no more than 4 inches in height and must be located no more than 12 inches above the bottom of the barrier segment.

12-3.20B(2) Temporary Concrete Barriers

12-3.20B(2)(a) General

Temporary concrete barrier segment must:

1. Be precast concrete with a minimum 4,000-psi compressive strength.
2. Have reinforcement steel that complies with section 52.
3. Have a finished surface that complies with section 51-1.03F(2).
4. Include the manufacturer's name, lot number, and month and year of manufacture stamped on the top of each barrier segment except for Type K temporary railing. The stamped information must be:
 - 4.1. No more than 6 inches in height.
 - 4.2. From 3/16 to 1/4 inch in depth.
 - 4.3. Centered on the top width of the barrier segment.
5. Use one of the following segment connections:
 - 5.1. Loop and pin
 - 5.2. "J" hook
 - 5.3. Cross bolt

6. Comply with the tolerances shown in the following table:

Precast Barrier Tolerances	
Quality Characteristic	Tolerance($\pm$)
Length (in)	1
Insert Placement (in)	1/2
Horizontal Alignment (in)	1/8 per 10 feet of length
Deviation of Ends (in): Horizontal Skew Vertical Batter	1/4 1/8 per foot of depth

Reinforcement steel must:

1. Comply with ASTM A615 or ASTM A706, Grade 60
2. Be galvanized under section 52-3, when required

Combinations of reinforcing steel and welded wire reinforcement are allowed. Welded wire reinforcement must comply with ASTM A1064.

Stake must:

1. Comply with ASTM A36/A36M-14 or ASTM A529-14, Grade 50
2. Be 1-1/2-inch-diameter-by-48-inch-long
3. Have a plate 1/2-by-3-1/2-by-3-1/2-inch welded 2 inches down from the upper end using a 1/4-inch fillet weld under AWS D1.1 or D1.4

Anchor bolt must:

1. Be a threaded rod, 1-1/8-inch-diameter-by-10-1/2-inch-long
2. Comply with ASTM A307
3. Include a nut complying with ASTM A563
4. Include a plate washer:
 - 4.1. 1/2-by-3-1/2-by-3-1/2-inch with a 1-1/4-inch diameter hole in the center
 - 4.2. Complying with ASTM A36/A36M
 - 4.3. Galvanized post fabrication under section 75-1.02B

Epoxy adhesive must have a bond strength of minimum 1,650 psi, except for temporary concrete barrier with "J" Hook.

12-3.20B(2)(b) Temporary Concrete Barriers with Cross Bolt

Cross bolt hardware includes:

1. Cross bolt
2. Nut complying with ASTM A563
3. Hardened washer complying with ASTM F436, Type 1
4. Plate washer complying with ASTM A36/A36M and galvanized post fabrication under section 75-1.02B

Cross bolt must:

1. Be a 7/8-inch bolt or threaded rod and comply with one of the following:
 - 1.1. HS threaded rod ASTM A193, Grade B7
 - 1.2. HS threaded rod ASTM A449, Type 1
 - 1.3. HS nonheaded anchor bolt ASTM F1554, Grade 105, Class 2A
2. Have a permanent grade symbol and manufacturer's identifier

12-3.20B(2)(c) Temporary Concrete Barriers with Loop and Pin

12-3.20B(2)(c)(i) General

Steel bar loop must comply with ASTM A36/A36M.

Connecting bolt must comply with ASTM A307, be 1-1/4-inch in diameter, and a minimum 26-inch length. A round bar of the same diameter and length may be substituted for the connecting bolt. The round bar must:

1. Comply with ASTM A36/A36M
2. Have a 3-inch-diameter, 3/8-inch-thick plate welded on the upper end using a 3/16-inch fillet weld

12-3.20B(2)(c)(ii) Type K Temporary Railings

Anchor bolt must:

1. Be a threaded rod, 1-inch-diameter-by-15-1/2-inch-long
2. Comply with ASTM A307
3. Include a nut complying with ASTM A563
4. Include a plate washer:
 - 4.1. 3/8-by-2-1/2-by-3-inch with a 1-1/8-inch diameter hole in the center
 - 4.2. Complying with ASTM A36/A36M
 - 4.3. Galvanized post fabrication under section 75-1.02B

12-3.20B(2)(d) Temporary Concrete Barriers with "J" Hook

"J" hook must:

1. Comply with ASTM A36/A36M
2. Be 3/8-inch-thick steel plate
3. Be a minimum 18 inches in height

Anchor hardware must include:

1. Anchor bolt insert 1-inch diameter, 6-inch long
2. Hex head bolt 1-inch diameter with a minimum length of 11 inches plus thickness of asphalt overlay
3. Plate washer 3/8-inch by 3-inch by 3-inch
4. Retainer ring

12-3.20B(3) Temporary Steel Barriers

Temporary steel barrier segment must:

1. Be galvanized steel.
2. Have a joint connection.
3. Include permanent identification information with no more than 6 inches in height and 12 inches in length and centered on the top width of the segment. The identification information must include:
 - 3.1. Manufacturer's name.
 - 3.2. Serial number.
 - 3.3. Lot number.
 - 3.4. Month and year of manufacture.

Temporary steel barrier 19-foot segment must be filled to a depth of 11-13/16 inches with concrete ballast.

12-3.20C Construction

12-3.20C(1) General

Clean temporary barrier segments at time of installation and at least every 6 months thereafter.

Install temporary barrier systems based on the requirements shown in the following table:

Minimum Clear Area Width

Barriers	Configuration	Height differentials 3 feet or less (ft)	Height differentials greater than 3 ft up to 8 feet (ft)	Edge of deck or height differentials greater than 8 feet (ft)	Fixed objects, falsework members, or temporary supports ^a (ft)
10-foot, 20-foot & 30-foot temporary concrete barriers with cross bolt	Freestanding	1	2	5	5
12.5-foot temporary concrete barriers with "J" hook	Freestanding	3	4	8	7
	3 stakes per segment traffic side	1	1	2	3
	2 anchor bolts per segment traffic side	1	1	2	3
20-foot temporary concrete barriers with "J" hook	Freestanding	3	4	8	7
	4 stakes per segment traffic side	1	1	2	3
	3 anchor bolts per segment traffic side	1	1	2	3
50-foot temporary steel barriers	Staked or anchored at both ends only	6	7	9	10
	Staked or anchored every 250 feet	5	6	8	9
	Staked or anchored every 33 feet	1	1	3	4
19-foot temporary steel barriers	Freestanding	4	5	7	8
20-foot Type K temporary railings	Freestanding	2	3	8	7
	2 stakes or 2 anchor bolts per segment traffic side	1	1	3	4
	4 stakes or 4 anchor bolts per segment	N/A	N/A	3	3

^aThe minimum clear area width to a falsework or temporary support footing can be 2 feet less than the clear area width shown. Measure clear area width to the footing edge closest to traffic.

Stake temporary barrier systems when placed on an asphalt concrete surface.

Anchor temporary barrier systems when placed on a concrete surface. For bridge decks, confirm the anchor will not penetrate closer than 1-1/2 inches from the bottom of the deck before placement. When temporary barrier is not shown, request the Engineer to verify the bridge deck thickness.

For installations on concrete surfaces, drill holes and bond threaded rods or dowels under section 51-1.03E(5). Do not drill the top of supporting beams or girders, bridge expansion joints, or drains.

Install stakes and anchor bolts so the heads do not project above the top of the temporary barrier pocket profile.

In addition to the temporary barrier minimum length required, for the approach zone before the protected area, place a minimum:

1. 60 feet temporary barrier on facilities with a posted speed of 45 mph or less
2. 100 feet temporary barrier on facilities with a posted speed greater than 45 mph

Offset the approach end of temporary barrier systems a minimum of 15 feet from the edge of an open traffic lane, use the offset rate shown in the following table:

Temporary Barrier System Offset Rate	
Posted speed (mph)	Rate ^a
0 to 45	10:1
46 to 60	15:1
61 to 70	20:1

^aRate is longitudinally to transversely with respect to the edge of the traveled way

If a 15-foot minimum offset cannot be achieved, offset temporary barrier systems the maximum distance available and install an authorized temporary crash cushion system at each barrier approach end.

Install a reflector on the top or face of barrier segments placed within 10 feet of a traffic lane. Space reflectors at approximately 20-foot intervals. Apply adhesive for mounting the reflector under the reflector manufacturer's instructions.

Install a Type P marker panel complying with section 82 at:

1. Each end of a temporary barrier system placed adjacent to a two-lane, two-way highway
2. The end facing traffic for a temporary barrier system installed adjacent to a one-way roadbed
3. The end of the skew nearest the traveled way when a temporary barrier system is placed on a skew

Maintain a minimum height of 31-1/2 inches above surface for temporary barrier systems. For paving activities adjacent to temporary barriers, do not pave within 2 feet of the barrier segments unless authorized. For paving under the temporary barrier, remove and reset the barrier.

Remove temporary barrier systems when no longer required for the work. Remove stakes and anchor bolts so that minimal damage is done to surface.

After removing the temporary barrier systems:

1. Restore the area to its previous condition or construct it to its planned condition if temporary excavation or embankment was used to accommodate the temporary barrier.
2. Remove all threaded rods or dowels to a depth of at least 1 inch below the top of a concrete surface. Fill the resulting holes with mortar under section 51-1 except cure the mortar by the water method or by the curing compound method using curing compound no. 6.
3. Repair a damaged asphalt surface by providing a clean, smooth edge around the damaged area. Repair any heaving caused by stake removal to provide a uniform surface. Remove loose debris and use compressed air to clean out the stake hole. Comply with manufacturer's requirements except fill the stake hole with grout to existing pavement elevation under section 51-1.

If the Engineer orders a lateral move of a temporary barrier system and repositioning is not shown, the lateral move is change order work except for work area access, clear area width compliance, or because of your means and methods to perform the work.

12-3.20C(2) Temporary Concrete Barriers

12-3.20C(2)(a) General

Before placing temporary barrier systems on the job site and after each described relocation, paint the exposed surfaces of the segments with white paint complying with specifications for acrylic emulsion paint for exterior masonry.

Place and maintain the abutting ends of segments in alignment without substantial offset from each other.

Install temporary barrier systems with the last segment extending a minimum of 60 feet past the length of the protected area.

Stake or anchor a minimum 20 feet of barrier at each end of temporary barrier systems. For:

1. Temporary concrete barriers with "J" hook, place a minimum of 6 stakes or anchors at each end, 3 on each side.
2. Temporary concrete barriers with cross bolt, place a minimum of 6 stakes or anchors at each end, 3 on each side.
3. Type K temporary railings, place 4 stakes or anchors at each end, 2 on each side.

Maintain a minimum 1-foot set back distance for temporary barrier systems placed between two-way traffic.

12-3.20C(2)(b) Temporary Concrete Barriers with Cross Bolt

Install a minimum 210 feet of temporary concrete barrier with cross bolt.

Place temporary barrier systems on a concrete or asphalt concrete surface.

Do not stake or anchor temporary barrier systems, except for 20 feet at each end.

Intermix segments of different lengths within a temporary barrier system when necessary.

For temporary barrier systems placed on a curved layout, maintain the minimum curve radius shown in the following table:

Minimum Curve Radius	
Segment length (ft)	Curve radius (ft)
10	125
20	265
30	400

12-3.20C(2)(c) Temporary Concrete Barriers with Loop and Pin

12-3.20C(2)(c)(i) General

Not Used

12-3.20C(2)(c)(ii) Type K Temporary Railings

Do not install Type K temporary railings on projects advertised after December 31, 2026.

Install a minimum 160 feet of Type K temporary railing.

Excavate and backfill under section 19-3.

Do not compact earth fill placed behind Type K temporary railings in a curved layout.

Place temporary barrier systems on a firm, stable surface. Grade the area to provide a uniform bearing surface throughout the entire length of the system.

Anchor or stake the first and last segment and every other segment with four stakes as shown, when placed between two-way traffic.

12-3.20C(2)(d) Temporary Concrete Barriers with "J" Hook

Install a minimum 200 feet of temporary concrete barrier with "J" hook.

Place temporary barrier systems on a concrete or asphalt concrete surface. The asphalt concrete surface must have a minimum 2 inches of asphalt concrete over 6 inches of compacted subbase.

When temporary barrier systems are placed between two-way traffic, install on each side as shown:

1. 2 anchors or stakes for 12.5-foot segments
2. 3 anchors or stakes for 20-foot segments

12-3.20C(3) Temporary Steel Barriers

12-3.20C(3)(a) General

Install temporary barrier systems under manufacturer's instructions.

Maintain a minimum 2-foot set back distance on both sides of temporary barrier systems placed between two-way traffic.

12-3.20C(3)(b) 50-Foot Temporary Steel Barriers

Use 50-foot temporary steel barriers with or without rubber pads.

Install a minimum 250 feet of 50-foot temporary steel barrier. The last segment must extend a minimum 25 feet past the length of the protected area.

Place temporary barrier systems on a concrete or asphalt concrete surface. Do not place systems on a dirt surface.

Anchor or stake the first and last segment of temporary barrier systems.

Maintain a minimum radius of 800 feet for segments placed on a curved layout. For tighter curves down to a 250-foot radius, contact the manufacturer before installation and provide manufacturer's written recommendation for the installation.

12-3.20C(3)(c) 19-Foot Temporary Steel Barriers

Install a minimum 323 feet of 19-foot temporary steel barrier.

Place temporary steel barrier on a concrete or asphalt concrete surface.

Maintain a minimum radius of 262 feet for segments placed on a curved layout.

12-3.20D Payment

The payment quantity for temporary barrier systems is the length measured along the top of the barrier segments.

Replace section 12-3.22 with:

12-3.22 TEMPORARY CRASH CUSHIONS

12-3.22A General

12-3.22A(1) Summary

Section 12-3.22 includes specifications for installing, repairing, replacing, maintaining, and removing temporary crash cushions.

12-3.22A(2) Definitions

Not Used

12-3.22A(3) Submittals

At least 10 days before installation, submit as informational submittal for each temporary crash cushion model:

1. Certificate of compliance
2. Two copies of the following:
 - 2.1. Manufacturer's installation and maintenance manual
 - 2.2. Department approved manufacturer's drawings from the Department's Division of Safety Programs website
3. Record of training provided by manufacturer for each person installing the temporary crash cushion

12-3.22A(4) Quality Assurance

12-3.22A(4)(a) General

Not Used

12-3.22A(4)(b) Quality Control

Temporary crash cushion must be installed under the manufacturer's instructions by personnel trained by the manufacturer.

Keep a copy of the manufacturer's drawings, and installation and maintenance manual for each temporary crash cushion model at the job site during installation.

12-3.22B Materials

12-3.22B(1) General

Temporary crash cushion must:

1. Be on the Authorized Materials List for highway safety features
2. Comply with MASH:
 - 2.1. Test Level 3 (TL-3) for a posted speed limit 45 mph or greater
 - 2.2. Test Level 2 (TL-2) for a posted speed limit less than 45 mph

Water-filled temporary crash cushion must:

1. Include all components required for attachment to temporary barrier or protected obstacle
2. Comply with the manufacturer's drawings shown on the Department's Division of Safety Programs website

Each sand-filled temporary crash cushion module must be:

1. Colored standard yellow with a black lid
2. Free from structural flaws and objectionable surface defects

Sand for filling module must:

1. Be commercial quality, washed concrete sand
2. Contain no more than 5 percent water under California Test 226
3. Be clean when placed in the module

12-3.22B(2) Temporary Crash Cushions TL-3

Temporary crash cushion TL-3 must:

1. Comply with MASH TL-3
2. Be one of the following:
 - 2.1. Water-filled temporary crash cushion gating, non-redirective system
 - 2.2. Sand-filled temporary crash cushion module array

12-3.22B(3) Temporary Crash Cushions TL-2

Temporary crash cushion TL-2 must:

1. Comply with MASH TL-2
2. Be one of the following:
 - 2.1. Water-filled temporary crash cushion gating, non-redirective system
 - 2.2. Sand-filled temporary crash cushion module array

12-3.22C Construction

12-3.22C(1) General

When activities expose traffic to fixed obstacles, protect traffic from the obstacles with temporary crash cushions.

You may use NCHRP Report 350 compliant temporary crash cushions before December 31, 2026.

Install temporary crash cushions under the manufacturer's instructions before:

1. Starting activities requiring the crash cushions.
2. Opening to traffic lanes adjacent to the protected obstacles.

Temporary crash cushions must not impede the flow of traffic or encroach on the traveled way.

Attach a Type R or Type P marker panel to the front of temporary crash cushions if the closest point to the traveled way is within 12 feet of the traveled way. Fasten marker panels firmly to crash cushions with commercial quality hardware or by other authorized methods.

Maintain temporary crash cushions in place at each location, including when work is not in progress. Keep the area behind the temporary crash cushions clear of obstructions.

Repair damaged temporary crash cushions immediately. Remove and replace temporary crash cushions damaged beyond repair. Repair and replacement of temporary crash cushions damaged by traffic is change order work.

12-3.22C(2) Water-Filled Temporary Crash Cushions

Place temporary crash cushions and a minimum 20 feet of temporary barrier at a parallel 2-foot offset from edge of traveled way.

Install temporary crash cushions adjacent to a:

1. Barrier at an offset distance 1 to 2 feet or 7 feet or greater from the barrier
2. Dike or curb at an offset distance 0 to 4 feet or 7-1/2 feet or greater from the dike or curb

12-3.22C(3) Sand-Filled Temporary Crash Cushions

Do not use sand-filled temporary crash cushions for permanent installations.

Use the same type of sand-filled modules for each array. Fill each sand-filled module with sand under the manufacturer's instructions.

Securely fasten the top edge of a seal to the wall of the sand-filled module with a continuous strip of heavy-duty tape, when a seal is required.

You may place sand-filled temporary crash cushion modules on movable pallets or frames complying with the dimensions shown. The pallets or frames must provide a full-bearing base beneath the modules. Do not move the modules and supporting pallets or frames by sliding or skidding along the pavement or bridge deck.

Attach a Type R marker panel such that the top of the panel is 1 inch below the module lid. Attach a Type P marker panel such that the bottom of the panel rests upon the roadway surface or pallet surface when used.

You may remove sand-filled modules during the work shift for access to the work area if the exposed fixed obstacle is 15 feet or more from the nearest lane carrying traffic. Reset the modules before the end of the work shift.

A lateral move of a temporary crash cushion module is change order work if ordered and the repositioning is not shown.

Remove sand-filled temporary crash cushion modules, including sand, pallets or frames, and marker panels, at Contract acceptance.

12-3.22D Payment

Not Used

Add to the beginning of section 12-3.32C:

Place PCMSs at the locations shown and in advance of the 1st warning sign for each:

1. Stationary lane closure
2. Ramp closure
3. Speed reduction zone

Inform motorists of a temporary closing of an entrance or exit ramp using a PCMS. Place the PCMS and start displaying the message at least 7 days and no more than 10 days before the closure. Place at least one PCMS at each ramp to be closed. Section 12-4.02C(8)(b) does not apply.

For ramp closures of 1 day the PCMS must display the message "RAMP CLOSED" and the date and time of day of the closure. The date must match the approved closure date in LCS.

For ramp closures of more than 1 day the PCMS must display the message "RAMP CLOSED", the dates of the closure, and the time of day. Only display dates approved in LCS or authorized.

Provide accurate closure information so traffic can plan for closures. Update the PCMS message daily.

Replace the 5th paragraph of section 12-3.32C with:

When in operation, place the bottom of a PCMS at least 7 feet above the roadway. Place the top of the PCMS no more than 14.5 feet above the roadway.

Add between the 9th and 10th paragraphs of section 12-3.32C:

Start displaying the message on the sign 15 minutes before closing the lane or ramp or when directed by the Engineer.

Replace section 12-3.37 with:

12-3.37 PORTABLE RADAR SPEED FEEDBACK SIGN SYSTEMS

12-3.37A General

Section 12-3.37 includes specifications for placing, maintaining, and removing portable radar speed feedback sign systems.

The system must be able to operate on a continuous, 24- hour basis.

12-3.37B Materials

A portable radar speed feedback sign system consists of a vehicle speed feedback sign, a power source, and signs R2-1 and G20-5aP.

The portable radar speed feedback sign system must comply with section 87-14, except:

1. System must be mounted on a trailer.
2. LED character display must remain blank when no vehicles are detected or when the detected vehicle speed is 10 miles less than the preset speed

Sign panels must comply with section 12-3.11B(2).

12-3.37C Construction

Configure the portable radar speed feedback sign system to detect traffic only in the approach direction of travel.

Place the portable radar speed feedback sign system.

1. As far from the traveled way as practicable where it is visible and legible to approaching traffic. Where practicable, place the sign behind a barrier or guardrail.
2. At or before the crest of roadway vertical curvatures that restrict sight distance.
3. At or before the curve of horizontal roadway curvatures that restrict sight distance.

12-3.37D Payment

Not Used

Add to the end of section 12-4.01A:

Notify the following local authorities at least 10 business days before the start of job activities:

Authority	Contact Information	Remarks
Humboldt County Public Works	707-445-7652	Regarding traffic impacts to Benbow Drive, Hooker Creek Road, and Newton Road
Humboldt Transit Authority	707-443-0826	Regarding traffic impacts to the Redwood Transit System and Southern Humboldt Intercity bus line

Add to section 12-4.02A(2):

special days: Special days are shown in the following table:

Special Days		
Event	Event date	Special days
Cal Poly Humboldt Graduation	Second weekend in May	Friday through Sunday
Redwood Run and Music Festival	Second weekend in June	Friday through Monday
Kate Wolf Memorial Music Festival	Last weekend in June	Thursday through Monday
Northern Nights Music Festival	Second weekend in July	Friday through Monday
The Days Between Festival	First weekend in August	Friday through Monday
Reggae on the River Festival	First weekend in August	Saturday and Sunday
Tour of the Unknown Coast	Third Saturday in September	Saturday

Verify the dates before starting work.

Add between the 1st and 2nd paragraphs of section 12-4.02A(3)(c):

Submit a contingency plan for each of the following activities:

1. Drainage system work
2. Excavations encroaching on the traveled way not protected by temporary barrier system or other authorized protective system

Replace the 3rd paragraph of section 12-4.02C(1) with:

Do not close entrance ramps or exit ramps servicing 2 consecutive local street interchanges in the same direction of travel. Do not simultaneously close 2 entrance ramps or 2 exit ramps in the same interchange.

Add to the end of section 12-4.02C(1):

Keep the full width of the traveled way open to traffic when no active construction activities are occurring in the traveled way or within 6 feet of the traveled way.

Keep the full width of the ramp traveled way open for use by traffic on designated holidays and special days.

Replace section 12-4.02C(3)(d) with:

12-4.02C(3)(d) Local Street Closure Requirements

You may close local-street lanes using a one-way-reversing traffic-control lane closure as shown on chart no. M1.

Replace section 12-4.02C(3)(f) with:

12-4.02C(3)(f) Closure Restrictions for Designated Holidays and Special Days

Closure restrictions for designated holidays and special days are shown in the following table:

Lane Closure Restrictions For Designated Holidays And Special Days											
Thu	Fri	Sat	Sun	Mon	Tues	Wed	Thu	Fri	Sat	Sun	Mon
xx	H xx	xx	xx								
	SD xx										
	xx	H xx	xx								
		SD xx									
	xx	xx	H xx	xx							
			SD xx								
	xx	xx	xx	H xx	xx						
				SD xx							
				xx	H xx						
					xx	H xx					
						xx	H xx	xx	xx	xx	
							SD xx				
Legend:											
	Refer to lane requirement charts.										
xx	One lane in each direction of travel must be open for use by traffic. No ramp closures allowed.										
H	Designated holiday										
SD	Special day. One-way-reversing traffic control is not allowed.										

Replace section 12-4.02C(3)(g) with:

12-4.02C(3)(g) Freeway or Expressway Lane Requirement Charts

Freeway and expressway lane closures must comply with the requirements shown in the following chart:

Chart No. G1 Freeway and Expressway Lane Requirements																										
County: Humboldt									Route/Direction: 101/NB, SB									Post Mile: R5.6/M54.0								
Closure limits:																										
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Mon–Thu	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Fri	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Sat	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Sun	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Legend:																										
1	Provide at least 1 through freeway or expressway lane and an adjacent 5-foot-wide shoulder open in the direction of travel. The maximum lane closure length is 2 miles.																									
REMARKS:																										

Replace section 12-4.02C(3)(j) with:

12-4.02C(3)(j) Complete Ramp Closure Hour Charts and Ramp Lane Requirement Charts

Comply with the requirements for the complete ramp closure shown in the following chart:

Chart No. J1 Complete Ramp Closure Hours																										
County: Humboldt							Route/Direction: 101/NB/SB							Post Mile:R17.5, R17.7, R17.8, 33.1, 33.4, R39.5 R39.6, R43.1/R43.5, 47.0, and R52.5												
Closure limits: Route 101 SB exit and entrance ramp and NB exit ramp at Route 254 Route 101 NB entrance and SB exit ramps at Weott Route 101 NB exit and SB entrance ramp at Redcrest Route 101 ramps at Barkdull Road UC Route 101 SB exit ramp at Vista Point Route 101 exit ramp at Davis Street																										
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Mon–Thu	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	
Fri	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C											
Sat																										
Sun																					C	C	C	C	C	
Legend:																										
C	Ramp may be closed completely.																									
	No ramp closures allowed.																									
REMARKS: A ramp detour plan must be approved prior to ramp closures except at PM 47.0 Route 101 SB exit ramp at Vista Point.																										

Replace section 12-4.02C(3)(k) with:

12-4.02C(3)(k) Conventional Highway Lane Requirement Charts

Comply with the requirements for the conventional highway lane closures shown in the following chart:

Chart No. K1 Conventional Highway Lane Requirements																											
County: Humboldt										Route/Direction: 101/NB, SB 254/NB, SB										Post Mile: 2.4 19.9, 24.0							
Closure limits:																											
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		
Mon– Thu	I	I	I	I	I	I	I	I	I	I	R	R	R	R	R	R	R	R	R	R	I	I	I	I	I		
Fri	I	I	I	I	I	I	I	I	I	I	R	R	R	R	R	R											
Sat																											
Sun																					I	I	I	I	I		
Legend:																											
I		Provide at least 1 through traffic lane not less than 11-feet in width for use by both directions of travel (Reversing Control). The maximum lane closure length is 1, 500 feet. At Hum-101-2.4, you may stop traffic in both directions for a period not to exceed 10 minutes (intermittent closure). After each intermittent closure, all accumulated traffic must pass through the work zone before another closure is allowed.																									
R		Provide at least 1 through traffic lane not less than 11 feet in width for use by both directions of travel (Reversing Control). The maximum lane closure length is 1, 500 feet.																									
		No lane closures allowed.																									
REMARKS:																											

Replace section 12-4.02C(3)(m) with:

12-4.02C(3)(m) Local Street Lane Requirement Charts

Comply with the requirements for a local street lane closures shown in the following chart:

Chart No. M1 Local Street Lane Requirement																											
Location: Benbow Drive, near Hum-101-7.5 Hooker Creek Road, near Hum-101-17.5 Newton Road, near Hum-101-33.2														Direction: N/S N/S E/W													
Closure limits:																											
Hour	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		
Mon– Thu	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R		
Fri	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R												
Sat																											
Sun																					R	R	R	R	R		
Legend:																											
No lane closures allowed.																											
R Provide at least 1 through traffic lane not less than 11 feet in width for use by both directions of travel (Reversing Control).																											
REMARKS:																											

Replace the 1st paragraph of section 12-4.02C(7)(a) with:

Control traffic using stationary closures, except you may use a moving closure during traffic striping and pavement marker placement using a bituminous adhesive. Do not use a moving lane closure when grinding for recessed striping and recessed markers.

Add to the end of section 12-4.02C(7)(b):

Do not place portable transverse rumble strip arrays or C45(CA) signs for one-way-reversing traffic control. Use advance warning sign distance "C" between signs W20-1 and W20-4.

Not more than 2 stationary closures are allowed in each direction of travel at one time.

Concurrent stationary closures in the same direction of travel must be spaced no closer than 5 miles apart.

Closure spacing is the distance between the last cone of the upstream closure and the temporary sign W20-1 of the downstream closure. The number of lanes open in the upstream closures must be less than or equal to the number of lanes open in the downstream closures. For multiple closures in each direction of travel, pick up the downstream closures first.

For a stationary one-way-reversing traffic-control lane closure, you may stop traffic in 1 direction for periods not to exceed 15 minutes. After each stoppage, all accumulated traffic for that direction must pass through the work zone before another stoppage is made.

Instruct bicyclists to join the traffic queue through the one-way-reversing traffic-control work zone.

You may use a pilot car to control traffic. If a pilot car is used to control traffic, the cones shown along the centerline are not required. Pilot cars must have cellular or radio contact with other pilot cars and personnel in the work zone. The maximum speed of the pilot cars conveying or controlling traffic through the traffic control zone is 25 mph. Pilot cars must only use traffic lanes open to traffic.

Provide a stationary impact attenuator vehicle.

For a traffic control system with multiple work areas, place a stationary impact attenuator at each work area with a separation distance of at least 500 feet from the adjacent work area.

Replace section 12-4.02C(9)(c)(i) with:

12-4.02C(9)(c)(i) General

Provide advance flaggers for the lane requirement charts shown in the following table:

Chart number	Additional requirements
K1, M1	Use advance flagger during hours of daylight. Use PCMS in the place of an advance flagger during the hours of darkness. Do not place the SW60(CA) sign.

Provide additional flaggers for lane requirement charts K1 and M1 at all intersections and driveways within the one-way-reversing traffic control area. You may use sign C36(CA) (TRAFFIC CONTROL WAIT AND FOLLOW TRAFFIC) in place of an additional flagger. Use sign C37(CA) (TRAFFIC CONTROL WAIT AND FOLLOW PILOT CAR) when a pilot car is used.

Replace section 12-4.02C(12) with:

12-4.02C(12) Construction Work Zone Speed Limit Reduction

12-4.02C(12)(a) General

Section 12-4.02C(12) includes specifications for providing, installing, maintaining, and removing traffic control devices for reducing the speed limit for the construction work zones.

Speed limit reduction is limited to 10 mph from the posted speed limit in construction work zones unless a greater speed limit reduction is specified. Construction work zone speed limit reduction can either be required when construction activities are active in a closure as a temporary condition or 24 hours a day, 7 days a week based on the roadway conditions when specified.

Construction work zone speed limit reduction is required for lane closures when construction activities require workers to be present within the lane closures. Construction work zone speed limit reduction is not required for closures of 1 hour or less.

Construction work zone speed limit reduction is required when construction activities require lane closures for the locations shown in the following table:

Lane requirement chart no.
G1
K1

12-4.02C(12)(b) Materials

Not Used

12-4.02C(12)(c) Construction

Cover all existing speed limit signs while the construction work zone speed limit reduction is in effect. Remove covers when construction work zone speed limit reduction is no longer in effect.

For construction work zone speed limit reduction for 24 hours a day, install portable radar speed feedback system instead of temporary radar speed feedback systems.

Add between the 4th and 5th paragraphs of section 13-11.01D(2):

Test the receiving water under the test methods for the WQOs shown in the following table:

Water Quality Objectives

Quality characteristic	Test method	Detection limit (min)	Requirement
pH	Field test with a calibrated portable instrument	0.2	Lower WQO = 6.5 Upper WQO = 8.5 And any change greater than 0.5 units from natural background
Turbidity (NTU)	Field test with a calibrated portable instrument	1	20 percent above natural background
Temperature (°F)	Field test with a calibrated portable instrument	1	1 degree above natural background

Replace section 13-12 with:

13-12 TEMPORARY CREEK DIVERSION SYSTEMS

13-12.01 GENERAL

13-12.01A Summary

Section 13-12 includes specifications for constructing, maintaining, reconstructing, and removing temporary creek diversion system (TCDS), and restoring creek bed to original condition. The temporary diversion system is used to divert upstream water flows to allow construction in a dry or dewatered location.

13-12.01B Definitions

Not Used

13-12.01C Submittals

13-12.01C(1) General

Not used.

13-12.01C(2) Temporary Creek Diversion System Plan

Within 20 days of Contract approval, submit 3 copies of the Temporary Creek Diversion System Plan (TCDSP). The TCDSP must include:

1. Installation and removal process, including equipment, platforms for equipment, and access locations.
2. Anticipated flow rates.
3. Calculations supporting the sizing of piping, channels, pumps, or other conveyance by using FHWA HY-8 or other equivalent method. Calculate the discharge water flow rate and velocity anticipated where it discharges on any erodible surface, so its conveyance does not cause erosion within the project or at the discharge to the water body. Temporary culverts attached to banks, walls, or other locations must be designed to hold the full weight of the culvert at capacity and restrain the culvert for any expected hydraulic forces.
4. Plans showing locations of diversion, including layouts, cross sections, and elevations.
5. Materials proposed for use, including SDS if applicable.
6. Operation and maintenance procedures for the TCDS.
7. Restoration plans showing before and after conditions, including photos of existing conditions for areas disturbed during the installation, operation, and removal of the TCDS.
8. Monitoring and reporting plan to ensure applicable water quality objectives are met. This includes schedule of work including Temporary BMP implementation as part of the Construction Site BMP strategy, and SWPPP or WPCP as applicable. Use with section 13-3.01A.
9. Details of the pumping system, if used, including power source, debris handling, fish screens, and monitoring requirements.

- 10 The TCDS design must demonstrate how it will comply with section 13-12.03A, water tightness, and prevent seepage.
11. Contingency plan to remove workers, equipment, materials, fuels, and any other work items that will cause pollution or violation of PLACs during a rain event out of the flow area. Develop the contingency plan for when a 12-inch freeboard cannot be maintained and overtopping of the coffer dams may occur.
12. Measures to comply with aquatic species protection and PLACs.

If revisions are required, the Engineer notifies you of the date when the review stopped and provides comments. Submit a revised TCDSP within 15 days of receiving the comments. The Department's review resumes when a complete TCDSP has been resubmitted.

Submit an electronic copy on a read-only CD, DVD, or other Engineer-authorized data storage device and 4 printed copies of the authorized TCDSP.

If the RWQCB or other regulatory agency requires review of the authorized TCDSP, the Engineer submits it to the RWQCB for review and comment. If the Engineer orders changes to the TCDSP based on the RWQCB's comments, submit a revised TCDSP within 10 days.

All submittals which include plans, specifications, and calculations must be sealed and signed by a civil engineer registered in the State.

13-12.01D Quality Assurance

Not Used

13-12.02 MATERIALS

13-12.02A General

Not Used

13-12.02B Gravel

Gravel must:

1. Be river run gravel obtained from a river or creek bed with gradation of 100 percent passing a 3/4-inch sieve and 0 percent passing a 3/8-inch sieve
2. Be clean, hard, sound, durable, uniform in quality, and free of any detrimental quantity of soft, thin, elongated or laminated pieces, disintegrated material, organic matter, or other deleterious substances
3. Be composed entirely of particles that have no more than 1 fractured face
4. Have a cleanliness value of at least 85, as determined by California Test 227

13-12.02C Impermeable Plastic Membrane

Impermeable plastic membrane must be:

1. Single ply, commercial quality, polyethylene with a minimum thickness of 10 mils complying with ASTM D2103. You must use stronger plastic membrane if required as part of design to resist hydraulic forces.
2. Free of holes, punctures, tears or other defects that compromise the impermeability of the material.
3. Suitable for use as an impermeable membrane.
4. Resistant to UV light, retaining a minimum grab breaking load of 70 percent after 500 hours under ASTM D4355.

13-12.02D Gravel-Filled Bags

Gravel-filled bags must comply with section 13-5.02G.

The 2nd paragraph of section 13-5.02G does not apply.

13-12.02E Plastic Pipes

Plastic pipe must comply with section 61-3.01 and must:

1. Be clean, uncoated, in good condition free of rust, paint oil dirt or other residues that could potentially contribute to water pollution
2. Be adequately supported for planned loads

3. Use watertight joints under section 61-2.01.
4. Be made of a material or combination of materials that are suitable for clean water and which do not contain banned, hazardous or unlawful substances
5. For temporary pipes not reused on the project you may use the following materials:
 - 5.1. PVC closed-profile wall pipe must comply with ASTM F1803
 - 5.2. PVC solid wall pipe must comply with ASTM D3034, ASTM F679, AWWA C900, AWWA C905, or ASTM D2241 and cell class 12454 defined by ASTM D1784
 - 5.3. HDPE solid wall pipe must comply with AASHTO M 326 and ASTM F714
 - 5.4. Polyethylene large-diameter-profile wall sewer and drain pipe must comply with ASTM F894

13-12.02F Rock

Rock layer must comply with the table titled *Rock Gradation for 7-inch-thick Layer*, in section 72-4.02.

13-12.02G Pumping System

Pumping system must:

1. Comply with section 74-2.02B
2. Be equipped with secondary containment
3. Be free of fuel and oil leaks
4. Meet intake screen regulatory requirements

13-12.02H Seepage Pumping System

If seepage occurs in the dewatered work area, the water must be removed by sump pumps as part of the TCDS.

Seepage pumping system must:

1. Comply with section 74-2.02B
2. Ensure discharge water conform with PLACs or is treated on site
3. Be free of fuel and oil leaks

13-12.02I Discharge Water Energy Dissipation and Erosion Control

Discharge water from pumps, pipes, ditches, or other conveyances must have BMPs to dissipate the flows and velocity of water discharged from the temporary diversion system if erosion would otherwise occur.

Energy dissipation measures:

1. May be plastic sheeting, flared end sections, rubber matting, or other materials appropriate for the design hydraulics
2. Must be anchored to prevent movement by expected flows
3. Must be removed when the TCDS is removed

13-12.03 CONSTRUCTION

13-12.03A General

Construction, use and removal of the TCDS is restricted to the time period from June 15 to October 15. If the work cannot be completed during the initial restricted time period, remove TCDS, restore the creek to original flow condition, and reconstruct the TCDS after June 14 of the following year. No work is allowed within the stream except during the restricted time period.

Do not use motorized equipment or vehicles in areas of flowing or standing water for the construction or removal of the TCDS in compliance with section 13-4.03.

Remove vegetation to ground level and clear away debris.

Place temporary or permanent fill as allowed by PLACs.

Place rock at outlet of diversion pipe under section 72-4.03, except motorized vehicles and equipment must not be used in areas of flowing or standing water.

Do not construct or reconstruct TCDS if the 72-hour forecasts predict a 50 percent or greater chance of rain in the project area.

Stop all work and remove all material and equipment from the creek between upstream and downstream cofferdams if the 72-hour forecasts predict a 50 percent or greater chance of rain in the project area and the predicted rainfall is estimated to produce a flow rate exceeding the design capacity of the TCDS.

If the required freeboard cannot be maintained and overtopping may occur, implement contingency plan to remove all workers, equipment, and potential sources of pollution from the dry working area of the creek bed.

The TCDS must be constructed within the temporary impact footprint as described in the environmental commitments.

Lap and join joints between the edges of impermeable plastic membrane with commercial-quality waterproof tape with minimum 4-inch lapping at the edges.

Seal openings or penetrations through the impermeable plastic membrane with commercial quality waterproof tape.

The TCDS must be watertight to keep the work area dry for construction and prevent the creation of pollutants. Maintain all portions of the TCDS and fix leaks as soon as they are discovered.

Contact water agencies that discharge to the construction area to ensure that unexpected water is not discharged during construction which could compromise the TCDS.

13-12.03B Maintenance

Maintain the TCDS to provide a minimum freeboard of 12 inches between the water surface and the impermeable top of the cofferdams.

Do not discharge runoff from existing or proposed drainage systems into the dry work area between the cofferdams. Runoff from these systems may be connected to the diversion pipe or conveyed by pipes downstream of the cofferdam.

Prevent leaks in the TCDS. Provide seepage pumps as necessary and keep the work area dry to prevent the creation of sediment-laden water.

Repair holes, rips and voids in the impermeable plastic membrane with commercial-quality waterproof tape. Replace impermeable plastic membrane when patches or repairs compromise the impermeability of the material.

Repair TCDS within 24 hours after the damage occurs.

Prevent debris from entering the TCDS and receiving water.

Remove and immediately replace gravel, gravel-filled bags, impermeable plastic membrane, or plastic pipes contaminated by construction activities.

Remove sediment deposits and debris from the TCDS as needed. If removed sediment is deposited within project limits, it must be stabilized and not subject to erosion by wind or water, under sections 19-1.01 and 19-2.03 B.

13-12.03C Removal

When no longer required, remove all components of TCDS. Return the creek bed and banks to the original condition.

Do not excavate the native creek material. Backfill ground disturbance, including holes and depressions caused by the installation and removal of the TCDS with gravel. Maintain the original line and grade of the creek bed.

13-12.04 PAYMENT

Not Used

AA

14 ENVIRONMENTAL STEWARDSHIP

Add to the end of section 14-1.02:

An ESA exists on this project.

Before starting job site activities, install temporary high visibility fence to protect the ESA and mark its boundaries.

Limited access to the ESA is allowed for placing and removing Temporary High Visibility Fence, biological monitoring and release of encroaching wildlife. Notify the Engineer 5 business days or less before the planned entry date. Any other access to the ESA is prohibited.

Add to the 1st paragraph of section 14-6.03A:

This project is within or near habitat for the regulated species shown in the following table:

Regulated Species

Amphibians
Coho salmon
Chinook salmon
Marbled murrelet
Migratory and resident nesting birds
Northwestern pond turtle
Northern spotted owl
Raptors
Steelhead

This project includes the sensitive habitats shown in the following table:

Sensitive Habitats

Wetlands
Large diameter redwoods
Critical habitat
Riparian habitat
Essential fish habitat

Replace item 1 in the list in the 2nd paragraph of section 14-6.03A with:

1. Stop all work within a 100-foot radius of the discovery except as shown in the following table:

Regulated species	Protective radius (feet)
Marbled murrelet	328: March 24 through August 5
Northern spotted owl	328: February 1 through July 31

Add to section 14-6.03A:

Species protection areas within the project limits are as specified in the following table:

Species Protection Areas

Identification name	Location
MAMU/NSO protection area	PMs 2.4, 7.51, 19.43, 20.63, 23.46, 26.16, 26.78, 27.52, 34.77, 35.00, 37.64, 39.01, 39.65, 40.38, 41.45, 41.69, 42.12, 42.34, 43.17, 43.54, 44.17, 45.25, 47.01, and 49.86.
Tree protection area	PMs 19.43, 23.46, 34.77, 35.00, 37.64, 39.01, 40.38, 41.45, 41.69, 43.54, 44.17, and 45.25
Aquatic protection area	PMs 2.4, 7.51, 15.64, 16.09, 16.44, 17.54, 17.76, 18.25, 19.43, 20.63, 26.16, 33.23, 34.77, 35.00, 37.64, 39.01, 39.65, 40.38, 41.45, 42.34, 44.17, 47.01, and 49.86
Raptor protection area	Entire project area

Within MAMU/NSO Protection Area, implement the following protection measures:

1. For marbled murrelet and northern spotted owl, noise level restrictions apply as follows:
 - 1.1. February 1 through August 5: Do not perform work that generates noise levels above 90 dBA LMax
 - 1.2. August 6 through September 15: Work that generates noise levels above 90dBA LMax is restricted to the time period from 2 hours after sunrise to 2 hours before sunset.
 - 1.3. September 16 through January 31: Comply with section 14-8.
 - 1.4. Measure the noise level at 50 feet from the source of the noise generating activity. Backup alarms are excluded from the noise requirements.
2. Remove vegetation between September 16 and January 31

Within Raptor Protection Area, implement the following protection measures:

1. For raptors, perform preconstruction survey for active raptor nests within one-quarter mile of the project area. If active nests are found, notify the Engineer and provide protection measures.

Within Tree Protection Area, implement the following protection measures:

1. For work within the tree structural root zone (SRZ) of redwood trees 4 feet or larger in diameter at breast height (DBH):
 - 1.1. Notify the Engineer at least 10 working days prior to working in the SRZ of a tree. The SRZ is a circular area with a radius 3 times the tree DBH.
 - 1.2. Perform excavation with hand tools unless mechanized equipment is permitted for use by the Contractor supplied arborist.
 - 1.3. Cut roots cleanly with sharp instruments to promote healing.
 - 1.4. Do not cut roots larger than 2 inches in diameter unless authorized
 - 1.5. A Contractor-supplied arborist must be onsite for the commencement of clearing and grubbing and the initiation of excavation work

Within Aquatic Species Protection Area, implement the following protection measures:

1. For aquatic species:
 - 1.1. Do not work below OHWM from October 15 to June 15
 - 1.2. Comply with section 14-6.03C

Replace the 2nd paragraph of section 14-6.03B with:

The Department anticipates nesting or attempted nesting by migratory and nongame birds from February 1 to September 15.

Replace the 8th and 9th paragraphs of section 14-6.03B with:

Do not perform tree removal or vegetation trimming during nesting or attempted nesting unless authorized.

To obtain authorization submit the bird survey and protection measures. Allow 2 days for review.

A Contractor-supplied biologist experienced in identification, and monitoring must:

1. Perform bird surveys within 5 calendar days before tree or vegetation removal and after any lapse in work activity of 7 days or more in a work area
2. Determine appropriate measures to protect migratory and nongame birds, their occupied nests, and their eggs

Replace the heading of section 14-6.03C with:

14-6.03C Aquatic Species Protection

14-6.03C(1) General

Add to section 14-6.03C(1):

Regulated aquatic species are anticipated adjacent to culverts at Post Miles 15.64, 17.54, 33.23, 34.77, 39.65, 42.34, and 47.01. Implement the following protection measures:

1. Install exclusionary material, a cofferdam, or a combination of both
2. Provide a Contractor-supplied biologist to relocate the regulated aquatic species

Handle regulated aquatic species to minimize stress by:

1. Keeping the aquatic species in water to the maximum extent possible during relocation
2. Keeping the aquatic species in cool, shaded, and aerated water while in captivity
3. Protecting the aquatic species from excessive noise, handling, temperature variation, jostling, or overcrowding while in captivity
4. Removing the aquatic species from water only when releasing them
5. Segregating aquatic species into separate containers by size to minimize predation
6. Do not use an electrofisher to herd aquatic species.

Exclusion material must be 0.25-inch stretched mesh.

Exclude aquatic species from the work area in the following sequence:

1. Install exclusion materials and remove as many aquatic species as possible
2. Install a cofferdam or water bypass
3. Gradually dewater the work area
4. Remove the remaining aquatic species using one or a combination of seining, baited minnow traps, and dip net and hand removal

Maintain exclusion material and cofferdams such that regulated aquatic species are prevented from entering the work area.

Add to the end of section 14-6.03C:

14-6.03C(2) Aquatic Species Protection and Relocation Plan

Submit the Aquatic Species Protection and Relocation Plan (ASPRP). Allow 45 days for review.

The ASPRP must include:

1. Schedule of work and protocols for dewatering and aquatic species relocation before and during any work in aquatic habitat. Protocols must include the location of the habitat where the aquatic species are to be relocated.

2. Description of avoidance and protection measures including:
 - 2.1. Aquatic species survey. Perform survey within 7 days prior to drainage system work. Identify habitat units and action area/area of effect. Submit survey results within 12 hours of the survey.
 - 2.2. Installation of exclusionary material, a cofferdam, or a combination of both
 - 2.3. Relocation performed by a contractor-supplied biologist
 - 2.4. Aquatic species survey prior to and during dewatering work to determine if animals can be herded or relocated outside the area of effect of drainage system work.
3. Post dewatering report. The report must include:
 - 3.1. Description of the relocation process, amount of time spent capturing aquatic species, and amount of time captured aquatic species spent out of the stream.
 - 3.2. Date, time and description of environmental conditions present before relocation efforts begin
 - 3.3. Photographs of removal site, species captured, and release site
 - 3.4. Number and type of regulated aquatic species captured, including any mortality or injury
 - 3.5. Narrative of the circumstances surrounding injuries or mortalities of regulated aquatic species
4. Description of compliance with:
 - 4.1. NMFS *Fish Screening Criteria for Anadromous Salmonids* (1997)
 - 4.2. PLACs

Add to section 14-6.03D(1):

A Contractor-supplied biologist who performs specialized activities must have demonstrated field experience working with the regulated species or performing the specialized task. The biologist must have experience that complies with the requirements shown in the following table:

Specialized activity/species	Requirements
General construction monitoring	Bachelor of Science in Wildlife Biology or related field and 2 years experience in the biology and natural history of local fish and wildlife resources at the project site. The Project may have more than one Qualified Biologist if specialized skills are needed (e.g., nesting bird surveys, fish identification/relocation, etc.).
Nesting bird survey, active raptor nest survey	Bachelor of Science in Wildlife Biology or related field, 2 years experience conducting bird surveys, knowledge and experience performing nesting bird surveys, and ability to identify birds by sight and sound.
Aquatic species relocation	Bachelor of Science in Wildlife Biology or related field and 2 years experience in the biology and natural history of local aquatic species. Relocation activities shall be conducted by a qualified biologist with demonstrated experience handling aquatic species.

Survey the job site for regulated species and submit a preconstruction survey report within 14 days before starting work.

The preconstruction survey report must include one of the following:

1. Detailed observations and locations where regulated species were observed
2. Statement that no regulated species were observed

Submit an initial monitoring report as an informational submittal within 12 hours after starting ground-disturbing activities.

Submit a biological resource incident report within 24 hours of the incident.

The incident report must include:

1. Description of any take of regulated species or any violation of a biological resource PLAC
2. Species name and number taken
3. Details of required notifications with contact information
4. Corrective actions proposed or taken
5. Disposition of taken species

Submit an annual monitoring report no later than January 15 during each year of construction.

The annual monitoring report must include:

1. Start and end dates of construction
2. Project impacts on the regulated species
3. Species protection measures and implementation details
4. Incidental take details, including species name, number taken, people contacted, contact information, and disposition of taken species
5. Assessment of the effectiveness of the species protection measures in mitigating project impacts
6. Recommendations for improving species protection measures

Submit a final monitoring report no later than 20 days after completion of the project. If the report requires revisions, the Department provides comments. Submit a revised report within 7 days of receiving comments. The final monitoring report must be a cumulative report including:

1. Start and end dates of construction
2. Project impacts on the regulated species
3. Species protection measures and implementation details
4. Incidental take details, including species name, number taken, people contacted, contact information, and disposition of taken species
5. Assessment of the effectiveness of the species protection measures in mitigating project impacts
6. Recommendations for improving species protection measures

Add to section 14-6.03:

14-6.03E Contractor-Supplied Arborist

14-6.03E(1) General

Section 14-6.03E applies if a bid item for a contractor-supplied arborist is shown on the Bid Item List. The contractor-supplied arborist monitors work activities to protect trees.

The contractor-supplied arborist must:

1. Monitor work activities that could impact trees in Tree Protection Area
2. Ensure that construction activities do not result in damage to trees that jeopardize tree survival or longevity
3. Immediately notify the Engineer of an activity that has jeopardized a tree
4. Prepare, submit and sign incident reports

An incident is substantial damage to a tree which could jeopardize the tree survival or longevity. Incident reports must include:

1. Name of arborist conducting monitoring
2. Dates and times of monitoring
3. Locations and activities monitored
4. Representative photographs
5. Findings
6. Name of the arborist who prepared the report
7. Signature of the arborist certifying the accuracy of the report
8. Details of damage to the tree and an assessment of the effects on the survival and health of the tree

Submit incident reports within 24 hours of the incident.

Submit the name, resume, and statement of qualifications for a Contractor-supplied arborist within 7 days after contract approval. Allow 30 days for review. If the submittal is incomplete, the Department provides comments. Submit a revised statement of qualifications within 7 days of receiving comments.

A Contractor-supplied arborist must have a certification from the International Society of Arboriculture and at least 3 years' experience assessing tree survival and health.

Do not start work activities that could impact trees until the Contractor-supplied arborist is approved.

Replace *Reserved* in section 14-6.05 with:

14-6.05 INVASIVE SPECIES CONTROL

14-6.05A General

14-6.05A(1) Summary

Section 14-6.05 includes specifications for preventing the introduction and spread of invasive species to and from the job site.

Vehicle equipment and cleaning must comply with section 13-4.03E(3).

14-6.05A(2) Definitions

Susceptible areas: Job Site.

14-6.05A(3) Submittals

Submit an invasive species control plan prepared by the contractor supplied biologist. The plan must describe measures for preventing the introduction and spread of invasive species.

Include the following protection measures in the plan:

1. For work at the job site:
 - 1.1. At least 2 business days before using equipment and vehicles on the job site, submit a signed statement that the equipment and vehicles have been cleaned of soil, seeds, vegetative matter, and other such debris that may introduce or spread invasive species. The statement must include:
 - 1.1.1. Equipment identification number and description for equipment and vehicles that could introduce or spread invasive species
 - 1.1.2. Date of cleaning for each piece of equipment and vehicles
 - 1.1.3. Description of the cleaning process
 - 1.1.4. Measures to ensure the equipment and vehicles remain clean until operation at the job site
 - 1.1.5. Measures to ensure runoff resulting from the cleaning of equipment and vehicles will not flow into waterways, drainage inlets, or sensitive habitat.
 - 1.1.6. Update the list of equipment and vehicles as needed
 - 1.1.7. Verification that the equipment and vehicles have not been operated in waters known to be infested by aquatic invasive species
 - 1.4. Equipment cleaning station and proposed location of station. Typical equipment used in Forest Service operations is included in the *Information Handout*, Missoula Technology and Development Center (MTDC) Portable Vehicle Washer. Propose equivalent or better.
2. For work in susceptible areas:
 - 2.1. Pressure wash equipment and vehicles before entering with hot water (minimum temperature of 140 degrees F).
 - 2.2. Decontaminate personal equipment used in water at the job site, such as boots, waders, or hand tools, in accordance with CDFW Aquatic Invasive Species Decontamination Protocol 2022.
 - 2.3. Before exiting, ensure equipment and vehicles are free of soil and plant material.
 - 2.4. Re-clean any equipment or vehicle that is removed from the susceptible area before operating it again in the susceptible area.
4. Procedures that will be taken to verify that equipment and vehicles have been cleaned when moving offsite, to prevent cross contamination to other locations.

14-6.05B Materials

Not Used

Manage work activities to comply with the approved plan.

14-6.05D Payment

Add to section 14-8.02:

Submit a noise control plan (NCP) within 15 days of contract approval to minimize construction noise. Do not start affected work until the plan has been authorized.

1. Identify the activities, equipment, and processes that may cause noise levels in excess of the permitted limits
2. Include a noise abatement strategy for each activity, equipment, or process identified
3. Have a contingency plan for each activity, equipment, or process that describes and lists alternative equipment or processes to be used in the event the required noise levels are not achieved

The sound-level meter must:

1. Be calibrated and certified by the manufacturer or an independent acoustical laboratory before delivery to the Department
2. Be capable of taking A-weighted measurements and have slow response settings
3. Have a microphone fitted with a windscreen
4. Be recalibrated annually by the manufacturer or an independent acoustical laboratory

Provide noise monitoring equipment training to 1 Department employee designated by the Engineer. The person delivering the training must be trained in noise monitoring.

The Department returns the equipment to you at contract acceptance.

Add to the 1st paragraph of section 14-12.02:

This project is located within the limits of Richardson Grove State Park, Benbow State Recreation Area, and Humboldt Redwoods State Park.

AA

Add between the 6th and 7th paragraph of section 15-1.03A:

The loop detectors and piezoelectric axle sensors shown in the following table must remain in place and operational.

Co-Route-PM	Location	No. of loops	No. of piezos ^a
Hum-101-33.306	Census #928 @ WEOTT		0
Hum-101-R49.787	Census #801 @ SCOTIA		0

Test the replaced equipment in the presence of the Engineer before Contract acceptance.

DIVISION III EARTHWORK AND LANDSCAPE

17 GENERAL

19-16.01C(2) Contractor Qualifications

Submit:

1. Summary of the contractor's experience that demonstrates compliance with section 19-16.01D(3)
2. List of at least 5 projects completed in past 5 years that demonstrate the contractor's qualifications and ability to perform the work. For each project includes:
 - 2.1. Project description
 - 2.2. Project owner's name, email address, and phone number
 - 2.3. Project completion date

19-16.01C(3) Shop Drawings

Submit 5 copies of shop drawings. After review, submit 8 copies for authorization and use during construction.

Shop drawings must be stamped and signed by an engineer who is registered as a civil engineer in the State.

Shop drawings must include:

1. Your name, address, telephone number, and email address
2. Plans showing work site layout, cross sections and profile of construction equipment, shield cut, overcut, pipes, and construction operation and sequence
3. Details of automated electronic system for excavation and alignment monitoring that can produce continuous record
4. Details of pipe and pipe joints that can carry and uniformly distribute the thrust of jacking forces, if applicable, and other construction loads in addition to overburden, earth and hydrostatic pressures

19-16.01C(4) Calculations

Submit 5 copies of calculations. After review, submit 8 copies for authorization and use during construction.

Calculations must be stamped and signed by an engineer who is registered as a civil engineer in the State.

Calculations must include:

1. Bracing, shoring, and thrust block design
2. Thrust forces and distribution of the forces for trenchless construction
3. Groundwater and surface water flow, and placement and capacity of the dewatering system
4. Estimated ground surface movement profile, and contour on highway pavement and 2 horizontal planes between highway surface and the elevation of trenchless construction.
5. Anticipated quantity of spoils by volume

19-16.01C(5) Work Plans

Submit 5 copies of work plans. After review, submit 8 copies for authorization and use during construction.

Work plans must be stamped and signed by an engineer who is registered as a civil engineer in the State.

Work plans must include:

1. Your name, address, telephone number, and email address
2. Description of the trenchless construction method, sequence of operations, type of excavated face support, and spoil removal
3. Manufacturer and type of construction equipment for excavation, boring, spoil removal, lubrication, jacking, and grouting, related operating system proposed, and capability of equipment chosen
4. Additional subsurface exploration, including locations, drilling or sounding methods, and testing methods
5. Dewatering system, and plan to divert, control, and dispose of surface water and groundwater
6. Plans to reduce movement to be less than the maximum values under sections 19-16.01D(5)(c) and 19-16.01D(5)(d)

7. Contingency plan for: failed excavated face, damaged pipe, excessive ground surface or subsurface movement, deviation of alignment exceeding tolerance, and flooding
8. Methods for inspecting and grouting voids immediately outside of the completed culvert or pipeline
9. Mitigation Plan for restoring the pavement and ground surface

19-16.01C(6) Monitoring Plans

Submit monitoring plans for:

1. Culvert or pipeline grade and alignment control, including monitoring instruments, layout of instrumentation points, construction details, and monitoring frequency
2. Logging of excavated materials, including anticipated volume of excavation and measured volume of removed spoil
3. Critical operations of applicable trenchless construction, including excavation, boring, spoil removal, lubrication, jacking, installation, and grouting
4. Ground surface movement for highway section with Annual Average Daily Traffic (AADT) volume:
 - 4.1. Greater than or equal to 100,000 Include digital surface survey method, survey data processing and analysis method, and digital surface file.
 - 4.2. Less than 100,000 Include survey method, and survey points shown.
5. Subsurface movement, including monitoring instruments and equipment, layout of monitoring alignment, method, and frequency of monitoring
6. Pavement smoothness monitoring, including monitoring instruments and equipment, method, and frequency of monitoring, report templates

19-16.01C(7) Culvert or Pipeline Control Line Survey

Submit control line survey. Control line may be on a local coordinate system if the control is referenced to the project coordinate system.

19-16.01C(8) Daily Construction Record

Submit daily construction record before noon the day after the completion of each work shift.

Daily construction record must include:

1. Date and time of operation.
2. Names of key personnel.
3. Length of constructed culvert or pipeline, including coordinates and elevation of the beginning and ending (latitude, longitude and northing, easting, elevation) of the culvert or pipeline advanced during each work shift. Record must reference the project coordinate system designated by the Department.
4. Rate of advance.
5. Jacking force.
6. Problems encountered, possible causes, and mitigation performed.
7. Geological log of excavated face and materials, with the logging performed by a geologist who is registered as an engineering geologist in the State.
8. Records and field note of:
 - 8.1. Any visible cracks.
 - 8.2. Culvert or pipeline line and grade control.
 - 8.3. Anticipated and actual volumes of spoil removed and causes of the volume discrepancy.
 - 8.4. Groundwater table elevation if dewatering is required.

19-16.01C(9) Ground Surface Movement Monitoring Records

For highway section with AADT volume is equal to or greater than 100,000, submit:

1. Before construction:
 - 1.1. Survey data
 - 1.2. Surface model
 - 1.3. Comparison between your and the Department's surface model demonstrating compliance with the Department's Survey Manual and supplemental guidance
2. During and after construction:
 - 2.1. Survey data
 - 2.2. Surface model

- 2.3. Vertical movement based on the comparison between current and preconstruction surface model

For highway section with AADT volume is less than 100,000, submit:

1. Before construction:
 - 1.1. Survey data
 - 1.2. Comparison between your and the Department's survey demonstrating your survey's compliance with the quality requirements
2. During and after construction:
 - 2.1. Survey data
 - 2.2. Vertical movement based on the comparison of current and the Department's preconstruction surface data

19-16.01C(10) Subsurface Movement Monitoring Records

If subsurface movement monitoring is required under section 19-16.01D(5)(c), submit:

1. Before construction:
 - 1.1. Calibration and verification report of the monitoring system
 - 1.2. Baseline reading
2. Monitoring records during and after construction

19-16.01C(11) Pavement Smoothness Report

If pavement smoothness monitoring is required under section 19-16.01D(5)(d), submit:

1. Inertial profiler certification under section 36-3.01D(2)
2. Pavement smoothness reports in the following schedule:
 - 2.1. Pre-construction report before work starts
 - 2.2. Monthly report by the 5th day of each month during construction
 - 2.3. Post-construction report after trenchless construction and pavement restoration are completed

Pavement smoothness report must include:

1. Inertial profile data with the values of 5 inertial profiler runs for each lane and average values for each lane. The begin and end positions of each inertial profile must be within 1.0 foot of the monitored section.
2. ALR.
3. Faulting at each concrete pavement joint.

Inertial profiler data must comply with section 36-3.01C(2).

ALR values of each inertial profile of the monitored section must be presented as a spreadsheet that includes:

1. Current and previously recorded values
2. Calculated change from pre-construction value to the nearest whole percentage.
3. Values increased by more than
 - 3.1. 10 percent must be highlighted in yellow
 - 3.2. 20 percent must be highlighted in orange
4. Values greater than 180 inches/mile must be highlighted in red.

Faulting at each concrete pavement joint must be presented as a spreadsheet that includes:

1. Current and previously recorded values
2. Values greater than 0.75 inches must be highlighted in red.
3. Profile with the number of faulting greater than 0.1 inches increased by more than
 - 3.1. 10 percent must be highlighted in yellow
 - 3.2. 20 percent must be highlighted in red

19-16.01C(12) Contact Grouting Record

Submit contact grouting record before noon the day after the completion of each work shift.

Contact grouting record must include:

1. Injection locations
2. Grout quantity
3. Grout pressure
4. Measurements and observations, including heave, casing or carrier pipe movement, grout loss quantity, communication between grout ports, ground surface, and nearby utilities and storm drains
5. Problems encountered, possible causes, and mitigation performed

19-16.01C(13) Post-Construction Record

Submit a copy of the completed culvert or pipeline construction inspection records, including video recording and photographs.

Submit as-built plans showing details and alignment of the constructed culvert or pipeline, horizontal and elevation survey based on project coordinate system, any problems encountered, and mitigation actions performed.

Submit as-built plans showing details of pavement restoration work performed.

19-16.01D Quality Assurance**19-16.01D(1) General**

Not Used

19-16.01D(2) Pre-construction Meeting

Hold a pre-construction meeting at least 10 days before the start of culvert or pipeline construction. The Engineer conducts the meeting.

Attendees must include:

1. The Engineer
2. Your project manager
3. Your project superintendent
4. The subcontractor for trenchless construction

Provide and present:

1. Culvert or pipeline construction shop drawings, work plans, and calculations
2. Mitigation plans for both during and after construction
3. Construction timeline and critical path activities

19-16.01D(3) Contractor Qualifications

The contractor for trenchless construction of culvert or pipeline must:

1. Have successfully completed at least 5 projects in the past 5 years involving trenchless construction of culvert or pipeline in similar lengths and diameters in similar geotechnical conditions
2. Employ a superintendent, who has successfully completed at least 5 of such projects

19-16.01D(4) Quality Control

Not Used

19-16.01D(5) Department Acceptance**19-16.01D(5)(a) General**

Not Used

19-16.01D(5)(b) Ground Surface Movement Monitoring

19-16.01D(5)(b)(i) General

Mark monitoring points on critical structures and utilities at locations shown. Include these points in monitoring surveys. Perform monitoring surveys before noon and at ambient temperature below 85 degrees F.

Perform ground surface survey under the Department's Survey Manual and supplemental guidance.

Notify the Engineer at least 15 days before trenchless construction for the Department to perform a pre-construction ground surface survey.

19-16.01D(5)(b)(ii) Highway section with AADT volume equal to or greater than 100,000

The Department will provide:

1. Control points for ground surface movement monitoring.
2. Pre-construction digital surface model. The model will be based on a grid of points spaced not more than 1-foot apart and will extend 50 feet in each direction from the culvert or pipeline centerline.

Use the provided control points for ground surface movement monitoring. Use the provided surface model to determine the movement of ground surface and embankment slope.

Before starting trenchless construction, perform baseline ground surface survey.

Perform ground surface movement monitoring survey

1. Weekly during construction
2. Biweekly for 1 month after completion of each installation

Produce the surface model based on the monitoring survey data and calculate the movement of monitoring points using pre-construction surface model. Each monitoring survey may have different grid points. Digital survey file must be in Civil 3D or Land XML format.

Each ground surface horizontal and vertical measurement must be accurate to ± 0.03 feet on pavement and ± 0.1 feet on unpaved surfaces at the 95 percent confidence level. Vertical movement produced by comparing current surface model with pre-construction surface model must be accurate to ± 0.01 feet on pavement and ± 0.1 feet on unpaved surfaces at the 95 percent confidence level.

If ground surface movement in the pavement above the advancing pipe meets the requirements for 2 consecutive weeks, you may reduce the frequency of monitoring survey to biweekly.

Notify the Engineer at completion of each installation. The Department will perform ground surface survey 1 month after completion of each installation.

Ground surface vertical movement must comply with the following requirements:

Ground Surface Vertical Movement

Quality characteristic	Requirement
Critical Structure Monitoring Points – Horizontal or Vertical (max, feet)	0.02
Highway surface (max, feet)	0.04
Embankment slope (max, feet)	0.2

If ground surface vertical movement requirements are not met:

1. Immediately stop work
2. Notify the Engineer
3. Submit an alternative construction method
4. Submit a mitigation plan that includes methods to fill the voids created under the ground surface and restore the density of subsurface materials
5. Monitor ground surface movement in the area above the advancing pipe:

- 5.1. Daily until no additional vertical movement is detected in the areas that exceed the movement requirements
- 5.2. Every two days until the vertical movement meets the requirements for 2 consecutive weeks

19-16.01D(5)(b)(iii) Highway section with AADT volume less than 100,000

The Department will provide pre-construction survey as the baseline survey.

Use the provided baseline survey for ground surface movement monitoring.

Before trenchless construction, perform ground surface survey under the Department's Survey Manual and supplemental guidance.

Establish monitoring points to be surveyed:

1. At critical structures as shown
2. In a grid extending 20 feet on each side of the culvert or pipeline centerline, including embankment slopes, and spaced at:
 - 2.1. 5 feet along culvert or pipeline centerline
 - 2.2. 10 feet along the roadway centerline

Perform ground surface movement monitoring survey

1. Daily during construction
2. Biweekly for 1 month after completion of each installation

Notify the Engineer immediately of any discrepancies.

Ground surface survey and vertical movement produced by comparing the current survey with pre-construction survey must be accurate to ± 0.03 feet on pavement and ± 0.1 feet on unpaved surfaces at the 95 percent confidence level.

Notify the Engineer at completion of each installation. The Department will perform ground surface survey 1 month after completion of each installation.

Ground surface vertical movement must comply with the following requirements:

Ground Surface Vertical Movement

Quality characteristic	Requirement
Critical Structure Monitoring Points – Horizontal or Vertical (max, feet)	0.06
Highway surface (max, feet)	0.06
Embankment slope (max, feet)	0.2

If ground surface vertical movement requirements are not met:

1. Immediately stop work
2. Notify the Engineer
3. Submit an alternative construction method
4. Submit a mitigation plan that includes methods to fill the voids created under the ground surface and restore the density of subsurface materials
5. Monitor ground surface movement in the area above the advancing pipe twice each work shift until no additional vertical movement is detected in the areas that exceed the movement requirements

19-16.01D(5)(c) Subsurface Movement Monitoring

Subsurface movement monitoring is required if:

1. Bore diameter is greater than 48 inches
2. Minimum vertical distance between pavement surface and top of culvert is less than 8 times the bore diameter

Install horizontal in-place inclinometers with MEMS or ShapeArray to monitor subsurface movement at:

1. 6 feet above the crown and along the center line of the culvert or pipeline
2. 6 feet above the crown and 5 feet away horizontally from the spring lines of the culvert or pipeline.

Install movement monitoring instruments for critical underground utilities at locations shown.

Before trenchless construction, calibrate instruments and monitoring system. Produce calibration and verification report. Take readings at least 3 times to establish pre-construction baseline readings that is accurate to ± 0.01 inches.

Perform subsurface movement monitoring

1. Twice daily during construction
2. Twice weekly for 1 month after completion of each installation

Subsurface movement must comply with the following requirement:

Subsurface Movement

Quality characteristic	Requirement
Subsurface movement (max, inches)	0.2

If subsurface movement requirement is not met:

1. Immediately stop work
2. Notify the Engineer
3. Submit an alternative construction method
4. Submit a mitigation plan that includes methods to fill the voids created under the ground surface and restore the density of subsurface materials

19-16.01D(5)(d) Pavement Smoothness Monitoring

Pavement smoothness monitoring is required if:

1. AADT volume of the section of highway above the culvert or pipeline is greater than 100,000
2. Minimum vertical distance between pavement surface and the top of bore is less than 8 times the bore diameter

Perform inertial profiler verification tests under section 36-3.01D(3)(b)(ii).

Perform preconstruction, monthly during construction, and postconstruction pavement smoothness monitoring using inertial profiler.

Perform pavement smoothness monitoring under section 36-3.

Establish control points that can trigger the begin and end positions of each profile run for each lane. The control points must be located at least 0.1 miles from the culvert or pipeline center line measured along the highway center line. The same control points must be used throughout the construction period.

Perform 5 inertial profile runs of each traffic lane between the begin and end control points. The distance measuring instrument (DMI) direction of increase and the DMI station used to record the begin control point must be the same in all profiles of the lane.

For concrete pavement, faulting must be determined using ProVAL Automated Faulting module, with:

1. Nominal Joint Spacing set to the average slab length
2. Segment Length set to 528 feet
3. Joint Window set to 2 feet
4. Joint Detection Method set to "Step"
5. Cracks unchecked

Pavement smoothness of both asphalt concrete and concrete pavements based on the average ALR of 5 inertial profiles of the monitored section for each lane must meet the following requirements:

1. If preconstruction average ALR is equal to or less than 180 inches/mile, the final average ALR must not increase by more than 20 percent due to construction, with the maximum allowable ALR of 180 inches/mile.
2. If preconstruction average ALR is greater than 180 inches/mile, the final average ALR must not increase by more than 5 inches/mile compared to pre-construction values.

Pavement smoothness of concrete pavement must meet the following additional requirements:

1. If preconstruction average faulting is equal to or less than 0.75 inches, the final average faulting must not increase by more than 0.25 inches due to construction, with the maximum allowable faulting of 0.75 inches
2. If preconstruction average faulting is greater than 0.75 inches, the final average faulting must not increase by more than 0.10 inches compared to pre-construction values
3. Number of faulting with value greater than 0.1 inches must not increase by more than 20 percent due to construction

If pavement smoothness requirements are not met:

1. Immediately stop work
2. Notify the Engineer
3. Submit an alternative construction method
4. Submit a mitigation plan that includes methods to fill the voids created under the ground surface and restore the density of subsurface materials
5. Monitor pavement smoothness in the area above the advancing pipe:
 - 5.1. Weekly until pavement smoothness stops decreasing
 - 5.2. Biweekly until pavement smoothness meets the requirements for 2 consecutive measurements

19-16.01D(5)(e) Culvert or Pipeline Line, Grade, and Shape

For each culvert or pipeline:

1. Survey and record control lines at least 7 days before starting trenchless construction
2. Observe and adjust measurements of survey control lines weekly. Report discrepancies to the Engineer.

Survey and record the centerline of the constructed culvert or pipeline after each section is advanced, or every 5 feet of advancement, whichever is shorter.

Line and grade of a culvert or pipeline centerline must meet the following requirements throughout the entire culvert or pipeline alignment:

Line and Grade

Quality characteristic	Requirement
Line deviation (max, inches)	6
Grade deviation (max, inches)	1

The completed culvert or pipeline must have the shape and dimensions shown throughout the length of the culvert or pipeline. The flow line must be in the direction shown.

19-16.02 MATERIALS

19-16.02A General

For pipe with inside diameter greater than or equal to 24 and less than 48 inches and contact grouting is to be performed from inside of the pipe, the pipe sections must have prefabricated grout ports at 12 o'clock circumferential position every 6 feet along the pipe.

For pipe with inside diameter of 48 inches or larger, the pipe sections must have prefabricated grout ports at 2, 4, 8, and 10 o'clock circumferential positions every 6 feet along the pipe.

19-16.03A General

Repair or replace any damaged pipe sections.

Overcut must be less than

- Not Used

[illegible]

85

Replace section 21-2.02M with:

21-2.02M Hydraulic Biotic Growth Medium

Hydraulic biotic growth medium must be a hydraulically-applied material and may also include seed and fertilizer as shown.

Hydraulic biotic growth medium must be non-toxic

Hydraulic biotic growth medium must be certified weed free under the Department of Food and Agriculture.

Hydraulic biotic growth medium must be a blend of organic and natural fibers with fast-acting soil building and growth components and must be derived and contain three or more of a combination of the following materials:

1. Biochar
2. Humus/Humic Acid
3. Mycorrhizae Fungi
4. Seaweed Extract
5. Trace Elements
6. Growth Stimulators
7. Beneficial Microorganisms
8. Micronutrients
9. Organic Growth Medium

Hydraulic biotic growth medium must comply with the requirements shown in the following table:

Hydraulic Biotic Growth Medium		
Quality Characteristic	Test method	Requirement
Color		Colored to contrast application area, must not stain concrete painted surfaces or HMA
Organic Matter Content (min. %)	ASTM D586	85
Minimum Water Holding Capacity (%)	ASTM D7367	400
Acute Toxicity	EPA 2021.0	Non-Toxic
C:N Ratio	ASTM D1508	10:1100:1
pH	ASTM D1293	5.0-8.5
Moisture Content (% wet weight)	ASTM 586	10 -50
Minimum Vegetation Establishment (%)	ASTM D7322	400

Replace section 21-2.03L with:

21-2.03L Hydraulic Biotic Growth Medium

21-2.03L(1) General

Apply hydraulic biotic growth medium in the locations and rate as shown.

After hydraulic biotic growth medium installation apply all additional erosion control materials for a single area:

1. within 72 hours.
2. prior to a forecasted rain event.

21-2.03L(2) Delivery, Storage and Handling

Products that are not pre-packaged by the manufacturer will not be accepted.

Prior to application, bags of hydraulic biotic growth medium must be stored where they are protected from precipitation, construction operations, and temperatures above 100 degrees Fahrenheit and temperatures below 32 degrees Fahrenheit.

Handle hydraulic biotic growth medium per manufacturer's recommendations.

Clean any spills promptly.

Add to section 21-2.02H:

Straw must be certified weed free from the harvest site by the local County Agricultural Commission or the Department of Food and Agriculture.

Replace the 1st sentence in the 1st paragraph of section 21-2.02P with:

Fiber roll must be a premanufactured roll filled with rice or wheat straw, rice hull, wood excelsior, cotton or coconut fiber.

Add after the 1st sentence in the 1st paragraph of section 21-2.02P:

Straw must be certified weed free from the harvest site by the local County Agricultural Commission or the Department of Food and Agriculture.

Replace section 21-2.02K with:

21-2.02K Compost

Compost must be derived from one or a combination of the following types of materials:

1. Green material consisting of chipped, shredded, or ground vegetation or clean, processed, recycled wood products
2. Biosolids
3. Manure
4. Mixed food waste

Compost must not be derived from mixed municipal solid waste and must not contain paint, petroleum products, pesticides, or other chemical residues harmful to plant or animal life. Metal concentrations in compost must not exceed the maximum listed under 14 CA Code of Regs § 17868.2.

Process compost materials under 14 CA Code of Regs § 17868.3.

The particle size must comply with the requirements shown in the following table:

Compost Gradation

Quality characteristic	Test method ^a	Requirement	
		Min	Max
Gradation Fine:(dry weight % passing) 1-inch sieve 3/8-inch sieve	TMECC 02.02-B	100 95	-- --
Gradation Medium:(dry weight % passing) 2-inch sieve 3/8-inch sieve	TMECC 02.02-B	95 40	-- 55
Gradation Coarse:(dry weight % passing) 3-inch sieve 3/8-inch sieve	TMECC 02.02-B	95 25	-- 35

^aTMECC refers to *Test Methods for the Examination of Composting and Compost*, published by the United States Department of Agriculture and the United States Compost Council (USCC).

The quality characteristics of compost must have the values shown in the following table:

Compost

Quality characteristic	Test method ^a	Requirement	
		Fine	Medium/Coarse
pH	TMECC 04.11-A	6.0–8.0	6.0-8.0
Soluble salts (dS/m)	TMECC 04.10-A	0–10	0-10
Moisture content (% wet weight)	TMECC 03.09-A	25–60	25-60
Organic matter content (% dry weight)	TMECC 05.07-A	30–70	30-100
Maturity (seed emergence) (% relative to positive control)	TMECC 05.05-A	80 or above	80 or above
Maturity (seedling vigor) (% relative to positive control)	TMECC 05.05-A	80 or above	80 or above
Stability (mg CO ₂ -C/g OM per day)	TMECC 05.08-B	5 or below	8 or below
Pathogen Salmonella (most probable number per 4 grams dry weight basis)	TMECC 07.01-B	< 3	< 3
Pathogen Fecal coliform (most probable number per gram dry weight basis)	TMECC 07.01-B	< 1,000	< 1,000
Physical contaminants (% dry weight) Plastic, glass, and metal	TMECC 02.02-C	combined total: < 0.5	combined total: < 1.0
Film plastic (% dry weight)	TMECC 02.02-C	Combined total: < 0.1	Combined total: < 0.1

^a TMECC refers to *Test Methods for the Examination of Composting and Compost*, published by the United States Department of Agriculture and the United States Compost Council (USCC).

Replace section 21-2.02L with:

21-2.02L Fiber Reinforced Matrix

Fiber reinforced matrix must be a hydraulically applied material composed of bonded fiber by the manufacturer. When applied, fiber reinforced matrix must:

1. Form a lofty, flexible, interlocking blanket
2. Be colored to contrast application area
3. Not stain concrete painted surfaces or HMA
4. Have functional longevity of at least 18 months

Fiber for fiber reinforced matrix must contain at least 80 percent wood fiber and comply with the specifications for fiber in section 21-2.02D, except that at least 50 percent is retained when passed through a no. 25 sieve. Fiber must include mechanical reinforcing fibers.

Tackifier for fiber reinforced matrix must comply with section 21-2.02E and may include seed and fertilizer as shown. Tackifier must:

1. Be bonded to the fiber and prepackaged by the manufacturer
2. Contain a minimum of 10 percent of the combined weight of the dry fiber, activating agents, and additives
3. Be an organic, cross-linked, high viscosity colloidal polysaccharide with activating agents or a blended hydrocolloid-based binder

Fiber Reinforced Matrix must comply with the requirements shown in the following table:

Fiber Reinforced Matrix		
Quality characteristic	Test method	Requirement
Ground Cover (%)	ASTM D6567	95
Organic matter content (min. % dry weight)	ASTM D586	90
Water holding capacity (min. %)	ASTM D7367	1,250
Acute Toxicity	EPA 2021.0	Non-Toxic
Maximum Slope Application (H:V)	Observed	1.0:1.0
Cover Factor (C)	ASTM D6459, D8298, D8298M	≤ 0.01
Minimum Vegetation Establishment (%)	ASTM D7322	600

Replace section 21-3 with:

21-3 PERMANENT EROSION CONTROL ESTABLISHMENT WORK

21-3.01 GENERAL

21-3.01A Summary

Section 21-3 includes specifications for performing permanent erosion control establishment work.

Permanent erosion control establishment work consists of weekly inspections of the project site for deficiencies in erosion control features.

The permanent erosion control establishment period starts after permanent erosion control work has been completed.

The Engineer notifies you when the permanent erosion control establishment period starts and furnishes weekly statements regarding the number of working days credited to the permanent erosion control establishment period after the notification.

At the start of the permanent erosion control establishment period you may request relief from maintenance and protection for work items that are not associated with water pollution control and permanent erosion control establishment work.

Working days on which no work is required during the permanent erosion control establishment period are credited as permanent erosion control establishment working days, regardless of whether or not you performed permanent erosion control establishment work.

Working days on which you fail to adequately perform permanent erosion control establishment work as required are not credited as permanent erosion control establishment working days.

Working days that occur after you fail to meet a due date for a Permanent Erosion Control Establishment (PECE) Report submittal will not be credited as permanent erosion control establishment working days.

21-3.01B Definitions

Not Used

21-3.01C Submittals

Submit a Permanent Erosion Control Establishment (PECE) Report form as an informational submittal within 24 hours of completing a weekly inspection and within 24 hours of each qualifying rain event. The

1. Slides
2. Slipouts
3. Surface erosion
4. Damage to:
 - 4.1. Erosion control devices
 - 4.2. Water pollution control devices
5. Poor seed germination
6. Poor plant growth
7. Dead or damaged erosion control plant material
8. Misaligned features
9. Required repair work

Perform a final inspection of the permanent erosion control establishment work in the presence of the Engineer 20 to 30 days before the anticipated contract acceptance date provided by the Engineer.

Not Used

Perform work ordered from the PECE report. This work is change order work.

Not Used

[illegible]

Produce Type A HMA using a WMA additive technology.

Surface abrasion loss (max, g/cm ²)	California Test 360	0.4
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For Type A HMA using RAP substitution of 15 percent or less of the aggregate blend, the grade of the virgin binder must comply with the PG binder grade specified above.

Add to the beginning of section 39-2.02C:

Use a material transfer vehicle when placing Type A HMA if:

1. Quantity of HMA to be paved is greater than 1,000 tons.
2. Any of the following exists:
 - 2.1. Paving is allowed and the ambient air temperature is below 70 degrees F.
 - 2.2. Time from discharge to truck at the HMA plant until transfer to the paver's hopper is 90 minutes or greater.

AA

DIVISION VII DRAINAGE FACILITIES
70 MISCELLANEOUS DRAINAGE FACILITIES

Add to the list in 2nd paragraph of section 70-3.01A:

6. Trenchless culverts

Add to the end of section 70-3.01A:

Installation for trenchless method must comply with section 19-16.

Add to the end of section 70-3.02A:

Welded steel pipe for trenchless installation:

1. Must be spiral or straight seam welded pipe or seamless pipe
2. May be bare inside and out
3. Must be square cut with dead-even lengths compatible with trenchless culvert installation
4. Must have sufficient strength to sustain vertical and internal loads
5. Must comply with or exceed the following minimum wall thickness requirements:

Minimum Wall Thickness

Pipe diameter	Up to 150 foot length	Over 150 foot length
30 in – 38 in	3/8 in	1/2 in
40 in – 60 in	1/2 in	3/4 in

Welded steel pipe couplings and joints for trenchless installation must comply with the following requirements:

1. Couplings must be flush.
2. Joints must be airtight and continuous over the circumference of the pipe with a full bead weld equal to or larger than the pipe thickness when measured at a 45 degree angle to the pipe and coupling interface.
3. Welded coupling joints must be capable of transferring the jacking forces.
4. The welder must be qualified under AWS D1.1.
5. Mechanical press fit design with no internal or external bells or field welding may be used. Field welds must comply with section 49-2.02B(1)(b).

AA

71 EXISTING DRAINAGE FACILITIES

Replace section 71-3.01A(3) with:

71-3.01A(3) Submittals

71-3.01A(3)(a) General

For each culvert, submit a work plan in a 3-ring binder that has labeled dividers for each type of plan. The work plan must include:

1. Cleaning plan
2. Inspection plan
3. Contact grouting plan when this work is described
4. Annular-space grouting plan when this work is described
5. Inspection and evaluation report
6. Postrehabilitation inspection report

For each portion of the work plan, obtain the Department's authorization before you perform work based on that portion.

At the prerehabilitation meeting, submit the work plan. Include work plans for contact grouting and annular grouting when this work is described. Allow 7 days for the Department's review. If multiple binders are submitted simultaneously or a binder is submitted before the review of a previously submitted binder is complete, designate the sequence that the binders are to be reviewed. Allow review time specified plus 3 days for each additional binder.

Within 7 days of the inspection of a culvert, submit an inspection and evaluation report. Allow 7 days for the Department's review. If multiple reports are submitted simultaneously or a report is submitted before the review of a previously submitted report is complete, designate the sequence that the reports are to be reviewed. Allow the review time specified plus 3 days for each additional report.

Within 7 days of the completion of all culvert or work, submit a postrehabilitation inspection report as an action submittal. Allow 5 days for the Department's review. If multiple reports are submitted simultaneously or an additional report is submitted before the review of a previously submitted report is complete, designate the sequence that the reports are to be reviewed. Allow the specified review time plus 3 days for each additional reinspection report. Obtain written approval of the reinspection report.

To make a change to an authorized plan or report, submit it as specified for a new submittal. Note the changes.

Retain a copy of the culvert inspection videos and records until the postrehabilitation inspection report is authorized.

Within 5 days of completing annular space grouting at a culvert, submit the grouting records.

71-3.01A(3)(b) Cleaning Plan

The cleaning plan must include methods for:

1. Cleaning.
2. Controlling sediments dislodged during the cleaning.
3. Controlling and diverting the existing stream or groundwater flow. The bypass system must have adequate capacity and size. Include:
 - 3.1. Your calculations for the existing flows and the capacity of the bypass system.
 - 3.2. Your schedule for the required use of the bypass system.

71-3.01A(3)(c) Inspection Plan

The inspection plan must include:

1. Sample of the printed CCTV video log. The printed log must include digital photographs of material not dislodged during cleaning operations, locations where invert repairs were necessary, culvert joints, lateral connection joints, protrusions, and other portions of the culvert drain.
2. Sample CCTV video recording from previous culvert drain inspection work. The video must have audio commentary stating operational and structural defects in pipes that are the same size as those for this Contract. Sample video recording must be taken with the same camera and lighting equipment proposed for this work. Describe the camera type, model and transporter.

If human entry is possible, the plan must also include samples obtained from handheld video and still photographs. Include the following:

1. Sample of the printed video log. The printed log must include digital photographs of material not dislodged during cleaning operations, locations where invert repairs were necessary, culvert joints, lateral connection joints, protrusions, etc.
2. Sample video recording from previous culvert inspection work. The video must have audio commentary showing operational and structural defects in pipes that are the same size as those for this Contract. Sample video recording must be taken with the same camera and lighting equipment proposed for this work. Describe the camera type, model, and transporter.

71-3.01A(3)(d) Contact Grouting Plan

If contact grouting is described, ordered, or proposed as a repair method, submit a contact grouting plan.

The grouting plan must include:

1. Order of work.
2. Maximum injection pressures.
3. Details and data for drilling and grouting equipment.
4. Plans for controlling groundwater and existing culvert stream flows.
5. Pressure gauge, recorder, and field equipment certifications, including calibrations by an independent testing agency.
6. Sample printout of the form for recording grouting operations. Form must show the following tabulated information for each grout port:
 - 6.1. Port location.
 - 6.2. Pressure.
 - 6.3. Volume.
 - 6.4. Start and end time.
7. Schedule of grout port installations and method for obtaining probe-depth dimensions at grout ports; tabulation of locations and dimensions.
8. Culvert strut details as necessary.
9. Method for monitoring deformation of culvert or concrete lining.
10. Grout mix design, including:
 - 10.1. Densities and viscosity.
 - 10.2. Initial set time.
 - 10.3. Materials and the independent testing agency's test data as specified in section 41-2.
 - 10.4. Grout working time before 15 percent change in density or viscosity occurs.

71-3.01A(3)(e) Annular Space Grouting Plan

If a pipeliner with annular space grouting is described, submit an annular space grouting plan. Submit a separate plan for each kind of grout or variation in grouting procedure or pipeliner installation. For each plan, state the corresponding culvert location and applicable installation conditions. With authorization, you may change the grout mix or grouting procedure before starting grouting at the given culvert location. The grout plan must show grouting methods and procedures and include:

1. Grout mix design. Include details of and test results from an independent testing agency for the following:
 - 1.1. Components and proportions.
 - 1.2. Densities and viscosity.
 - 1.3. Initial set time of the grout.
 - 1.4. 28-day minimum grout compressive strength.
 - 1.5. Grout working time before a 15 percent change in density or viscosity occurs.
2. Maximum injection pressures, including those for the stage at the crown of the culvert.
3. Proposed grout lift heights and volumes (e.g., stage 1 to spring line; stage 2 is fully grouted).
4. Bulkhead designs and locations.
5. Calculations of the buoyant forces during grouting and details for holding the pipeliner on the invert of the culvert for a period of time long enough to allow the first lift of grout to set before proceeding to the second lift.
6. Details for re-establishing lateral connections and openings.

7. Pressure gauge, recorder, and field equipment calibration records from an independent testing agency.
8. Proposed number and location of vents relative to the pipeliner diameter, pipeliner stiffness, and depth of flow of grout in the pipe.
9. Culvert strut details as necessary.
10. Written confirmation that you have coordinated grouting procedures with the grout installer and pipeliner manufacturer.
11. Calculations or other documentation verifying that pipeliner joints will not leak, separate, or deform under the proposed grouting pressure.

71-3.01A(3)(f) Inspection and Evaluation Report

Report the inspection findings and recommended repairs. CCTV or video recording and photography must comply with section 71-3.01A(4).

The inspection and evaluation report must include:

1. 2 copies of the CCTV or video recording. On the outside of each copy include:
 - 1.1. Recording number.
 - 1.2. Inspection date.
 - 1.3. Current distance along the culvert measured by a count meter.
 - 1.4. Encoded text description of location, culvert size, type and length.
 - 1.5. Printed labels with location and date on a hard copy of the video recording.
 - 1.6. Audio portion stating:
 - 1.6.1. Inspection date.
 - 1.6.2. Confirmation of the orientation and origin of the tape counter meter.
 - 1.6.3. Description of culvert size, type and length.
 - 1.6.4. Description and location of each defect.
 - 1.6.5. Description and location of each lateral pipe opening.
 - 1.6.6. Description of flow direction.
2. 1 set of the digital photographs.
3. Documentation of the condition of the culvert, including:
 - 3.1. Any condition that might prevent proper installation of pipeliner or invert paving.
 - 3.2. Protrusions.
 - 3.3. Collapsed or crushed areas.
 - 3.4. Reduced cross-sectional areas.
 - 3.5. Each lateral pipe opening, including:
 - 3.5.1. Drainage system identification.
 - 3.5.2. Distance into culvert.
 - 3.5.3. Exact position and orientation within culvert wall.
 - 3.5.4. Size or dimensions of opening.
 - 3.5.5. Connecting joint.
 - 3.5.6. Flow direction.
 - 3.6. Each culvert joint.
4. Recommendation for repairs or statement that no repairs are needed.
5. If repairs are recommended, describe the location and conditions, including any sharp or protruding appurtenances that might snag or tear the pipeliner. If possible, include a detailed evaluation by human entry of the areas where repairs are needed. Describe any corrective action proposed to re-establish lateral pipe and service openings.

If the Engineer determines that the inspection documentation is not adequate, your inspection and evaluation report will be rejected and you must reinspect and resubmit your report. Inadequate documentation may include poor camera head position, poor camera focus, poor illumination, rapid rate of progression, and incomplete records and logs. No payment is made for recleaning and reinspecting.

71-3.01A(3)(g) Postrehabilitation Inspection Report

After the completion of the postrehabilitation inspection, submit a postrehabilitation inspection report. The written logs, videos, and photographs must comply with section 71-3.01A(4).

The report must document the repairs, rehabilitation, and any subsequent deficiencies including:

1. Defects
2. Discoloration
3. Irregularities
4. Surface discontinuities
5. Anomalies
6. Constrictions
7. Deformities

If there are deficiencies and human entry is possible, the report must include more detailed documentation of human-entry inspection, concentrating on the areas with deficiencies.

If there are deficiencies, the inspection report must include recommendations to correct the deficiencies.

If the Engineer determines that the inspection report is not adequate, your report will be rejected and you must reinspect and resubmit your report. Inadequate documentation may include poor camera head position, lack of focus, poor illumination, rapid rate of progression, and incomplete records and logs. No payment is made for recleaning and reinspecting.

If there are deficiencies, do not start corrections until you have obtained authorization of the report.

Replace section 71-3.02 with:

71-3.02 FILL CULVERT VOIDS

71-3.02A General

Section 71-3.02 includes specifications for filling voids below and around a culvert that have been found during cleaning and inspection.

Wherever pipeliners are shown, fill voids before installing pipeliners.

Wherever there are voids in the materials below the invert of the culvert and these voids are greater than 6 inches deep, fill the voids with slurry cement backfill.

Filling the voids below the invert of the culvert with slurry cement backfill is paid for as change order work.

Wherever there are voids in the materials around the culvert and these voids are greater than 6 inches deep, fill the voids with contact grout.

If grout is used to fill voids:

1. Submit a contact grouting plan under section 71-3.01A(3)(d)
2. Schedule and conduct a pregrouting meeting under section 71-3.01A(4)(b)(iii)

Filling the voids around the culvert with grout is paid for as change order work.

71-3.02B Materials

Not Used

71-3.02C Construction

After receipt of the inspection and evaluation report, the Engineer may order additional void-detection work including probing and hammer sounding. Additional void-detection work is change order work.

Prevent the flow of cementitious material and water from construction activities into waterways and drainage facilities.

If voids are found, install grout ports as ordered. Install valves or removable plugs at grout ports. Probe at each grout port location. The probe must be at least 4 feet long, fit through the grout ports, and be rigid enough to sense probe refusal. Extend grout ports through the invert paving using steel pipes or suitable packers. Grout ports must be watertight. If authorized, you may screw grout ports in place or attach them by other methods. Do not weld grout ports to galvanized surfaces.

Pump grout into voids until it appears that all water and air has been ejected. Plug grout ports or close port valves as soon as you stop pumping the grout.

The maximum injection pressure at the nozzle must not exceed 5 psi for fluid, unsanded grout mix.

Monitor the culvert for deformation and cracks. If cracking occurs in a concrete culvert or lining, reduce the grout injection pressure. If deformation of the existing structure exceeds 1/2 inch at any location, reduce the injection pressure.

Repair any permanent deformations or cracks resulting from your grouting work. The Department does not pay for these repairs.

71-3.02D Payment

Record the quantity of slurry cement backfill that is installed and submit this quantity. The Department does not pay for slurry cement backfill that leaks through to the inside of the culvert. The Department does not pay for slurry cement backfill that is wasted, disposed of, or remaining on hand after completion of the work.

Record the quantity of grout that is installed and submit this quantity. The Department does not pay for grout that leaks through to the inside of the culvert. The Department does not pay for grout material that is wasted, disposed of, or remaining on hand after completion of the work.

Replace section 71-3.04 with:

71-3.04 INVERT PAVING

71-3.04A General

Section 71-3.04 includes specifications for preparing the surface of the culvert invert, installing bar reinforcement and anchorage devices, and paving the invert with concrete, shotcrete, or other authorized material.

71-3.04B Materials

Concrete must comply with the requirements for minor concrete and be placed to the thickness shown.

You may use shotcrete under section 53 or other authorized material for invert paving from the Authorized Material List for cementitious pipeliners and concrete invert paving as alternatives to concrete. You must use the same thickness shown for concrete.

Reinforcement and welded wire fabric must comply with section 52.

Steel angles and studs must comply with section 75-1.02.

71-3.04C Construction

Before placing invert paving, perform a mock-up field test in the presence of the Engineer to demonstrate the installation method. If the Engineer accepts the field test, you may start invert paving activities.

Before paving the invert, remove all coatings, corrosion, and other surface material until only base steel is exposed by sandblasting the portion of the culvert to be paved.

Minimum thickness must be measured over the crest of the corrugation.

Placing concrete paving must comply with the specifications for placing, finishing, and curing concrete structures in section 51-1.03.

Placing products listed on the Authorized Material List for cementitious pipeliners and concrete invert paving as an alternative to concrete or shotcrete must comply with the manufacturers' procedures for placing, finishing, and curing.

Placing, finishing, and curing shotcrete must comply with section 53.

71-3.04D Payment

The Department does not deduct the volume of reinforcement and miscellaneous metal from the payment quantity for invert paving.

Replace section 71-3.05 with:

71-3.05 EPDM RUBBER PIPE JOINT SEAL

71-3.05A General

71-3.05A(1) Summary

Section 71-3.05 includes specifications for installing EPDM rubber pipe joint seals onto the interior of an existing pipe or culvert, creating a circumferential leak resistant seal at joints.

71-3.05A(2) Submittals

Submit a work plan for installation of EPDM rubber pipe joint seals.

The work plan must include:

1. Installation procedure.
2. Identification of manufacturer and authorized installer.
3. List of equipment to be used.
4. Technical data for proposed internal joint seal including material safety data sheets, application instructions and product data.
5. Certificates of compliance showing that products submitted meet requirements of standards referenced and pipe diameter shown.
6. Manufacturer's recommendations for delivery, storage, and handling.
7. Certificates of qualification of each technician, unless manufacturer's representative will be present for all seal installation activities.
8. Summary sheet for each culvert. Identify the summary sheet by the structure number shown for the corresponding culvert. Summary sheets must include:
 - 8.1. Width dimensions of EPDM rubber pipe joint seal for pipe diameter.
 - 8.2. Manufacturer's recommendations for:
 - 8.2.1. Minimum band dimensions for pipe diameter.
 - 8.2.2. Seal width.
 - 8.2.3. Maximum allowable pressures to ensure no damage to the seal or to the culvert. Assume the groundwater level is at 1/2 the culvert depth.
 - 8.2.4. Backing plate width as required.

71-3.05A(3) Quality Assurance

71-3.05A(3)(a) General

Sealed joints must be individually pressure-tested a minimum of 30 minutes after the final fitting of the seal in compliance with the following air test procedure steps:

1. For seal diameters up to 84 inches, apply a pressure of 10 psig to the seal through the test valve and maintain with a regulated air supply. For seal diameters from 84 to 96 inches, apply a pressure of 7 psig to the seal through the test valve and maintain with a regulated air supply. For seal diameters greater than 96 inches, apply a pressure of 5 psig. Apply a soap and water solution to the outer edge and entire body of the seal to detect leakage. Use a restraining device locked in its expanding position during testing to prevent excessive ballooning to the center membrane of the seal.
2. If the pressure test indicates leakage, determine cause and repeat step 1. If the pressure test indicates leakage a second time, replace with new seal and repeat step 1.
3. Repeat step 1 using a pressure of 5 psig without the restraining device.
4. Inspect for leakage. If the pressure test indicates leakage, remove seal and replace with new seal and repeat steps 1 through 4.
5. Seal the test valve of the seal with a counter sunk hex head completion plug and a nontoxic thread sealing compound on the threads.
6. Remove all installation hardware and pressure gauges.

71-3.05A(3)(b) Qualifications

Either a manufacturer representative must be present or internal seal installation technicians must be certified by the manufacturer to safely install seals of the types and sizes being used. Certification must be current on the actual date of installation.

71-3.05B Materials

71-3.05B(1) General

Except as amended under these special provisions, comply with the following:

1. ASTM D395: Standard Test of Rubber Compression Set
2. ASTM D1171: Standard Test Method for Rubber Deterioration Surface Ozone Cracking Outdoors or Chamber
3. ASTM D2000: Standard Classification System for Rubber Products in Automotive Application
4. ASTM D3568: Standard Test Method for Rubber-Evaluation of EPDM. Including Mixtures with Oil

71-3.05B(2) EPDM Rubber Membrane

Each EPDM width dimension in your submittal must comply with the manufacturer's recommendations in your work plan.

EPDM rubber membrane must be at least 1/4 inch thick and must:

1. Contain ingredients that comply with 21 CFR § 177.2600. The final rubber membrane must not support microbiological growth when used in potable water, sea water, or in humid aerobic conditions.
2. Have a volume change that does not exceed 3 percent when tested after immersion in fresh or sea water at 212 degrees F for 70 hours.
3. Have a stress relaxation that does not exceed 12 percent when tested from a time of 30 minutes to 24 hours.
4. Comply with the requirements in the following table:

Quality characteristics	Test method	Requirement
Tensile strength (psi, min)	ASTM D412	1450
Tensile strength change after heat aged for 70 hours at 70° C (max, %)	ASTM D573	30
Elongation (min, %)	ASTM D412	350
Elongation change after heat aged for 70 hours at 70° C (max, %)	ASTM D573	-50
Durometer hardness	ASTM D2240	65 ± 5
Durometer hardness after heat aged for 70 hours at 70° C (max)	ASTM D573	±15
Ozone resistance (min, %)	--	85
Permanent set after compression set for 22 hours at 70° C (max, %)	ASTM D395B	25

71-3.05B(3) EPDM Joint Splicing

The joint splice must be manufactured using a compression-molding method with virgin rubber of the same compound used to manufacture the seal. Maintain a uniform cross section with a minimum thickness of 1/4 inch at interface.

Vulcanize the joint splice at 330 degrees F minimum and a pressure of 2000 psi.

Do not use adhesive or glue.

71-3.05B(4) Bands, Backing Plates, Shims, Set Screws, and Test Valve

Bands, backing plates, spacers, shims, clips, and set screws for securing rubber membrane across pipe joints and test valve must be UNS S31603, Type 316L, stainless steel and comply with ASTM A240.

Push tabs must be from the same manufacturer lot number as the bands.

Bands must:

1. Meet ASTM A240
2. Have a minimum strength of 70,000 psi
3. Have a minimum yield strength of 25,000 psi in tension
4. Have a maximum Brinell hardness of 217 or a maximum Hardness Rockwell B value of 92
5. Have a minimum elongation of 40 percent or more when tested at a rate of 20 ± 2 inches per minute

Bands and shims must be rolled to the radius of the pipe being repaired. Chemical properties of base metal for retaining band must be compatible with the weld wire.

The minimum band dimensions must comply with the manufacturer's recommendations in your work plan.

Backing plates must be 16 gauge stainless steel.

71-3.05B(5) Cement Mortar

Cement mortar for pipe joint sealing must be fast setting type suitable for wet and dry conditions and comply with ASTM C150.

71-3.05B(6) Joint Lubricant

Joint lubricant must:

1. Be nontoxic and must not support bacteria growth
2. Not have any deteriorating effects on natural or synthetic rubber
3. Not contain any petroleum based oils or greases
4. Be suitable for intended use from 0 to 120 degrees F

71-3.05B(7) Epoxy

Epoxy must:

1. Be nontoxic and must not support bacteria growth
2. Not have any deteriorating effects on natural or synthetic rubber
3. Provide chemical resistance to common acids, bases, solvents, and oils
4. Not contain any petroleum based oils or greases
5. Contain 100 percent solids
6. Not contain any solvents
7. Not contain volatile organic compounds (VOCs)
8. Be temperature resistant up to 250 degrees F
9. Cure in approximately 4 to 8 hours at 70 degrees F
10. Provide a service temperature range of 50 to 200 degrees F
11. Comply with the requirements in the following table:

Quality characteristic	Test method	Requirement
Flexural strength (min, psi)	ASTM D790	6000
Flexural modulus (min, psi)	ASTM D790	550,000
Compressive yield strength (min, psi)	ASTM D695	4000
Tensile strength (min, psi)	ASTM D638	3700
Tensile ultimate elongation (max, %)	ASTM D638	1.4
Shore D hardness(min)	ASTM D2583	85
Pencil hardness	ASTM D3363	6H
IZOD impact (ft-lb/in of notch, min)	ASTM D256	0.19

71-3.05C Construction

Remove grease, dirt, dust, roots, scale, and other solid and semisolid debris from the host pipe walls a minimum of 3 inches beyond each side of proposed seal position. Remove all dirt and debris from the joint gaps, leaving a clean area for joint filling. Perform these cleaning operations with hand brushing and scraping, pneumatic wire brush, pneumatic grinder, or oil free air jet. Dispose of all material from area to be sealed.

Protect host pipe from damage during cleaning operations. Repair host pipe that is damaged by the cleaning operations.

Fill any joint gaps, low areas and deep imperfections of the pipe periphery with cement mortar or epoxy and on each side of the joint where the seating surface band location of the seal is to be located and render flush with the surrounding joint surface in compliance with the manufacturer's recommendations. Dispose of surplus material spillage.

Pack joint separations with cement mortar or epoxy using a hand trowel.

If host pipe has a concrete surface, apply a thin layer of cement mortar or epoxy to the preparation area where the seal will be placed.

If host pipe is corrugated metal, EPDM seal must have a smooth surface on both sides with no extrusions. Retainer bands must be installed over the rerolled ends near the joint.

Wait a minimum of 30 minutes after cement mortar or epoxy application to begin seal installation or backing plate placement.

Verify the sealing surface surrounding the joint area and the area where the seals are to be seated is free of debris and smooth.

Immediately before fitting the seal, lubricate the prepared seal area with a brush. Do not reintroduce debris from the brush, surrounding surfaces, or areas into the prepared seal area.

Position the seal parallel to the joint gap with the pressure test valve located at approximately 3 o'clock or 9 o'clock positions.

Install pipe joint seal per your authorized work plan.

Hydraulic expander for installation of steel and stainless steel expansion bands must be capable of providing hydraulic expansion pressures of 6,000 psi.

71-3.05D Payment

Not Used

Replace section 71-6.03 with:

71-6.03 CULVERTS AND PIPELINES

71-6.03A General

Abandon culverts or pipelines by removing portions of the culverts or pipelines, filling the inside, and backfilling the depressions and trenches to grade. As an alternative to abandoning a culvert or pipeline, you may remove the culvert or pipeline, dispose of it, and backfill.

Notify the Engineer before abandoning a culvert or pipeline.

71-6.03B Materials

Openings into existing structures that are to remain in place must be plugged with minor concrete under section 90.

71-6.03C Construction

Wherever culverts or pipelines intersect side slopes, remove them to a depth of at least 3 feet. Measure the depth normal to the plane of the finished side slope. Abandon the remaining portion of the culvert or pipeline.

Culverts or pipelines that are 12 inches or more in diameter must be completely filled by authorized methods. Backfill with sand that is clean, free draining, and free from roots and other deleterious substances. As an alternative to sand, you may backfill with one of the following:

1. Controlled low-strength material under section 19-3.02G
2. Slurry cement backfill under section 19-3.02E

Ends of culverts and pipelines must be securely closed by a 6-inch-thick, tight-fitting plug or wall of commercial-quality concrete.

71-6.03D Payment

If backfilling inside the culvert or pipeline is required, payment for backfilling inside the culvert or pipeline is paid for as sand backfill. Payment for backfilling outside the culvert or pipeline is included in the payment for abandon culvert or abandon pipeline.

AA

DIVISION VIII MISCELLANEOUS CONSTRUCTION

72 SLOPE PROTECTION

Replace section 72-8 with:

72-8 GRAVEL FILTER

72-8.01 General

Section 72-8 includes specifications for constructing gravel filter.

72-8.02 Materials

Gravel for gravel filter must consist of hard, durable, clean, washed gravel, cobble, crushed stone, crushed rock, or any combination of these. Gravel must be free from organic material, clay balls, and other deleterious substances.

Gravel must have a durability index not less than 40 and must contain at least 90 percent crushed particles when tested under California Test 205.

The percentage composition by weight of Type U gravel filter in place must comply with the gradation requirements shown in the following table:

Gradation Requirements	
Sieve size	Percentage passing
6 inch	95–100
4 inch	65–95
3 inch	30–65
2 inch	20–35
1 1/2 inch	10–25
1 inch	0–10

The percentage composition by weight of Type A gravel filter in place must comply with the gradation requirements shown in the following table:

Gradation Requirements	
Sieve size	Percentage passing
2 1/2 inch	85–100
2 inch	65–85
1 1/2 inch	30–65
1 inch	20–30
3/4 inch	10–20
1/2 inch	0–10

72-8.03 Construction

Deliver uniform mixture of gravel filter to the site. Place gravel filter on streambed subgrade. Spread uniform mixture in layers and shape to thickness and limits shown using suitable equipment.

Local surface irregularities of the gravel filter aggregate must not vary from the planned slope by more than 2 inches as measured at right angles to the slope.

72-8.04 Payment

Not Used